By: Nicolas Cote-Colisson and Charlie Rothbarth

July 2026 www.research.hsbc.com

SpaceX

Initiate at Hold: Rocket-centric

Space, Connectivity, and AI are designed to leverage a repeatable, vertically integrated, cost-efficient model; SpaceX's market leading rocket operation already effectively supports Starlink

Full vertical integration relies on as yet unproven technologies, including orbital compute power for its GenAI models, the outcome of which impacts the required Space launch cadence

We initiate at Hold with an SOTP-based target price of USD115, including an innovation premium of 2x. Our blue-sky scenario is USD293

Financials & valuation: SpaceX

Hold

Financial statements

Year to	12/2025a	12/2026e	12/2027e	12/2028e
Profit & loss summary (USDm)				
Revenue	18,674	38,208	49,783	62,272
EBITDA	4,112	15,426	22,728	33,671
Depreciation & amortisation	-6,701	-15,612	-24,194	-32,187
Operating profit/EBIT	-2,589	-186	-1,465	1,484
Net interest	-1,630	-3,040	-883	-958
PBT	-4,219	-3,226	-2,348	526
HSBC PBT	-1,747	238	3,022	5,877
Taxation	-718	-184	399	-89
Net profit	-4,937	-4,080	-1,949	436
HSBC net profit	-2,465	-617	3,421	5,787
Cash flow summary (USDm)				
Cash flow from operations	6,785	14,708	28,240	38,498
Capex	-20,737	-68,587	-53,847	-59,109
Cash flow from investment	-19,575	-75,204	-53,847	-59,109
Dividends	0	0	0	0
Change in net debt	-3,459	-30,702	26,477	21,521
FCF equity	-14,247	-53,962	-25,607	-20,612
Balance sheet summary (USDm)				
Intangible fixed assets	15,628	75,830	76,230	76,140
Tangible fixed assets	43,862	100,371	149,625	176,547
Current assets	30,952	101,964	55,197	36,714
Cash & others	24,747	87,818	41,316	19,795
Total assets	92,079	279,459	282,345	290,695
Operating liabilities	27,858	34,947	35,706	39,179
Gross debt	22,896	55,265	35,240	35,240
Net debt	-1,851	-32,553	-6,076	15,445
Shareholders' funds	41,325	182,198	204,349	209,227
Invested capital	16,184	155,401	204,029	230,428

Ratio, growth and per share analysis

Year to	12/2025a	12/2026e	12/2027e	12/2028e
Y-o-y % change				
Revenue	33.2	104.6	30.3	25.1
EBITDA	-4.1	275.2	47.3	48.1
Operating profit	-655.6			
PBT	-1843.4			
HSBC EPS	-875.9			69.1
Ratios (%)				
Revenue/IC (x)	0.9	0.4	0.3	0.3
ROIC	-14.5	-0.2	-0.7	0.6
ROE	-7.3	-0.6	1.8	2.8
ROA	-6.6	-1.8	-0.7	0.2
EBITDA margin	22.0	40.4	45.7	54.1
Operating profit margin	-13.9	-0.5	-2.9	2.4
EBITDA/net interest (x)	2.5	5.1	25.8	35.1
Net debt/equity	-4.5	-17.2	-2.9	7.1
Net debt/EBITDA (x)	-0.5	-2.1	-0.3	0.5
CF from operations/net debt				249.3
Per share data (USD)				
EPS Rep (diluted)	-1.69	-0.38	-0.15	0.03
HSBC EPS (diluted)	-0.84	-0.06	0.26	0.44
DPS	0.00	0.00	0.00	0.00
Book value	14.12	16.91	15.63	16.01

Key forecast drivers

Year to	12/2025a	12/2026e	12/2027e	12/2028e
Space revenues	4,086	3,757	4,970	5,680
Connectivity revenues	11,387	15,551	21,903	28,822
AI revenues	3,201	18,901	22,454	27,222
Space Adj EBITDA	653	-740	883	1,641
Connectivity Adj EBITDA	7,168	10,089	14,626	19,864
AI Adj EBITDA	-1,237	9,541	12,589	17,518

Valuation data

Year to	12/2025a	12/2026e	12/2027e	12/2028e
EV/sales	81.0	38.8	30.3	24.6
EV/EBITDA	368.0	96.1	66.4	45.5
EV/IC	93.5	9.5	7.4	6.6
PE*	nm	nm	440.4	260.3
PB	8.2	6.8	7.4	7.2
FCF yield (%)	-0.9	-3.6	-1.7	-1.4
Dividend yield (%)	0.0	0.0	0.0	0.0

* Based on HSBC EPS (diluted)

ESG metrics

Environmental Indicators	12/2025a
GHG emission intensity*	n/a
Energy intensity*	n/a
CO_2 reduction policy	Yes
Social Indicators	**12/2025a**
Employee costs as % of revenues	n/a
Employee turnover (%)	n/a
Diversity policy	Yes

Governance Indicators	12/2025a
No. of board members	8
Average board tenure (years)	12.9
Female board members (%)	12.5
Board members independence (%)	62.5
No. of board members	
No. of board members	

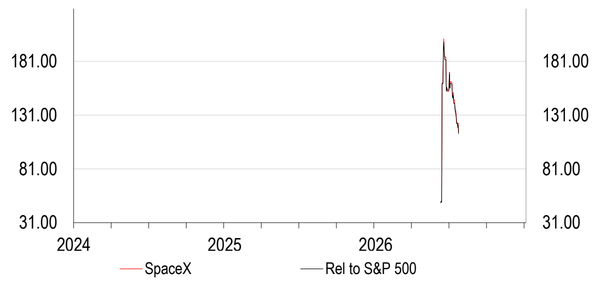
Source: Company data, HSBC

* GHG intensity and energy intensity are measured in kg and kWh respectively against revenue in USD '000s

Issuer information

Share price (USD)	115.26	Free float	5%
Target price (USD)	115.00	Sector	Internet Software & Services
RIC (Equity)	SPCX.N	Country/Region	United States
Bloomberg (Equity)	SPCX US	Analyst	Nicolas Cote-Colisson
Market cap (USDm)	1,516,729	Contact	+44 20 7991 6826

Price relative



Source: HSBC

Note: Priced at close of 22 Jul 2026

Meet our global tech analysts

Nicolas Cote-Colisson*
MD, Head of Global Tech Platforms
Subscribe to Nicolas's research
nicolas.cote-colisson@hsbcib.com

Charlie Rothbarth*
Global Tech Platforms Analyst
Subscribe to Charlie's research
charlie.rothbarth@hsbc.com

Paul Rossington*
Senior Global Tech Platforms Analyst
Subscribe to Paul's research
paul.rossington@hsbcib.com

Abhishek Shukla*, CFA
Senior Analyst, Technology
Subscribe to Abhishek's research
abhishek2.shukla@hsbc.com

Mohammed Khallouf*, CFA
Global Tech Platforms Analyst
Subscribe to Mohammed's research
mohammed.khallouf@hsbc.com

Madhvendra Singh*, CFA
Senior Analyst, TMT
Subscribe to Madhvendra's research
madhvendra.singh@hsbc.com

Phani Kanumuri
Analyst, TMT
Subscribe to Phani's research
phani.kanumuri@us.hsbc.com

Ali Naqvi*
Senior Global TMT Analyst
Subscribe to Ali's research
ali.naqvi@hsbc.com

Frank Lee*
Global Head of Tech Hardware
& Semi Research
Subscribe to Frank's research
frank.lee@hsbc.com.hk

Corey Chan*
(Reg. No. S1700518100001)
Head, A-share Industrials & Renewables Research
Subscribe to Corey's research
corey.chan@hsbcqh.com.cn

Mike Tyndall*, CFA
Senior Global Autos Analyst
Subscribe to Mike's research
michael.tyndall@hsbc.com

Pushkar Tendolkar*
Global Autos Analyst
Subscribe to Pushkar's research
pushkarnarendratendolkar@hsbc.co.in

Dylan Whitfield*, FCA
Head, Accounting Valuations & Forensic Accounting
Subscribe to Dylan's research
dylan.b.whitfield@hsbc.com

Yaryna Kobel
Corporate Governance Analyst
Subscribe to Yaryna's research
yaryna.kobel@hsbc.com

Stephen Bersey
Head of US Technology Research
Subscribe to Steve's research
stephen.bersey@us.hsbc.com

*Employed by a non-US affiliate of HSBC Securiites (USA) Inc, and is not registered/ qualified pursuant to FINRA regulations

Contents

We acknowledge the contribution of Bangalore Associates Sonam Chamaria (Lead AVP) and Pulkit Aggarwal (AVP) in the preparation of this report.

Why read this report?

- SpaceX holds a competitive, leading position in the commercial space launch market, underpinning its long-term strategy
- Strong innovation track record so far; we are prudent on pricing in SpaceX's futuristic calls including space datacentres, Terafab, and the lunar economy
- We apply an "innovation premium" to our fundamental valuation to derive our TP of USD115.00, which implies 0.2% downside, and we therefore initiate with a Hold rating; our blue-sky scenario implies a valuation of USD293.00

How the market views SpaceX

We believe there are two broad investor camps for SpaceX. There are those whose view the company as a foundation platform incorporating all of CEO Elon Musk's vision, which has the potential to change the world. The second camp may not fully dismiss Mr Musk's plans, but is not ready to pay ahead of delivery. We note that Tesla has not traded on classical fundamentals, reflecting the fact that there may be greater support from bullish investors than the more sceptical investors.

What we bring to the debate

SpaceX brings a host of technological innovations and futuristic opportunities. However, there are a series of questions that we believe investors should consider when evaluating company. We aim to analyse these in further detail in this report, alongside challenging some implications from the SEC Form S-1 prospectus, including the total addressable market (TAM) for connectivity, the necessity or feasibility of orbital datacentres, as well as Terafab, and therefore the degree of launch cadence required.

What we expect for 2030e

For the group, we estimate revenues of USD106bn, adjusted EBITDA of USD76bn, GAAP operating profit of USD16bn and FCF of USD12.9bn. From what we can learn from Visible Alpha consensus, we are 72% below expectations on revenues and adjusted EBITDA for 2030e. Ultimately, we may be wrong, but we could be right too – for these higher revenue estimates to come to fruition, a variety of challenges must be overcome, but we do not think this is feasible by 2030.

How we value SpaceX – a detailed SOTP and an "innovation premium"

We value each business segment separately before we apply an "innovation premium" of 2.0x (page 17).

When valuing a company, analysts may sometimes apply a discount (holding structures, dyssynergies, risks etc), but we also consider the case for applying a premium, especially when a strong founder has a proven track record of transforming some industries (including cars and rockets). We considered several valuation approaches, including those for valuing SPACs, mining stocks, or the biotech/pharma sector, but in our view none of these methodologies correctly capture the innovation multiple that we believe should be considered. Therefore, we have looked at Tesla's share price performance over the first 10-years since its IPO, as we believe this is the most appropriate proxy to determine a valuation premium given the commonality in CEO/founder, innovation approach, and developing disruptive technology into a sustainable business.

Based on our observation of Tesla's share price performance and consensus target prices set by sell side research analysts covering the stock, we conclude that over the first 10 years since its IPO, i.e. 2010-20 (we take this period to reflect the early stage and rate of innovation of the business), Tesla's share price was more than double (118%) the average target price the year before. This indicates that fundamental valuation methods do not reflect what the market may be ready to pay for management or 'flywheel' effects, i.e. incremental advances in technology and efficiency compound over time, eventually leading to a sustainable, growing business model. We also conducted the same calculation for the 'Magnificent 7' (Mag-7) peer group (Alphabet, Amazon, Apple, Meta, Microsoft, Nvidia, and Oracle) and derive a much lower premium of 18% over the same timeframe (2010-20). We therefore estimate a 100% (118% for Tesla minus 18% for the Mag-7) "innovation premium" (i.e. 2.0x) for Tesla vs the broader tech market. We think the same premium could prevail with SpaceX, given the similar management approach and possibly a stronger flywheel effect at SpaceX (considering the interconnected business segments). We apply this "innovation premium" to our fundamental sum-of-the-parts valuation to derive our target price of USD115.00. We see that our innovation premium is broadly the inverse of the more widely used "conglomerate discount". As with all valuation premiums, it is liable to change.

Blue sky scenario: USD293 per share
We define bull case scenarios for each segment and aggregate them to build a 'blue sky' scenario for SpaceX and derive a value of USD293 per share.

- Space segment: Our bull case assumes that Starship's commercial viability starts in 2027, with overall mass to orbit 2x what we assume in our base case (page 21).
- Connectivity segment: We increase the addressable market for both Starlink's broadband and mobile services and increase ARPU too, while applying higher EV/sales multiples (page 22).
- AI segment: We increase our valuation multiple by 20% for X, we change our Grok valuation to Palantir EV/Sales, an uplift of 4.2x, assume a higher value for future compute offtake valuation and lift our Cursor valuation by 20% to reflect higher growth opportunities (page 24).

Key questions that investors need to consider and our view

SpaceX's defined total addressable market (TAM): is it really addressable?
SpaceX's SEC Form S-1 prospectus indicates a TAM of USD28.5trn, most of which comprises AI (SpaceX estimates USD26.5trn), followed by Connectivity (voice and broadband communications, USD1.6trn) and Space launch (USD0.4trn).

- **Challenge:** The TAM considered by SpaceX is equivalent to a quarter of global GDP (USD126trn in 2026e World Bank, HSBCe).
- **HSBC's views:** For AI, we do not see SpaceX emerging as a top-tier competitor versus the incumbent AI labs and hyperscalers. xAI currently operates at a scale disadvantage in terrestrial AI compute and frontier model development. Furthermore, for the Connectivity segment (Starlink) we identify a smaller addressable opportunity (USD105-310bn vs USD1.6trn TAM reported by SpaceX) given that traditional telecom operators already address connectivity solutions for a significant portion of the population, especially in urban/semi-urban areas. Whilst we see the Space TAM as the most accessible, it is by far the smallest of the segments (11% of SpaceX's calculated TAM).

Can SpaceX's Grok catch up with top AI LLM labs?
Currently, Grok sits as a high-intelligence frontier model, but lacks the same enterprise and consumer penetration as other AI labs and Alphabet. The company targets Grok as a central part of the AI ecosystem and to serve as a "truth-seeking" AI.

- **Challenge:** Grok is highly loss-making, primarily focused on consumers with limited enterprise take-up. How does Grok break into the top three of Anthropic, OpenAI, and Gemini? xAI operates at a competitive disadvantage on scale in terrestrial AI compute and frontier model development. How will it gain this market share and compete against hyperscalers and the established AI labs?
- **HSBC's views:** For xAI to win AI compute and frontier model market share, we think it would have to spend like hyperscalers/leading AI labs. We forecast FY26e capex at cUSD70bn at SpaceX vs Alphabet at cUSD190bn, heavily weighted to Google Cloud.

Please see more on page 57.

Are orbital datacentres feasible and needed?
SpaceX plans to launch 100GW of annual compute power to orbit, which represents c20% of the annual power production in the US per annum. Amazon aims to add c8GW of compute capacity over a two-year period, to take total compute capacity to 16GW.

- **Challenge:** For orbital datacentres to become the dominant compute option, launch costs need to decline further, certain engineering challenges (cooling, battery, and latency, among others) must be overcome, and operational costs decrease.
- **HSBC's views:** We see no economic rationale for setting up largescale datacentres in space within the next decade, even though SpaceX expects to begin deploying its orbital AI compute satellites as early as 2028 (Google Research's labs is also "exploring a space-based, scalable AI infrastructure system design" with its Project Suncatcher).

Please see more on page 77.

Can Starlink capture market share from the traditional telecom businesses?
Through its Starlink satellite constellation, SpaceX intends to disrupt the incumbent telecom operators. In the prospectus, the company notes that the telecom operators have been reducing capital expenditures, meaning the sector has limited willingness to maintain historically high network deployment.

- **Challenge:** Challenges for the business to reach a global audience are centred around regulation, technology, and a lack of government support. Furthermore, Starlink's premium pricing may constrain adoption.
- **HSBC's views:** Given the challenges above, we estimate that the serviceable obtainable market is 78-92% lower than the company's estimated TAM, and that collaboration with the telecom operators is the most likely route to market. Direct and ubiquitous competition with terrestrial networks would face difficulties inherent to the law of physics (i.e. speed, latency, or indoor coverage).

Please see more on page 50.

The constraints for Terafab
SpaceX's Terafab project aims to create an integrated semiconductor facility, in collaboration with Tesla, in contrast to the highly fragmented value chain of the current semiconductor industry. The project aims to produce output of 1TW per annum in order to meet both internal and external AI demand.

- **Challenge:** This is overly ambitious in our view with few details about the project. Constraints include economical and financial viability, single-source Intel dependency, memory bottlenecks, semi-equipment bottlenecks, energy constraints, and talent deficiency.
- **HSBC's views:** We see this as a long-term aspiration rather than a short-term catalyst, noting timeline difficulties shown by TSMC for a similar project in the US.

Please see more on page 75.

SpaceX's high cash burn

There are a number of short-term cash requirements for SpaceX.

- **Challenge:** First, it needs to spend in-line with peers for the same amount of compute capacity to reach its goals, particular in AI. Second, it is not yet vertically integrated through Terafab, and therefore it has to pay for external compute power (chips) prior to leasing. Third, LLM training and deployment uses capital.
- **HSBC's views:** We expect the business to become free cash flow positive in 2030, and for the company to use USD106bn of cash until then.

Please see more on page 98.

Are investors comfortable with the accounting policies?

The company's financial information relies on subjective judgements, estimation, and management's views about the future aspired operations of the company.

- **Challenge:** The early stage of development of substantial portions (although not all) of the business and the unconventional approach of the company's management should be considered when thinking about the financial reporting of the company and the control environment, which ultimately produces the information and disclosures.
- **HSBC's views:** Although we do not have any significant immediate concerns about the company's accounting, our forensic accountant advises caution, and that particular care is taken when considering the financial information and the company's provided financial disclosures.

Please see more on page 86.

Are the governance policies acceptable?

SpaceX's governance has its idiosyncrasies versus other large tech companies. These include; 1) the only person who can remove the CEO is the CEO himself, 2) 82% of the post-IPO voting rights are held by the CEO and founder and, 3) the company will require mandatory arbitration for shareholder disputes.

- **Challenge:** With 82% of the post-IPO voting rights, Elon Musk is effectively in charge. Therefore, shareholders can only count on the founder/CEO to deliver a strategy that aligns with their interests. In addition, the company's strategy is subject to policy changes in the US administration.
- **HSBC's views:** We do not take a view on governance policies, but we explain these in more detail in this note. However, investors should be aware of them.

Please see more on page 92.

The message from Space(X)

- SpaceX holds a competitive, leading position in the commercial space launch market, underpinning its strategy and the satellite internet industry
- The long-term investment case relies on a vertical AI strategy comprising frontier models, terrestrial and orbital AI compute, and semis with Terafab, while some of these technologies have yet to be developed
- We initiate on SpaceX with a Hold rating and a target price of USD115.00, implying 0.2% downside

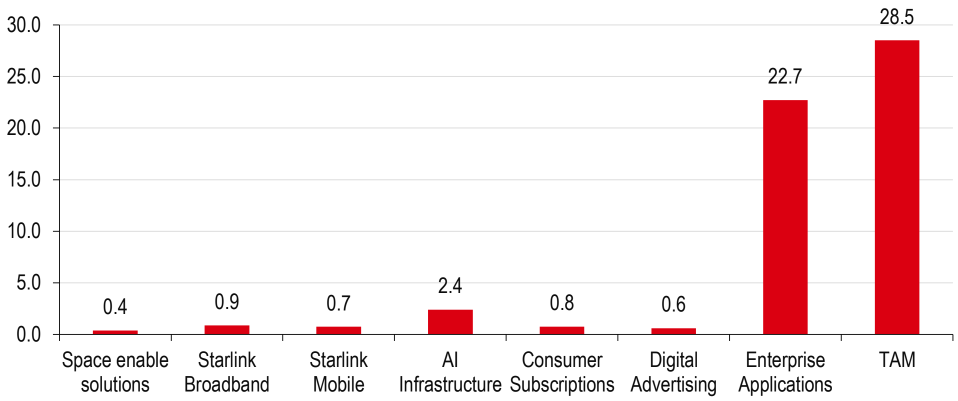
The mission is clear, but is it achievable?

Full vertical integration depends on technological innovation
Founded in 2002 by Elon Musk, SpaceX today is formed of three key pillars of Space, Connectivity, and AI that are designed to leverage a repeatable, vertically integrated business model for the anticipated benefit of cost efficiencies. This promotes space launch capacity supporting Starlink's communication business and in the future, if it can be achieved, orbital compute power to feed its genAI models, while building what may be the highest barrier to entry in the AI value chain, the semiconductors. The company also has ambitions for a potential lunar economy and asteroid mining. If all of these ambitions were to be realised, we think there would be a significant and transformative business 'flywheel'.

- **Space:** SpaceX is the global market leader in space launch; it estimates a TAM of USD0.4trn. 2025 revenues were USD4.1bn, 22% of total revenues.
- **Connectivity:** Starlink is the global leader in satellite connectivity; it estimates a TAM of USD1.6trn. 2025 revenues were USD11.4bn, 61% of total revenues.
- **AI:** The largest addressable TAM at USD26.5trn according to SpaceX; 2025 revenues were USD3.2bn, 17% of total revenues. Of this, USD1.8bn came from advertising. The AI segment comprises infrastructure (currently terrestrial, with ambitions to deploy compute to space), consumer subscriptions, advertising, and enterprise applications.

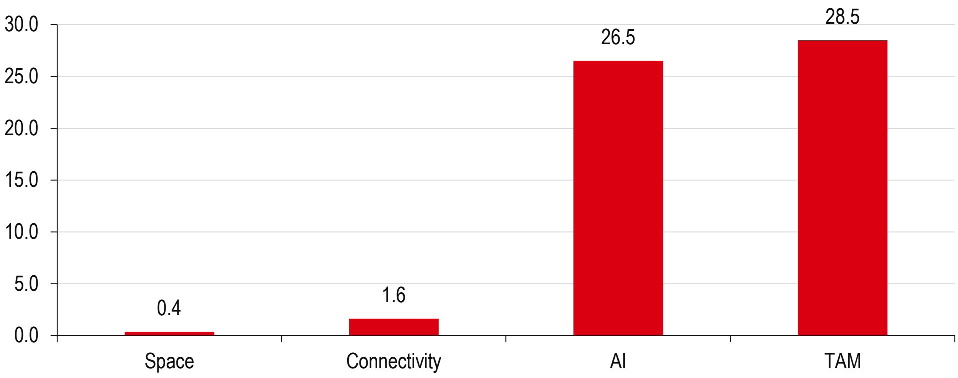
In our valuation process, we do not consider SpaceX's long-term space/interplanetary ambitions including long-haul terrestrial travel, space tourism, lunar/Mars passenger and cargo transportation and energy production, in-orbit manufacturing, or asteroid mining.

SpaceX's estimated total addressable market (TAM) by global industry segment (USDtrn)



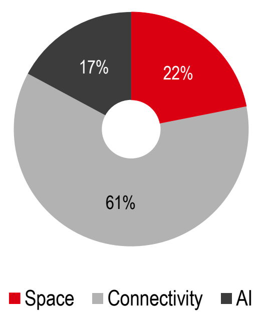
Source: Company data, HSBC

SpaceX's estimated total addressable market (TAM) by business segment (USDtrn)



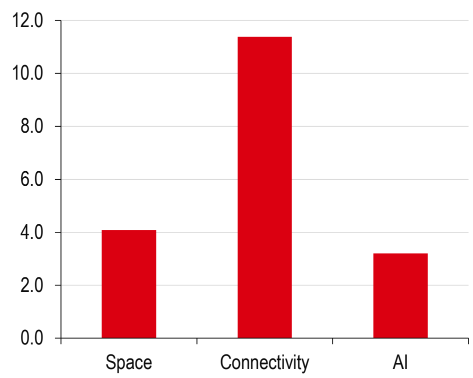
Source: Company data, HSBC

2025A revenue by segment



Source: Company data, HSBC

2025A absolute revenue by segment (USDbn)



Source: Company data, HSBC

A repeatable, vertically integrated business model…

The 'flywheel' explained
SpaceX notes that the cornerstone of its business model is built on its market leading launch capabilities, and a repeatable, engineering-driven framework focused on vertical integration, rapid iteration, and disciplined capital investment, of which the core principles are:

- **Leverage unparalleled launch capabilities,** enabling the deployment of space assets at scale that would not otherwise be economically viable.
  - **Launch business supports Starlink** (Connectivity segment), including its existing global broadband and mobile connectivity for consumers, enterprises, and governments.
  - **As well as AI** if computational infrastructure can be deployed in space, as terrestrial supply is constrained and if Terafab can produce semiconductors.
- **Design solutions with world-class engineering and first-principles thinking,** based on a physics-based, ground-up approach to drive 'step-function' improvements in performance, scalability, and cost.
- **Apply "The Algorithm" (make less dumb, delete, optimise, accelerate, automate),** a five-step iterative process applied across every aspect of the organisation, creating a cultural and operational standard that has defined SpaceX since its inception.
- **Vertically integrate to the end customer** via in-house production across a significant proportion of components (e.g. engines, avionics, structures, software, semis), enabling shorter technological development speed and cost efficiencies.
- **Continuously lower costs and increase throughput via** rocket reusability, manufacturing at scale, advanced automation, and operational discipline to reduce unit costs while increasing launch cadence, satellite network, and AI hosting capacity.
- **Generate significant cash flow and reinvesting in the future/nascent new markets,** driving a self-reinforcing cycle of constant innovation and creating additional value.

…with a demonstratable track record of technological innovation

We outline how SpaceX's repeatable, vertically integrated business model has already delivered tangible benefits to its Space and Connectivity segments, and why this is relevant to the much larger TAM by AI.

Space: market leading position underpinned by lower launch costs
SpaceX was the first private company to launch a liquid fuelled rocket into orbit (2008), dock a private spacecraft with the ISS (International Space Station, 2012), and propulsively land (2015) and re-fly orbital class rocket boosters (2017). Superior cost benchmarking and in-house production (SpaceX manufactures c85% of rocket components in-house) has allowed Falcon 9 to achieve 85% reduction in payload launch costs to USD2,700/kg (vs USD18,500/kg previously), according to NASA, on a 23t payload. Falcon Heavy with a larger payload reduced launch costs further to USD1,400/kg.

End-to-end control over the value chain has delivered cost efficiencies relative to peers, and a reliable and high cadence launch service. As of 31 March 2026, SpaceX had delivered c7,400t of mass into orbit via >650 launches with a 99% mission success rate. Focus is now on the development of Starship, a new fully reusable two-stage rocket that has completed 12 test launches to date, is expected to launch commercially in 2H 2026, targeting a further reduction in launch costs driven by higher payload of 100t.

Connectivity: market leading position supported by rapid LEO deployment

Launched in 2020, Starlink operates the world's largest low-latency space-based internet-broadband and mobile service with fibre-level download speeds at a median of 225Mbps during peak hours. Starlink can provide this service anywhere on Earth, enabled by a network of over c9,600 LEO satellites, with c3,100 launched in 2025, c5x more than the second largest LEO satellite network. Starlink operates c75% of all active, manoeuvrable, satellites in orbit.

Around 85% of Starlink satellite components are produced in-house including specialised chips and power systems (i.e. solar). This has allowed the rapid scaling of production at its Bastrop, Texas facility that, as of late-2025, was reportedly producing over 70 satellites per week. Starlink expects to deploy next-generation V3 satellites via Starship from H2. These are designed to offer 1Tbps of downlink capacity per satellite, improved power capacity and network efficiency, with SpaceX targeting total cost reduction of nine times vs V1 generation satellites via vertical integration.

Compute infrastructure/Grok/consumer and enterprise applications

Through the COLOSSUS and COLOSSUS II facilities, xAI aims to provide 1GW of compute capacity, with additional power capacity for datacentre operations. These facilities were reportedly scaled at speed and lower cost than peers, built in 91-122 days vs industry lead times of two years for 100MW greenfield datacentres. xAI has also launched the 'Terafab' project in conjunction with Tesla and Intel to build production facilities capable of producing 1TW of AI compute hardware per year, thus reducing hardware related bottlenecks to compute growth. SpaceX has an agentic AI collaboration with Cursor, and call option to acquire Cursor in Class A common stock offer for USD60bn.

SpaceX describes Grok as a frontier model that emphasises intellectual freedom and real time social context. It is a "core-pillar" of its mission to advance universe understanding through "truth-seeking" artificial intelligence. Since launch in November 2023, it has released four versions culminating with Grok-4.3 in April 2026, with Grok-5 under development. Grok's development benefits from a vertically integrated AI stack that includes COLOSSUS and COLOSSUS II compute capacity and integration with 'X' social media enables real time access to c350m daily posts, which enhances relevance and contextual awareness.

SpaceX's business in short

Although SpaceX's future aspirations are heavily weighted to AI, it comprises three key businesses, two of which enjoy scale and technological competitive advantages in what could be described as the relatively niche space launch and satellite connectivity markets. The other is xAI, which operates at a scale disadvantage to other hyperscalers, and whose long-term orbital compute ambitions are tethered to technologies not yet available, even if the addressable market offers substantial potential.

Space – SpaceX: unrivalled launch capabilities

75-80% of SpaceX's launches are for its own internal needs

In 2025, SpaceX launched 170 rockets, sending c2,200t into space, the vast majority using its Falcon 9 rocket, accounting for 80% of total global mass deployed into orbit. Based on Q1 data, SpaceX is on track to exceed that launch cadence in 2026. Around 75-80% of SpaceX's launches are for its own internal needs (e.g. Starlink), with commercial and government contracts making up the remainder. SpaceX is currently not profitable but with a pathway to profitability via the development of the next generation Starship rockets.

Space segment forecasts

USDm	2025A	2026E	2027E	2028E	2029E	2030E
Revenue	4,086	3,757	4,970	5,680	6,390	7,450
EBITDA	100	(1,281)	256	1,073	1,839	2,895
Capex	3,832	4,202	4,622	4,853	5,096	5,351
EBITDA - capex	(3,732)	(5,483)	(4,366)	(3,781)	(3,257)	(2,455)

Source: Company data, HSBC estimates

Connectivity – Starlink: leading space-based internet broadband service

As of 31 March 2026, Starlink serves 10.3m Starlink subscribers across 164 markets and other territories, and satellite to mobile services to c7.4m customers in c30 countries. It is a "critical partner" to an array of enterprise customers in the construction, agriculture, retail, telecom, hospitality, aviation, maritime, and land mobility industries, and government contracts focused on public services, social impact, humanitarian efforts, and disaster response in even the most remote and challenging environments. Starlink has commercial agreements in places with c30 mobile network operators (MNOs) including T-Mobile in the US and One NZ, Optus, Telstra, Rogers, KDDI, Salt, Entel, Kyivstar, and VMO2 in international markets where it supplements terrestrial networks to reduce mobile 'dead zones'. Starlink is currently the highest revenue and only profitable segment in SpaceX.

Connectivity segment forecasts

USDm	2025A	2026E	2027E	2028E	2029E	2030E
Revenue	11,387	15,551	21,903	28,822	37,917	48,797
EBITDA	6,799	9,543	13,914	18,999	25,755	33,826
Capex	4,178	6,662	9,225	14,256	15,703	17,225
EBITDA - capex	2,621	2,881	4,689	4,743	10,052	16,602

Source: Company data, HSBC estimates

AI – Compute infrastructure/Grok/consumer and enterprise applications

xAI currently monetises its investments in compute infrastructure via the leasing of surplus AI compute capacity to third parties, notably Anthropic, and Grok through consumer and enterprise applications and advertising on X. Grok and X had over 1.3bn supported accounts in the past 12 months as of end 31 March 2026, compared with over 1.1bn accounts in the 12 months ending 31 December 2025. Over the same period, monthly active users (MAUs) increased to 550m (from 520m) including 117m MAUs using Grok AI features. xAI is currently loss-making.

AI segment forecasts

USDm	2025A	2026E	2027E	2028E	2029E	2030E
Revenue	3,201	18,901	22,909	27,770	36,083	50,015
EBITDA	(2,787)	7,164	8,558	13,600	22,067	31,383
Capex	12,727	57,723	40,000	40,000	40,000	40,000
EBITDA - capex	(15,514)	(50,559)	(31,442)	(26,400)	(17,933)	(8,617)

Source: Company data, HSBC estimates

At the group level, we expect SpaceX to achieve USD38.2bn of revenue in 2026, an increase of 105% compared with 2025. We also expect SpaceX revenue to increase 30% in 2027. Our 2026 revenue forecast is 6% below Visible Alpha consensus and 37% below for 2027.

We assume a significant increase in EBITDA and EBIT in 2026 vs 2025 due to the nature of the AI segment changing with the addition of generous compute deals with Anthropic and Alphabet. Our group free cash flow forecasts indicate breakeven in 2030 and a limited net debt peak of USD22bn in 2029 (0.5x adjusted EBITDA).

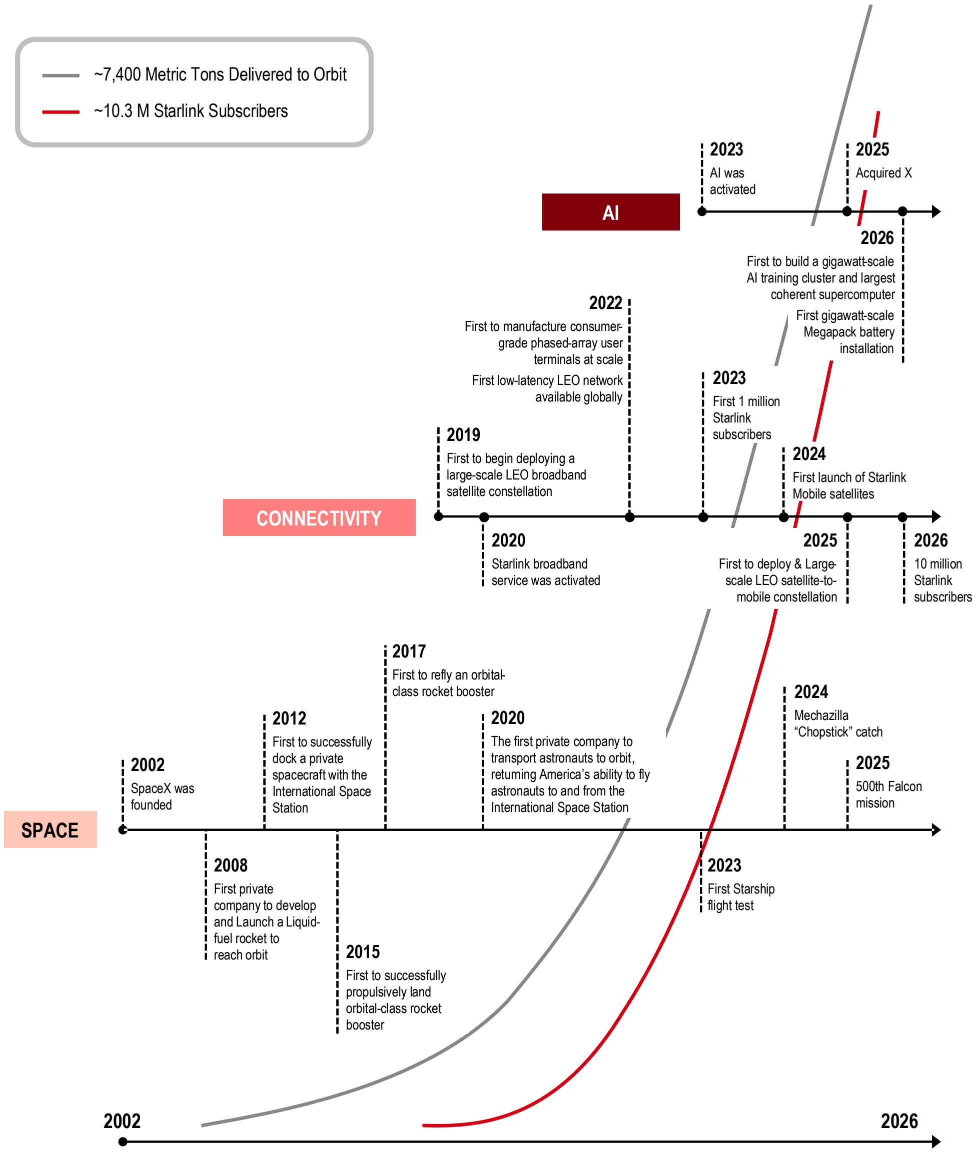
HSBC estimates

USDm	2023A	2024A	2025A	2026E	2027E	2028E	2029E	2030E
Space	3,557	3,796	4,086	3,757	4,970	5,680	6,390	7,450
Connectivity	3,869	7,599	11,387	15,551	21,903	28,822	37,917	48,797
AI	2,961	2,620	3,201	18,901	22,909	27,770	36,083	50,015
Total revenues	10,387	14,015	18,674	38,208	49,783	62,272	80,390	106,262
Space	570	658	100	(1,281)	256	1,073	1,839	2,895
Connectivity	1,353	3,514	6,799	9,543	13,914	18,999	25,755	33,826
AI	(2,793)	118	(2,787)	7,164	8,558	13,600	22,067	31,383
Total EBITDA	(870)	4,290	4,112	15,426	22,728	33,671	49,662	68,105
Margin	-8.4%	30.6%	22.0%	40.4%	45.7%	54.1%	61.8%	64.1%
Space	997	1,154	653	(740)	883	1,641	2,478	3,640
Connectivity	1,602	3,849	7,168	10,089	14,626	19,864	26,893	35,290
AI	1,222	347	(1,237)	9,541	12,589	17,518	26,250	36,625
Total adj. EBITDA	3,821	5,350	6,584	18,890	28,099	39,022	55,621	75,556
Margin	36.8%	38.2%	35.3%	49.4%	56.4%	62.7%	69.2%	71.1%
Space	1,497	2,032	3,832	4,202	4,622	4,853	5,096	5,351
Connectivity	2,455	3,498	4,178	6,662	9,225	14,256	15,703	17,225
AI	463	5,633	12,727	57,723	40,000	40,000	40,000	40,000
Total capex	4,415	11,163	20,737	68,587	53,847	59,109	60,799	62,576
Space	(1)	21	(657)	(2,098)	(1,133)	(876)	(614)	(19)
Connectivity	469	2,006	4,423	5,741	8,342	11,379	15,502	20,632
AI	(3,973)	(1,561)	(6,355)	(3,829)	(8,674)	(9,019)	(7,209)	(4,235)
EBIT	(3,505)	466	(2,589)	(186)	(1,465)	1,484	7,679	16,378
Margin	-33.7%	3.3%	-13.9%	-0.5%	-2.9%	2.4%	9.6%	15.4%
Net other income	(1,486)	(224)	(1,630)	(3,040)	(883)	(958)	(1,287)	(1,294)
Profit before tax	(4,991)	242	(4,219)	(3,226)	(2,348)	526	6,393	15,083
Tax	(363)	(549)	718	184	(399)	89	1,087	2,564
Net profit	(4,628)	21	(4,937)	(4,080)	(1,949)	436	5,306	12,519
FCF	105	(5,387)	(13,952)	(53,880)	(25,607)	(20,612)	(5,817)	12,906
Net debt (cash)		1,608	(1,851)	(32,553)	(6,076)	15,445	22,229	10,507

Source: Company data, HSBC estimates

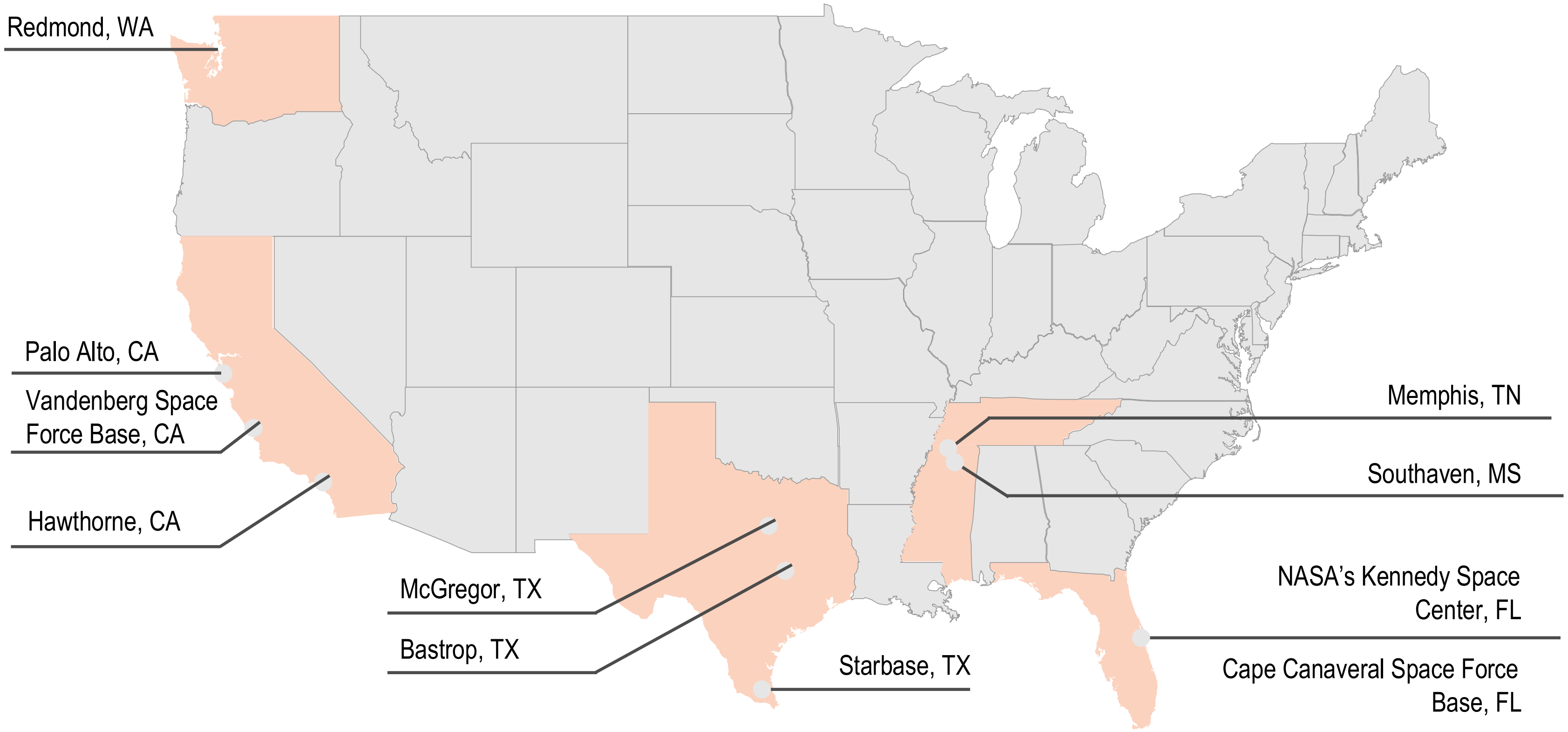
SpaceX's business history summary

Founded in 2002, SpaceX launched its first Falcon 1 rocket in 2008, the first privately developed liquid fuelled rocket to reach orbit, which secured critical NASA contracts at the time. In 2019, SpaceX was the first privately owned company to deploy and scale a low Earth orbit (LEO) broadband satellite constellation ahead of the launch of Starlink in 2020. A largescale mobile constellation deployment followed in 2025. Starlink now caters for 10.3m subscribers globally, up from 1.0m in 2023. In 2026 the group developed its first GW scale megapack battery installation and AI training clusters that, through the launch of Grok in 2023 and acquisition of X in 2026, form the backbone of SpaceX's vertically integrated, full stack AI strategy.



Source: Company data, HSBC

SpaceX infrastructure and facilities



Source: Company data, HSBC

Starbase, TX. SpaceX HQ and home to the development, manufacturing, testing and launch of Starship.

Hawthorne, CA. The original flagship SpaceX facility manufacturing Falcon 9, Flacon Heavy, Dragon Crew / Cargo spacecraft, Merlin / Raptor engines and Starlink terminals.

McGregor, TX. Rocket engine development and testing complex use for testing Merlin / Raptor engines and Starship hardware.

Bastrop, TX. Manufactures the majority of Starlink products including thousands of Starlink 'kits'. Management expects to double the size of Bastrop to include new Starlink products including Antenna's solar cells and AI compute satellites.

Redmond, WA. Starlink manufacturing facility produces c70 satellite per week / 3,640 per year incl. bus structures, phased array antenna's Propulsion, Solar arrays, inter-satellite lasers. Kennedy Space Centre / Cape Canaveral, FL. SpaceX operations in Florida span across NASA's Kennedy Space Centre and Capa Canaveral Space Force Station including two active launch sites, Falcon booster and Dragon refurbishment facilities, launch operations, payload processing. SpaceX also uses landing zones for Falcon Boosters. Management anticipates expanding operations in Florida to include forthcoming Starship launches with dedicated launch pads, catch and testing operations.

Vandenberg Space Force Base, CA. This is SpaceX's West Coast launch site critical for polar and high inclination orbit missions relating to Starlink constellation deployment. This site also operates as a dedicated Falcon 9 Booster landing zone.

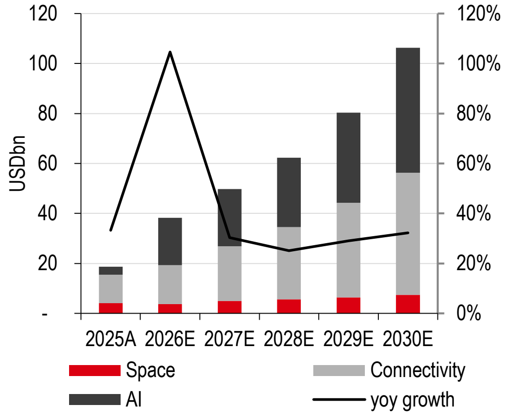
Memphis, TN. This is the home of COLOSSUS and COLOSSUS II high density datacentres to power training and inference for AI models including Grok and Anthropic.

Palo Alto, CA. Corporate HQ of AI operations following the acquisition of xAI in February 2026. This facility also undertakes advanced AI research, development and engineering situated in Silicon Valley to attract AI research talent.

In addition to infrastructure and facilities, SpaceX also operates a fleet of autonomous spaceport drone ships that form the backbone of the SpaceX's reusable rocket architecture, consisting of three operational vessels.

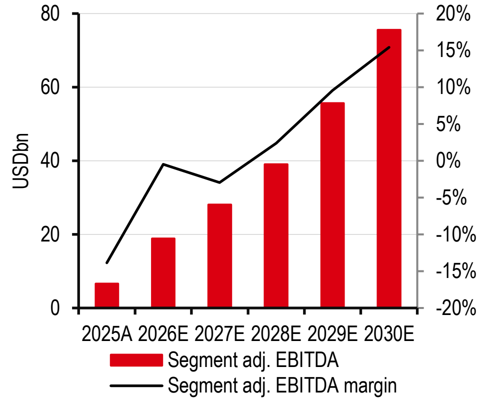
Our investment case in charts

Revenue and yoy growth (RHS)



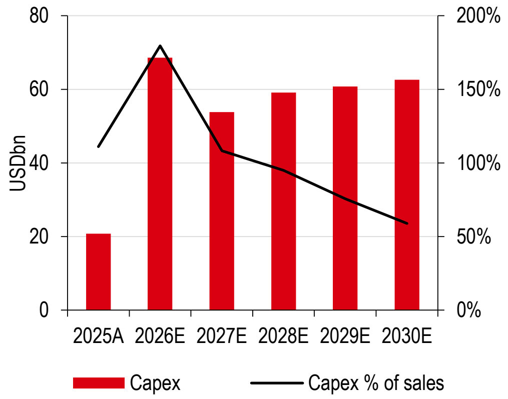
Source: Company data, HSBC estimates

Segment adj. EBITDA and margin (RHS)



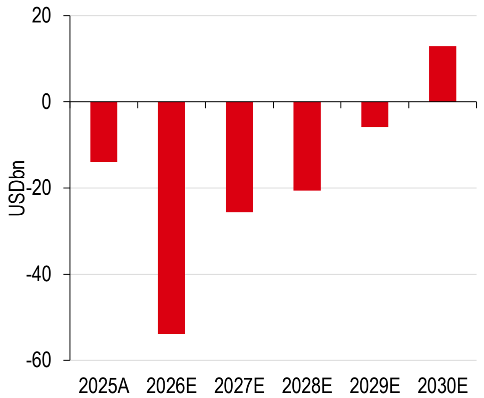
Source: Company data, HSBC estimates

Capex and as percentage of sales



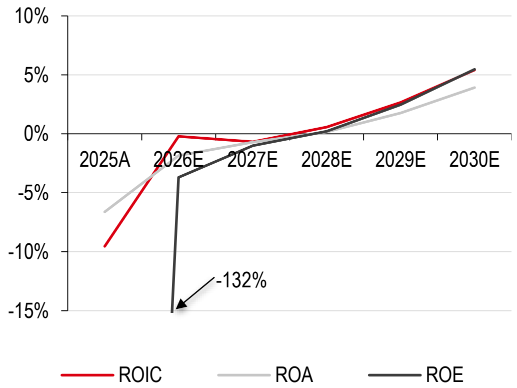
Source: Company data, HSBC estimates

Free cash flow



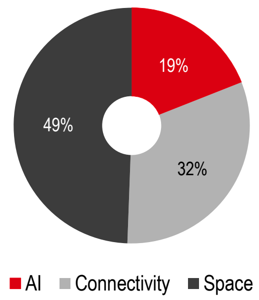
Source: Company data, HSBC estimates

Returns



Source: Company data, HSBC estimates

HSBC valuation by segment (EV USD1.78trn including our innovation premium)



Source: HSBC estimates

Valuation premium warranted

- Given SpaceX's unique offering and management, we think comparison with peers is challenging and undervalues the business
- We use SOTP methodology, valuing each segment separately, before we apply a 2.0x "innovation premium" to reflect the premium observed at Tesla
- We derive a TP of USD115.00, implying 0.2% downside; our 'blue sky' scenario implies a value of USD293.00

How to value SpaceX?

SOTP plus an "innovation premium"
We value each business segment separately before we apply an "innovation premium" of 2.0x.

- Given the nascent nature of SpaceX, a focus on purely cash metrics (i.e. DCF valuation) would not correctly value the opportunity available to the company. Furthermore, the 'Magnificent 7' (Mag-7) stocks (Nvidia, Alphabet, Apple, Microsoft, Amazon, Meta, and Tesla) have performed well, +28% versus the S&P500 at +23% since the start of 2025, despite FCF yields falling on the back of the AI capex spend and higher valuations.
- SpaceX is a unique business comprising distinct segments, which makes it difficult to compare against an appropriate peer group. Nonetheless, we form a large-cap AI-orientated peer group to provide some comparison basis.

We therefore opt for a SOTP approach, where we select the most relevant peer for each subsegment of SpaceX's reporting business lines.

Where does the "innovation premium" of 2.0x come from?
When valuing a company, analysts may sometimes apply a discount (holding structures, dyssynergies, risks etc), but we also consider the case for applying a premium, especially when a strong founder has a proven track record of transforming some industries (including cars and rockets). We considered several valuation approaches:

- **Special purpose acquisition companies (SPACs):** Given SpaceX's current high-in-comparison multiples (vs the Mag-7), it would be able to perform M&A for relatively low share dilution (merger arbitrage) compared with peers. This would be like a SPAC (also referred to a 'blank check' company). However, prior to acquisitions, SPACs broadly trade close to cash alongside a management premium to a business with neither revenues nor potential earnings. We calculated that the premium to cash of the 10 largest SPACs in the US is only 10%. We concluded that this would only value management and not the growth or unknown opportunities within the business, and therefore not applicable to SpaceX, in our view.
- **Mining explorers:** Given the commentary around lunar mining in the prospectus, we investigated the option of valuing the opportunity cost. This was not viable as, whilst Earth-based mining explorers have EV/resource multiples available, applying these to potentially viable asteroids or the Moon with unknown timelines and costs lead to entirely untethered valuations.

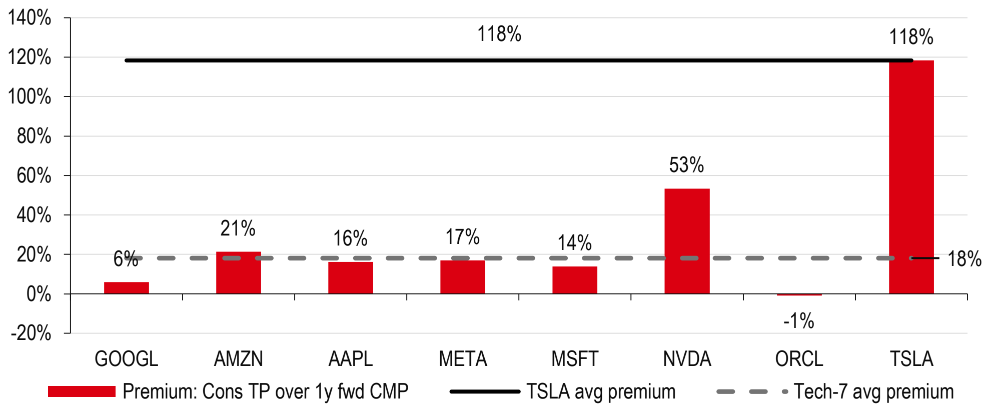
- **Biotech companies:** As speculative businesses looking to solve immensely difficult issues, we thought about comparing SpaceX with the biotech industry. However, this was not viable due to a lack of both a clear issue (equivalent to a disease) and historical data to apply to SpaceX. Therefore, we concluded this was not viable.

In our view, none of these methodologies correctly capture the innovation multiple that we believe should be considered. Therefore, we have looked at Tesla's share price performance in the first 10 years since its IPO, as we believe this is the most appropriate proxy to determine a valuation premium given the commonality in CEO/founder, innovation approach, and developing disruptive technology into a sustainable business.

Based on our observation of Tesla's share price performance and consensus target prices set by sell side research analysts covering the stock, we conclude that over the first 10 years since its IPO, i.e. 2010-20 (we take this period to reflect the early stage and rate of innovation of the business), Tesla's share price was more than double (118%) the average target price the year before. This indicates that fundamental valuation methods do not reflect what the market may be ready to pay for management or 'flywheel' effects, i.e. incremental advances in technology and efficiency compound over time, eventually leading to a sustainable, growing business model.

We also conducted the same calculation for the 'Magnificent 7' (Mag-7) peer group (Alphabet, Amazon, Apple, Meta, Microsoft, Nvidia, and Oracle) and derive a much lower premium of 18% over the same timeframe (2010-20). We therefore estimate a 100% (118% for Tesla minus 18% for the Mag-7) "innovation premium" (i.e. 2.0x) for Tesla vs the broader tech market. We think the same premium could prevail with SpaceX, given the similar management approach and possibly a stronger flywheel effect at SpaceX (considering the interconnected business segments). We apply this "innovation premium" to our fundamental sum-of-the-parts valuation to derive our target price of USD115.00.

Tesla's premium vs the Mag-7 stocks (2010-20)



Source: Refinitiv Workspace, HSBC

Investors should also consider potential share release post lock-up

SpaceX's IPO prospectus indicated that 555,555,555 shares would be issued to constitute the free float. We understand that the underwriters have exercised their option to purchase additional shares of Class A common stock in full, so the free float would have extended to 638,888,888 shares.

We identify 4,678m locked up shares and another 8,160m shares subject to an extended lock-up. Based on the information provided by the SpaceX prospectus dated 12 June 2026, we calculate that 912m shares could be available for sale in the public market from 6 August 2026, compared with 640m shares constituting the free float at present. The free float would increase from 4.9% at present to 11.8%.

Another release event could occur on the same day depending on SpaceX shares trading above USD175.5 for at least five of 10 consecutive trading days ending on 4 August 2026 (i.e. between 22 July and 4 August 2026).

The table below provides further event/date triggers for subsequent share releases.

Those restricted shares are currently owned by funds and individuals that have participated in the private rounds of financing and may be inclined to keep their shares. But we think investors should be aware of this.

Locked up shares: timetable for earliest release

Earliest date available for sale in the public market	Class A shares (m)	% total shares	o/w locked-up (m)	% locked-up shares	Extended locked-up (m)
The second full trading day on Nasdaq immediately following the First Earnings Release Date (4 August 2026)	912	7%	912	19%	
If the reported closing price of our Class A common stock on Nasdaq is at least 30% greater than the public offering price of USD135 for at least five of the ten consecutive trading days ending on, and including, the First Earnings Release Date, the second full trading day immediately after the First Earnings Release Date	456	3%	456	10%	
20-Aug-26	319	2%	319	7%	
09-Sep-26	319	2%	319	7%	
10-Sep-26	59	0%	59	1%	
24-Sep-26	328	3%	328	7%	
09-Oct-26	328	3%	328	7%	
24-Oct-26	328	3%	328	7%	
The second full trading day on Nasdaq immediately following the public release of our quarterly financial results for the quarter ended September 30, 2026	1300	10%	1300	28%	
08-Dec-26	328	3%	328		
The second full trading day on Nasdaq immediately following the public release of our quarterly financial results for the quarter ended December 31, 2026	352	3%			352
18-Mar-27	176	1%			176
The second full trading day on Nasdaq immediately following the public release of our quarterly financial results for the quarter ended March 31, 2027	352	3%			352
17-May-27	176	1%			176
21-Jun-27	352	3%			352
12-Jun-27	6400	49%			6400
The second full trading day immediately following the public release of our quarterly financial results for the quarter ended June 30, 2027	352	3%			352
Total shares available for sale	**12838**		**4678**		**8160**
Total ex Elon Musk	**6438**		**4678**		**1760**

Source: Company data

Group SOTP valuation methodology

We value each business segment separately, using a variety of timelines to best reflect different development stages. Please refer to the table below.

From this we generate a valuation of USD391bn for Space, USD250bn for Connectivity, and USD150bn for AI. We apply the innovation premium of 2.0x to reach a TP of USD115.00.

We do not include further potential M&A in our estimates, but note that with SpaceX trading at higher-than-“normal” company multiples, it should allow the business to undertake accretive M&A due to multiples arbitrage.

For each of SpaceX’s segments, we take a positive stance on valuation and then apply the innovation premium at the group level. For example:

- **AI:** We apply the multiples afforded to both Anthropic and OpenAI at recent funding rounds. This is despite SpaceX’s AI not enjoying the market share of the leading AI labs.
- **Connectivity:** We apply a 2x or 3x multiple versus that of the US telecoms for connectivity businesses and 4x multiple for the futuristic ventures.
- **Space:** We use GLV revenues instead of only client revenues to best capture the functionality of the business. We argue that this is a ‘premium’ as it includes theoretical revenues from Starlink launches.

HSBC SOTP valuation

USDm	EV	Year taken and multiple	Rationale
AI			
X	6,236	2x 2026e revenue	In-line with tier-2 social media peers such as SNAP and PINS
Grok	90,656	3.63x 2030e revenue	In-line with OpenAI's recent funding round
Compute offtake by Anthropic	4,535	Present value	Good deal, but likely to be short lived. Six months' EBIT discounted to today
Compute offtake by Google	4,472	Present value	Good deal, but likely to be short lived. Eight months' EBIT discounted to today
Compute offtake by others	20,081	10x 2030e non-GAAP EBIT	Year-taken to allow contracts to mature
Cursor	24,450	2.83x 2030e revenue	In-line with recent Anthropic funding raise. Acquisition cost removed at group level
Terafab	-		Still at a framework agreement stage
Orbital AI	-		Not necessary in the foreseeable future
AI valuation	**150,430**		
Connectivity			
Residential	93,090	5x 2028e revenues	2x that of US telecoms
Enterprise	76,532	7.5x 2028e revenues	3x that of US telecoms
Residential mobile	24,887	5x 2040e revenues discounted back	In-line with residential above, 2040 taken for business to reach maturity
Futuristic venture	55,407	10x 2040e revenues discounted back	10x to reflect high growth from new business opportunities, 2040 taken for business to reach maturity
Connectivity valuation	**249,916**		
Space			
Total launches incl. Starlink	92	2030e	Take total launches to correctly value optionality to wider business
Implied GLV revenues	40,018	2030e	
Segment adj. EBITDA margin	48.9%	2030e	HSBC forecast margin
GLV EBITDA	19,555	2030e	
EBITDA multiple	20.0x	2030e	Reflects high-market share
Space valuation	**391,103**		
Total EV	**791,449**		
Innovation premium	**2.0x**		
Valuation EV	**1,582,898**		
Less upcoming acquisitions	(76,900)		
Net debt (cash)	(1,851)		
Implied market cap	1,504,147		
Shares issued (m)	638.9		
Free-float commentary	4.89%		
Number of shares (bn)	13,076		
TP (USD)	**115.0**		

Source: HSBC estimates

Space segment SOTP

To best represent the capabilities of the launch platform, we base our valuation on gross metrics, i.e. including inter-segment revenues where Connectivity (notably Starlink) is treated as a client. The company charges no inter-segmental revenues, and the launch costs for Starlink satellites are incurred at cost in Connectivity. We include hypothetical inter-segmental revenues to best capture the potential of the launch business.

For our valuation we take an adjusted EBITDA multiple of 20.0x 2030e GLV (gross launch value) adj. segment EBITDA of USD19.5bn. This places the segment at the top-end with the range of the large-cap AI peer group, which we think is justified given the market dominance of the Space business.

This leads to an EV of cUSD391bn.

Space segment valuation

	Unit	
Launches 2030e		92
Implied GLV revenues	USDm	40,018
EBITDA margin	%	48.9%
GLV EBITDA	USDm	19,555
EBITDA multiple	x	20.0x
Valuation	**USDm**	**391,103**

Source: HSBC estimates

HSBC Space valuation multiple implication

EV=USD391,103m	Unit	2026E	2027E	2028E	2029E	2030E
GLV	USDm	10,724	36,008	32,933	35,536	40,018
GLV EBITDA	USDm	(2,113)	6,399	9,513	13,782	19,555
EV/GLV	x	36.5x	10.9x	11.9x	11.0x	9.8x
EV/GLV EBITDA	x	(185.1)x	61.1x	41.1x	28.4x	20.0x

Source: HSBC estimates

What are the bull and base cases?

- **Bear case:** No further contract wins, leading to flat revenues to 2030e as fixed price contracts fade out. The Starship rocket never reaches commercial viability leading to flat segment adj. EBITDA margin of 16%. The Starship impairment of USD15bn would lead to negative overall EBITDA and operating income. Total GLV revenues would be USD11.4bn, with adj. GLV EBITDA of USD1.8bn (i.e. 2025A values).
- **Bull case:** We assume that Starship's commercial viability starts in 2027, with overall mass to orbit 2x what we assume in our base case. We retain our flat pricing assumption. This leads to 2030e GLV adj. EBITDA of USD47.7bn, with margins of 59.7%.

Space segment – bear, bull, and base case valuation scenarios

Space		Bear	Base	Bull
Total launches 2030e	#	170	92	184
Implied GLV	USDm	11,395	40,018	80,036
Adj. EBITDA margin	%	16.0%	48.9%	59.7%
GLV adj. EBITDA	USDm	1,821	19,555	47,745
EBITDA multiple	USDm	20.0x	20.0x	20.0x
Valuation	**USDm**	36,420	391,103	954,903

Source: Company filings, HSBC estimates

*Please note the bear case assumes no Starship, thus higher payload per launch assumptions are never realised. Therefore, Falcon spacecraft remains the only option.

Connectivity segment SOTP

Given the relative nascency of the Starlink operations and growth potential, especially in the enterprise segment, we use SOTP-based EV/Sales on 2028e revenues for the residential broadband and enterprise segments. In addition, we use 2040e EV/Sales multiple discounted to current for not yet launched residential mobile and futuristic B2B services to derive our valuation of around USD250bn for Starlink. Our SOTP approach comprises:

- 5x 2028 EV/Sales for the Residential broadband segment (two times that of US telecoms.
- 7.5x 2028 EV/Sales for the Enterprise segment (three times that of US telecoms multiple).
- 10x 2040 EV/Sales for the "futuristic ventures" segment. We reflect Starlink's ambitions regarding autonomous driving, airlines, ships. and IoT market. as well as the data backbone for robots, to derive potential 2040e revenue from these ventures. We then assign a 10x EV/Sales multiple before discounting the EV back to 2026 using a WACC of 8.5%.
  We assign a 5% probability of success to temper our ambitious revenue forecasts and to take into account possible restrictions from government and regulatory headwinds.

- Similarly for the residential mobile segment, we use a 5x 2040e EV/Sales multiple discounted to 2026.
- To derive our WACC for calculating the net present value for the futuristic ventures and the residential mobile activities, we use a cost of equity of 11.0% (risk free rate of 4.25% and enterprise risk premium of 4.20%, both based on HSBC strategy team estimates; see *Cost of equity 2026*, 13 July 2026, beta of 1, and an execution risk premium of 2.5%), and cost of debt of 6%, rounded to the closest 0.5%.

We apply our innovation premium at the group level to the sum of the EVs from each segment.

Starlink: SOTP-based valuation on EV/Sales

	Revenue – 2028	EV/Sales (x)	EV
Residential broadband	18,618	5.0	93,090
Enterprise	10,204	7.5	76,532
Residential mobile		5.0	24,887
Futuristic venture		10.0	55,407
EV			**249,916**
EV/EBITDA – 2027			**17.1x**
EV/EBITDA – 2028			**12.6x**

Source: HSBC estimates

What are the bull and base case valuation scenarios?

- **Bear case:** For residential broadband, we assume that Starlink subscribers as a percentage of addressable homes is 2%, 2.2ppt below that of our base case assumption of 4.2%. For residential mobile, we assume that mobile subscribers as a percentage of the total addressable population reached is 0.5%, or 0.4ppt below our base case. We lower our ARPU growth rate assumptions for both residential mobile and residential broadband by 2ppt per year and we lower our enterprise growth rates by 5ppt lower than our base case. We lower residential broadband 2028e EV/Sales multiple to 2.5x (vs 5.0x in base case), enterprise segment 2028e EV/Sales multiple to 5.0x (vs 7.5x in base case), Residential mobile 2040e EV/Sales multiple to 2.5x (vs 5.0x in base case), and Futuristic ventures 2040e multiple to 7.5x (vs 10.0x in base case). We also lower the probability of futuristic ventures to 2.5% (from 5.0% in base case). All the above assumptions lower our fair value to USD100bn in a bear case scenario.
- **Bull case:** For residential broadband, we assume that Starlink subscribers as a percentage of addressable homes is 6.0%, 1.8ppt above that of our base case assumption of 4.2%. For residential mobile, we assume that mobile subscribers as a percentage of the total addressable population reached is 2.0%, or 1.1ppt above our base case. We increase our ARPU growth rate assumptions for both residential mobile and residential broadband by 2ppt per year and we increase our enterprise growth rates by 5ppt above our base case assumptions. We increase residential broadband 2028e EV/Sales multiple to 7.5x (vs 5.0x in base case), enterprise segment 2028e EV/Sales multiple to 10.0x (vs 7.5x in base case), Residential mobile 2040e EV/Sales multiple to 7.5x (vs 5.0x in base case), and Futuristic ventures 2040e multiple to 12.5x (vs 10.0x in base case). We also increase the probability of futuristic ventures to 12.5% (vs 5.0% in base case). All the above assumptions increases our fair value to USD550bn in a bull case scenario.

Starlink: Bear, bull, and base case valuation scenarios

Assumptions	Bear case	Base case	Bull case
Residential broadband			
Starlink broadband subscribers as a percentage of addressable homes	2.0	4.2	6.0
ARPU change vs base case (per year, ppt)	-2.0	0.0	2.0
Residential mobile			
Residential mobile subscribers as a percentage of the addressable population	0.5	0.9	2.0
ARPU change vs base case (per year, ppt)	-2.0	0.0	2.0
Enterprise			
Enterprise growth vs base case (per year, ppt)	-5.0	0.0	5.0
Futuristic ventures			
Probability of success %	2.5	5.0	12.5
Multiples			
Residential broadband revenue multiple (x) - 2028e	2.5	5.0	7.5
Residential mobile revenue multiple (x) - 2040e	2.5	5.0	7.5
Enterprise revenue multiple (x) - 2028e	5.0	7.5	10.0
Futuristic revenue multiple (x) - 2040e	7.5	10.0	12.5
Implied EV (current, USDbn)	**100**	**250**	**550**

Source: HSBC estimates

AI segment SOTP

- **X social media platform:** We value X at USD6.2bn using 2x 2026e EV/Sales. This is in-line with second tier social media peers such as SNAP and PINS.
- **Grok:** We value Grok at USD90.7bn, using 3.63x 2030e EV/Sales, in-line with the OpenAI post-money valuation of USD852bn from March 2026 (source: OpenAI).
- **Compute infrastructure:** SpaceX rents its compute infrastructure to AI competitors. We value the Anthropic contract at USD4.5bn using eight months EBIT and the Alphabet contract at USD4.5bn using six months EBIT. The short timelines limit the contribution to our valuation. In addition, we assume that further deals are signed in the future. We value these future deals at USD20,081m, which is 10x 2030e non-GAAP EBIT.
- **Cursor: SpaceX has agreed to acquire this agentic AI company for USD60bn with SpaceX shares.** We cannot reconcile such a high price and value it at USD24.5bn, using 2.83x 2030e EV/Sales, in-line with Anthropic's latest financing round with a valuation of USD965bn from May 2026 (source: Anthropic). Closing of the acquisition could occur in Q3 2026.
- **Terafab and Orbital compute:** We value these both at zero value. We believe that the company remains far from recreating the complex hardware supply chain and will face hefty costs. We do not assume costs either.
- This generates a total EV of around USD150bn.

AI total valuation

USDbn	EV	Rationale
X	6,236	2x 2026e revenue, in-line with tier 2 social media peers such as SNAP and PINS
Grok	90,656	3.63x 2030e revenue, in-line with OpenAI
Compute offtake by Anthropic	4,535	Good deal, but likely to be short lived. Six months' EBIT discounted to today
Compute offtake by Google	4,472	Good deal, but likely to be short lived. Eight months' EBIT discounted to today
Compute offtake by others	20,081	10x 2030e non-GAAP EBIT
Cursor	24,450	Gross value is 2.83x 2030e revenue (in-line with Anthropic). USD60bn future acquisition cost is removed at group level
Terafab	-	Still at a framework agreement stage
Orbital AI	-	Not needed in the foreseeable future
AI total	**150,430**	

Source: HSBC estimates

What are the bull and base case scenarios?

- **Bear case:** 1) We trim our X multiple by 20% due to conservatism, 2) we discount our Grok valuation by 50% to reflect lower operating margins versus base case peer OpenAI, 3) future compute offtake valuation is trimmed by 20% to reflect uncertainties in future contract wins and, 4) we lower our Cursor valuation by 20% to reflect lower market share versus the base case peer Anthropic.
- **Bull case:** 1) We increase our X multiple by 20%, 2) we change our Grok valuation to Palantir EV/Sales, an uplift of 4.2x, 3) future compute offtake valuation is lifted by 20% to reflect higher certainties and, 4) we lift our Cursor valuation by 20% to reflect higher growth opportunities

AI Segment: Bear, base, and bull case valuation scenarios (USDm)

Segment	Bear	Base	Bull
X	5,197	6,236	7,483
Grok	45,328	90,656	382,105
Compute offtake by Anthropic	4,535	4,535	4,535
Compute offtake by Google	4,472	4,472	4,472
Compute offtake by others	16,734	20,081	24,097
Cursor	20,375	24,450	29,340
Terafab	-	0	-
Orbital AI	-	0	-
Total	**96,641**	**150,430**	**452,032**

Source: HSBC estimates

Putting together our 'blue sky' scenario

We take the bull case scenarios for each segment and aggregate them to build a 'blue sky' scenario for SpaceX and derive a value of USD293.00 per share.

HSBC 'blue sky' scenario valuation

USDm	Base case	Blue sky	Uplift	Comment
AI				
X	6,236	7,483	20%	20% premium to base case
Grok	90,656	382,105	321%	In-line with PLTR 2030e EV/Rev
Compute offtake by Anthropic	4,535	4,535	0%	In-line with base case
Compute offtake by Google	4,472	4,472	0%	In-line with base case
Compute offtake by others	20,081	24,097	20%	20% premium to base case
Cursor	24,450	29,340	20%	20% premium to base case
Terafab	-	-		In-line with base case
Orbital AI	-	-		In-line with base case
Total	**150,430**	**452,032**	**200%**	
Connectivity				
Starlink subscribers as percentage of addressable homes	4.2%	6.0%		1.8ppt higher than base case
ARPU change vs base case (per year, ppt)	0.0%	2.0%		2.0ppt higher than base case
Mobile subscribers as a % of addressable population	0.9%	2.0%		1.1ppt higher than base case
Enterprise growth change vs base case (per year)	0.0%	5.0%		5.0ppt higher than base case
Probability of futuristic ventures	2.5%	12.5%		5x our base case
Residential broadband revenue multiple (x) - 2028e	5.0x	7.5x	50%	Higher to adjust for higher growth
Residential mobile revenue multiple (x) - 2040e	5.0x	7.5x	50%	Higher to adjust for higher growth
Enterprise revenue multiple (x) - 2028e	7.5x	10.0x	33%	Higher to adjust for higher growth
Futuristic revenue multiple (x) - 2040e	10.0x	12.5x	25%	Higher to adjust for higher growth
Total	**249,916**	**550,000**	**120%**	
Space				
Total launches 2030e	92	184	100%	Assumes Starship commences 2027E
Implied GLV	40,018	80,036	100%	Assumes new contract wins & more Starlink launches
Adj. EBITDA margin	48.9%	59.7%	1,079bps	HSBC margin
GLV adj. EBITDA	19,555	47,745	144%	
EBITDA multiple (x)	20	20		Unchanged
Valuation	**391,103**	**954,903**	**144%**	
Total valuation	**791,449**	**1,956,935**	**147%**	
Innovation premium	2.0x	2.0x		Unchanged
Total valuation incl. premium	1,582,898	3,913,870	147%	
Adjustments	(78,751)	(78,751)		Unchanged
Fair values (USD)	**115**	**293**	**155%**	

Source: HSBC estimates

Peer group valuation

Peer valuation

Peers	RIC	LCY	CMP	Rating	Mkt Cap (USDm)	EV/sales FY 26	EV/sales FY 27	EV/EBIT FY 26	EV/EBIT FY 27	P/E FY 26	P/E FY 27
SpaceX	**SPCX.O**	**USD**	115.26	Hold	1,516,736	38.8x	30.3x	nm	nm	nm	nm
Space & defence											
Applied Aerospace	AADX.K	USD	17.29	N/R	2,952	5.7x	4.8x	42.8x	29.5x	NA	NA
Aevex	AVEX.K	USD	14.83	N/R	1,776	3.6x	3.0x	35.2x	23.3x	NA	46.3x
Arxis	ARXS.O	USD	42.08	N/R	17,267	NA	NA	NA	NA	NA	82.5x
Firefly Aerospace	FLY.O	USD	20.24	N/R	3,321	10.0x	6.4x	NA	NA	NA	NA
Gomspace	GOMX.ST	SEK	13.05	N/R	234	NA	NA	NA	NA	NA	NA
Hawkeye	HAWK.K	USD	19.14	N/R	1,875	0.5x	0.4x	NA	NA	NA	NA
Intutive Machines	LUNR.O	USD	14.09	N/R	3,064	5.4x	5.1x	NA	NA	NA	NA
MDA Space	MDA.TO	CAD	44.47	N/R	5,119	4.1x	3.6x	46.1x	39.2x	51.1x	32.4x
Rocket Lab	RKLB.O	USD	69.75	N/R	41,723	42.9x	19.7x	NA	NA	NA	NA
Redwire	RDW	USD	8.99	N/R	2,147	3.9x	3.4x	NA	NA	NA	NA
Voyager Technologies	VOYG.K	USD	26.39	N/R	1,564	4.2x	2.8x	NA	NA	NA	NA
Connectivity											
AST Spacemobile	ASTS.O	USD	61.95	N/R	24,044	NA	38.3x	NA	NA	NA	NA
Echostar	SATS.O	USD	92.28	N/R	26,744	3.3x	3.3x	30.1x	29.2x	NA	NA
Eutelsat	ETL.PA	EUR	2.15	N/R	2,884	3.2x	3.3x	NA	NA	NA	NA
Globalstar	GSAT.O	USD	79.94	N/R	10,295	30.5x	30.1x	NA	NA	NA	NA
Iridium	IRDM.O	USD	46.86	N/R	4,965	7.4x	5.5x	29.0x	21.5x	44.2x	40.7x
SES	SESFd.PA	EUR	7.40	N/R	5,231	2.6x	2.5x	40.2x	35.5x	NA	60.2x
Sky Perfect Jsat	9412.T	JPY	2299.00	N/R	4,196	NA	NA	NA	NA	28.0x	23.7x
Telesat	TSAT.O	USD	39.52	N/R	601	8.8x	12.2x	NA	NA	NA	NA
Viasat	VSAT.O	USD	74.37	N/R	10,157	2.9x	2.8x	NA	98.5x	NA	NA
Earth imaging/space software											
BlackSky Technology	BKSY.K	USD	22.49	N/R	835	8.0x	6.1x	NA	NA	NA	NA
Planet labs	PL	USD	22.60	N/R	8,055	11.9x	9.2x	NA	NA	NA	NA
Spire Global	SPIR.K	USD	11.43	N/R	442	NA	NA	NA	NA	7.1x	NA
Datacentres											
Alphabet	GOOGL.N	USD	342.09	Buy	4,113,004	7.8x	6.5x	22.0x	18.6x	24.0x	24.3x
Amazon	AMZN.OQ	USD	244.85	Buy	2,633,878	3.3x	3.0x	25.3x	20.0x	27.7x	24.9x
Meta	META.O	USD	627.17	Buy	1,592,413	6.4x	5.4x	18.8x	16.0x	20.0x	18.7x
Microsoft	MSFT.O	USD	390.34	Buy	2,899,616	8.7x	7.2x	18.6x	15.5x	22.4x	19.5x
Oracle	ORCL.OQ	USD	125.84	Buy	362,478	5.4x	4.1x	16.4x	13.0x	20.5x	15.5x
Social media											
Pinterest	PINS.N	USD	22.54	Hold	10,859	1.9x	1.7x	28.9x	17.8x	43.1x	29.6x
Reddit	RDDT.K	USD	170.38	N/R	32,800	10.5x	8.5x	35.7x	25.4x	65.0x	34.4x
Snap	SNAP.N	USD	4.47	Hold	6,384	1.1x	1.0x	6.0x	4.5x	NA	52.2x
Peer avg						**4.3x**	**4.0x**	**9.8x**	**9.5x**	**24.4x**	**20.5x**
Peer median						**5.4x**	**4.8x**	**29.0x**	**21.5x**	**27.7x**	**31.0x**

Source: LSEG DataStream, HSBC Research, *next FY estimates (FY27 in FY26) for Sky Perfect JSAT (Mar-end), Viasat (Mar-end), Planet Labs (Jan-end) and Oracle (May-end). Priced as of 22 July.

Upside risks to our valuation

Space

- Starship commercial deployment starts faster than expected.
- Starship expansion to 200t payload occurs faster than expected.
- Contract wins beyond our expectations from external customers.

Connectivity

- Greater than expected consumer take-up.
- Faster than expected Enterprise take-up.
- Faster delivery of a meaningful mobile segment.

AI

- Anthropic and Alphabet deals last longer than we forecast.
- X social media platform outperforms its peers and moves to a tier-1 status.
- Grok consumer take-up and revenues outperform our expectations.
- Terafab and orbital AI start to meaningfully support earnings faster than we expect.

Company basis

- Supportive fx scenario.
- Lower than expected capital costs.

Downside risks to our valuation

Space

- Starship commercial deployment delayed; failure of a Falcon launch.
- Competition including Blue Origin building up operation faster than expected.
- Large government contract losses.

Connectivity

- Regulatory constraints (spectrum) capping the B2C opportunity.
- Low enterprise take-up outside the US due to geopolitical factors.
- Further progress of terrestrial networks (including 6G).

AI

- Grok not gaining scale fast enough or spending too much capex to compete effectively against LLMs.
- X social media platform losing users due to low moderation or trust issues.
- Terafab and orbital AI projects discontinued.

Company basis

- Unfavourable political environment
- Unfavourable fx environment.
- Higher than expected cost of capital.

Space – the platform

- From launchpad to satellites; a build-your-own strategy
- Space currently supports Starlink; in the future it plans to deploy datacentres in space to fuel the development of xAI
- We estimate a valuation of USD391bn

Space enables all orbital revenue streams

Charlie Rothbarth*
Global Tech Platforms Analyst
HSBC Bank plc
charlie.rothbarth@hsbc.com
+44 203 2685284

** Employed by a non-US affiliate of HSBC Securities (USA) Inc, and is not registered/ qualified pursuant to FINRA regulations*

Launchpads, rockets, spacecraft, and more; a build-your-own strategy
The Space segment of SpaceX is the platform upon which the rest of the orbital opportunities are enabled. The Space segment designs, build, and launches rockets that place equipment into orbit. Space generates revenue from fixed price contracts for deploying payloads in orbit (one-to-five-year contract durations) and for the development of spacecraft, as well as launch and mission services (one-to-14-year contracts). The segment is the market leader in launch services, deploying 80-90% of global mass to orbit in 2025.

The segment's key client is Starlink (part of the connectivity section), which has placed over 9,600 satellites into low earth orbit (LEO). Externally, the key clients are space-based agencies and US government departments, including the Department of Defense. Over 2023-25, c30% of total launches were customer launches, with the rest Starlink satellite deployments or Starship test flights. We expect this number to fall to 11% by 2030 as the growth in internal launches exceeds that of external launches.

Space segment: HSBC long-term estimates

USDm, unless otherwise stated, FYE Dec	2023A	2024A	2025A	2026E	2027E	2028E	2029E	2030E
Launches (n)	98	138	170	186	89	80	86	92
Total mass to orbit (t)	1,210	1,699	2,213	2,415	8,900	8,000	8,600	9,200
Revenue	**3,557**	**3,796**	**4,086**	**3,757**	**4,970**	**5,680**	**6,390**	**7,450**
Cost of revenue	1,669	1,541	1,352	1,855	1,750	2,000	2,250	2,500
Research and development	1,538	1,835	3,004	3,720	3,906	4,101	4,306	4,522
Selling, general, and administrative	351	375	349	280	447	454	447	447
Restructuring charges	-	-	-	-	-	-	-	-
Impairment	-	24	38	-	-	-	-	-
Total costs and expenses	**3,558**	**3,775**	**4,743**	**5,855**	**6,103**	**6,556**	**7,004**	**7,469**
Income (loss) from operations	(1)	21	(657)	(2,098)	(1,133)	(876)	(614)	(19)
D&A	571	637	757	817	1,389	1,948	2,453	2,914
EBITDA	570	658	100	(1,281)	256	1,073	1,839	2,895
Adjustments	427	496	553	541	627	568	639	745
Segment adj. EBITDA	**997**	**1,154**	**653**	**(740)**	**883**	**1,641**	**2,478**	**3,640**
Share based compensation	427	472	515	541	627	568	639	745
Capital expenditures	1,497	2,032	3,832	4,202	4,622	4,853	5,096	5,351

Source: HSBC estimates

The company identifies a space-enabled solutions TAM of USD370bn, versus 2025 revenues of USD4.1bn. As SpaceX does not register any internal revenues, we do not expect this segment to show the same revenue growth as the Connectivity or AI verticals.

Long-term cost improvement towards the future Starship rocket's payload costs of USD185/kg (HSBCe USD1,000/kg currently) afford the segment strong long-term margin progression alongside the ability to lower cost to customers in order to increase demand for its services.

Valuation

We identify some complications for valuing the Space segment:

- A DCF-based approach would not correctly value the internal cash flows in the business as Starlink is not charged a margin for its own satellite deployment (30% of launches in 2023-25). Furthermore, terminal value would make up almost all the valuation given the early-stage of the business.
- SpaceX is too large to fairly compare with the nascent "space-based" peer group.

To best represent the capabilities of the launch platform, we base our valuation on gross metrics, i.e. including inter-segment revenues where Connectivity is treated as a client.

For the top-end of the range we take a multiple of 20.0x 2030e GLV (gross launch value) adj. segment EBITDA. This is at the top-end of the large cap AI peer group, which we think fairly reflects the market-leading position of the launch business.

We reach an EV valuation of around USD391bn.

Space segment valuation

	Unit	
Launches 2030e		92
Implied GLV revenues	USDm	40,018
EBITDA margin	%	48.9%
GLV EBITDA	USDm	19,555
EBITDA multiple	x	20.0x
Valuation	**USDm**	**391,103**

Source: HSBC estimates

HSBC space valuation multiple implication

EV=USD391,103m	Unit	2026E	2027E	2028E	2029E	2030E
GLV	USDm	10,724	36,008	32,933	35,536	40,018
GLV EBITDA	USDm	(2,113)	6,399	9,513	13,782	19,555
EV/GLV	x	36.5x	10.9x	11.9x	11.0x	9.8x
EV/GLV EBITDA	x	(185.1)x	61.1x	41.1x	28.4x	20.0x

Source: HSBC estimates

Large-cap AI peer group

Company	Currency	Current EV	2030 EBITDA	2030 EV/EBITDA
Alphabet	USD	4,052,037	468,250	8.7x
Amazon	USD	2,717,637	421,510	6.4x
Apple	USD	4,583,671	219,981	20.8x
ASML	EUR	611,247	41,251	14.8x
Broadcom	USD	1,933,134	260,885	7.4x
Meta	USD	1,597,612	267,801	6.0x
Nvidia	USD	4,911,562	497,416	9.9x
SK Hynix	KRW	1,344,532	454,925	3.0x
TSMC	USD	59,873,737	9,852,983	6.1x

Source: Refinitiv Workspace, HSBC estimates. Priced as of close 23 July 2026

Bear and bull case scenarios

Base case: We assume that the new Starship rocket moves into full operational deployment as of 2027. We assume that Space reaches 2030e external revenues of around USD7.5bn. This assumes further contract wins. We forecast a segment adjusted EBITDA margin of 48.9% as R&D and launch costs decline as a percentage of revenues.

Bear case: No further contract wins, leading to flat revenues over 2030e as fixed price contracts fade out. The Starship rocket never reaches commercial viability leading to flat segment adj. EBITDA margin at 16%. We assume a higher number of launches as current Falcon rockets have a lower payload than Starship. The Starship impairment of USD15bn would lead to negative overall EBITDA and operating income. Total GLV revenues would be USD11.3bn, with adj. EBITDA of USD1.8bn (i.e. 2025A values).

Bull case: We assume Starship commercial viability starts in 2027, with overall mass to orbit 2x what we assume in our base case. We retain our flat pricing assumption. This leads to 2030e GLV adj. EBITDA of USD47.7bn, with margin of 59.7%.

Space segment – bear, base, bull case valuation scenarios

Space		Bear	Base	Bull
Total launches 2030e	#	170	92	184
Implied GLV	USDm	11,395	40,018	80,036
Adj. EBITDA margin	%	16.0%	48.9%	59.7%
GLV adj. EBITDA	USDm	1,821	19,555	47,745
EBITDA multiple	USDm	20.0x	20.0x	20.0x
Valuation	**USDm**	**36,420**	**391,103**	**954,903**

Source: Company filings, HSBC estimates

*Please note the bear case assumes no Starship, thus higher payload per launch assumptions are never realised.

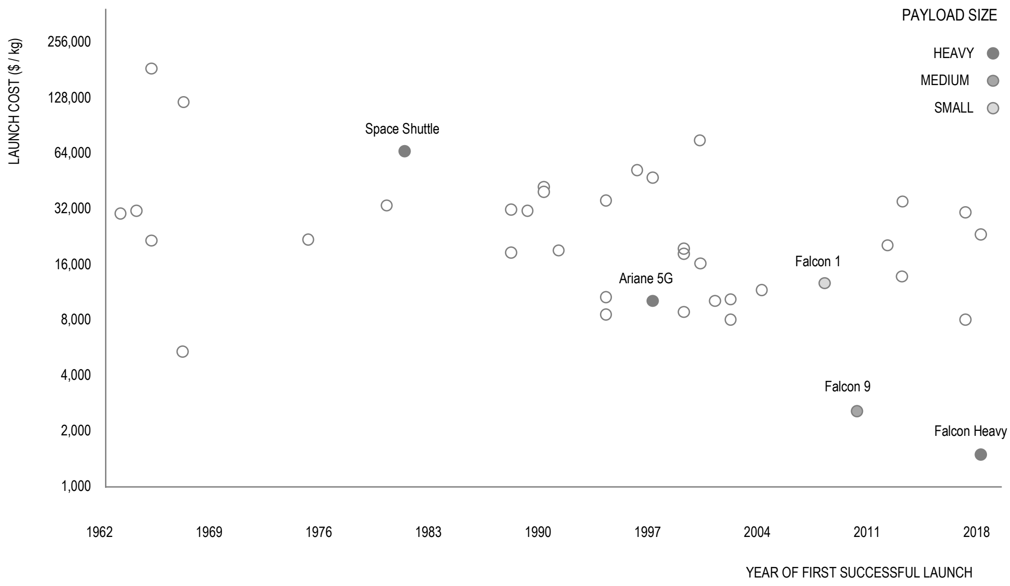
Space segment: the business explained

Falling launch cost drives the Space segment

The driving force of the Space segment is lowering launch costs through the use of rapidly reusable launch vehicles and mass-producing rocket engines at a lower cost than has historically been achievable.

The historical average cost for launch is USD18,500/kg, which the company has reduced to USD2,700/kg using Falcon 9 (including Falcon Heavy), which launched in 2010.

Launch costs per kg of payload



Source: Company data, HSBC

As of 31 March 2026, SpaceX had launched a total mass to orbit of 7,400t, with a success rate of over 99%. Around 650 orbital space launches have been completed, 540 of which were completed by a flight-proven Falcon rocket.

The next iteration will be Starship, which could lower launch costs to USD185/kg, according to the company, although it is not clear if this is for the 200t payload Starship or the current version undergoing flight testing.

Further reduction in launch costs is part of the plan beyond Starship, with historical CEO commentary aiming for USD2-10m per launch. For space-based datacentres to achieve largely superior returns, launch costs need to decline towards these long-term goals.

Our base case assumption is that Starship enters commercial use from 2027, but with most of the launches remaining focussed on Starlink in the short-term given the lack of economic viability of orbital datacentres, which requires the development of Terafab, which we currently assign zero value.

Launch vehicles and spacecraft: build-your-own strategy

SpaceX's competitively advantaged launch services are built around a low-cost, reliable, and frequent access service for commercial and government customers utilising a fleet of reusable rockets and spacecraft for satellite delivery, cargo delivery, and human spaceflight missions.

Fleet of launch vehicles

	Falcon 9	Falcon Heavy	Starship
Reusability	Partial	Partial	Full & Rapid Design
Height	70m / 299.6ft	70m / 229.6ft	124.4m / 488ft
Diameter	3.7m / 12.0ft	12.2m / 39.9ft	9m / 29.5ft
Mass	594,054kg / 1,207,920lbs	1,420,788kg / 3,125735lbs	5.533,000kg / 12,198,177lbs
Payload capacity to LEO	22,800kg / 50,265lbs	63,800kg / 140,660lbs	100+ Mt
Payload capacity to GTO	8,300kg / 18,300lbs	26,700kg / 58,860lbs	100+ Mt
Payload capacity to Mars	4,020kg / 8,860lbs	16,800kg / 37,040lbs	100+ Mt
Total number of flights	c620	11	12
First launch	June 4, 2010	February 6, 2018	April 20, 2023

Source: Company data, HSBC

Falcon 9: First launched in 2010, Falcon 9 is the world's first orbital-class rapidly reusable rocket and has a payload capacity to LEO of c23t when fully expendable. Falcon 9 has completed approximately 620 orbital space launches as of 31 March 2026, with over 99% mission success rate. According to NASA, the first version of Falcon 9 in 2010 reduced launch cost to USD2,700/kg, c85% less than the historical average launch costs of USD18,500/kg.

Falcon 9 is a two-stage rocket designed to transport satellites, scientific payloads, cargo, and crew to Earth orbit. Reusability allows SpaceX to re-fly the most expensive component of the rocket, i.e. its booster, which lands on either autonomous barges in the ocean or landing zones, while payload fairings (protective cone on top of the rocket) are recovered via parachute assisted splashdowns. The second stage of the rocket (the upper section of the rocket that takes the payload to orbit after the first stage is separated) safely deorbits after payload deployment and is expendable. Falcon 9 established SpaceX as a leading commercial provider via several notable cost achievements:

- **First orbital class rapidly reusable rocket:** Falcon 9 was the first rocket to achieve vertical landing of an orbital class booster in 2015.
- **Highest operational tempo in history:** Falcon 9 is the most frequently flown active orbital launch vehicle to date, providing insight into system performance not otherwise achievable.
- **In-house engine development and manufacturing:** Falcon 9 is powered by Merlin engines providing vertical integration across propulsion design, production, and testing, contributing to Falcon 9's performance and payload capacity.

Falcon Heavy: Falcon Heavy first launched in 2018 and is a partially reusable super-heavy lift launch vehicle designed to deliver large payloads into orbit. Falcon Heavy has a higher payload capacity to LEO vs Falcon 9 at c64t and is one of the most powerful operational rockets in the world measured by liftoff thrust. Falcon Heavy has completed 11 launches as of 31 March 2026 with a 100% mission success rate.

Falcon Heavy builds on existing architecture and uses three reusable Falcon 9 boosters. Like Falcon 9, the second stage is not reusable and safely deorbits after payload deployment. Falcon Heavy has contributed to lowering the cost of space access for large or high yield payloads but faces restrictions on government contracts with regard to booster use (i.e. certain US contracts prohibit the use of boosters that have flown over five times). Falcon Heavy has been selected by NASA to launch critical weather satellites, interplanetary probes including Europa Clipper (Jupiter) and Dragonfly (Saturn), and the upcoming Nancy Grace Roman telescope, demonstrating trans-Mars injection capability.

Starship: First tested in 2023, the strategic focus has shifted to Starship, which is a key enabler of the group's wider growth objectives. Starship is designed to be the first, fully reusable, super heavy lift launch vehicle, with a further step change in payload to over 100t into orbit, and delivery turnaround times equivalent to commercial aviation with multiple launches per day, based on the company's ambitions. This is expected to be more cost effective than Falcon 9, for which demand is expected to decline once Starship is operational. Furthermore, future generations of Starship are being designed to double this payload.

To date, SpaceX has successfully completed 12 Starship test flights, the most recent in May 2026. Given the increase in payload, Starship uses a super heavy booster powered by the next generation Raptor engine, and a newly designed launch pad and innovative new "Chopstick" capture system. Other technical advantages of Starship over Falcon 9 include:

- **Improved engine design increases payload capacity:** The next generation Raptor engine offers nearly triple the thrust and saves nearly 1 ton of vehicle mass compared with the Falcon 9 Merlin engine.
- **Orbital refuelling to support human exploration beyond Earth:** Starship's orbital refuelling capacity will allow tanker variants to refill the upper stage in LEO extending its range of deep-space missions beyond Earth's orbit, including to the Moon as part of NASA's Human Landing System for Artemis, and Mars.

Management expects Starship to enter commercial use/payload delivery in the second half of 2026. However, achieving the desired launch cadence will entail material progress on key operational milestones including developing high-rate launch sites across multiple locations alongside co-located air separation and methane liquification plants, scaling Starship production and securing the requisite regulatory approvals. On scaling Starship production, the group has made substantial investments, including Starfactory for high volume vehicle production, multiple scale vertical integration and refurbishment facilities, additional launch towers, test infrastructure, propellent production, and power generation capabilities.

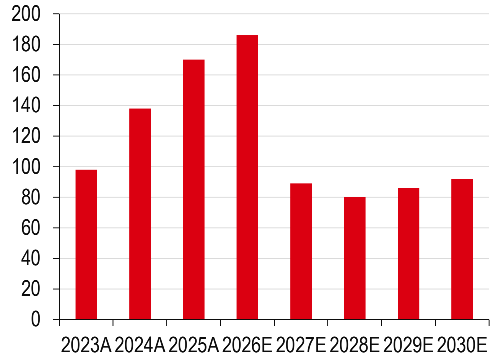
Dragon. Launched by Falcon 9 in 2012, Dragon is an unmanned commercial spacecraft designed to deliver cargo to and from the International Space Station (ISS) with a launch payload mass of 6,000kg, and return payload of 3,000kg. Eight years later, Dragon was the first privately built vehicle to fly humans to the ISS and since 2020 has safely flown 78 crew members to the ISS to date. Deployed by the Falcon 9, Dragon returns to Earth via parachute assisted splashdown after which it is recovered for re-use.

Spacecraft

	Dragon Cargo
Reusability	Partial
Height	8.1m / 26.7ft
Diameter	4.0m / 13.0ft
Spacecraft volume	9.3cbm / 328cbf
Trunk volume	37cbm / 1,300cbf
Lauch payload mass	6,000kg / 13,228lbs
Return payload mass	3,000kg / 6,614lbs
First flight	8 December 2010

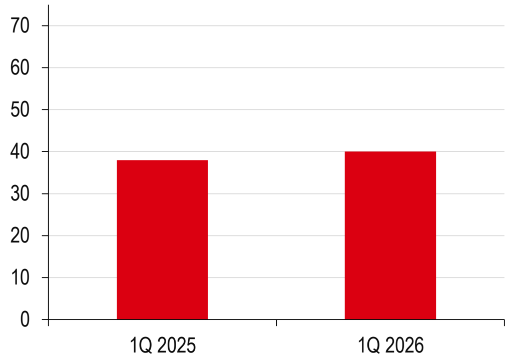
Source: Company data, HSBC

Total SpaceX launches; 2023-30e



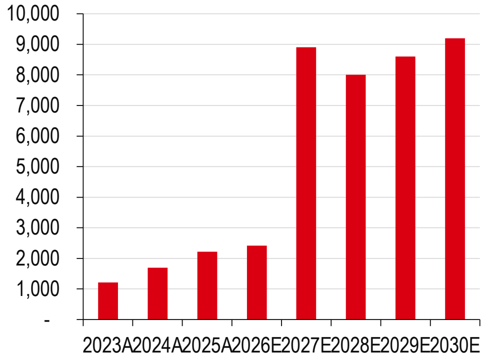
Source: Company data, HSBC estimates

Total SpaceX launches, 1Q25 vs 1Q26



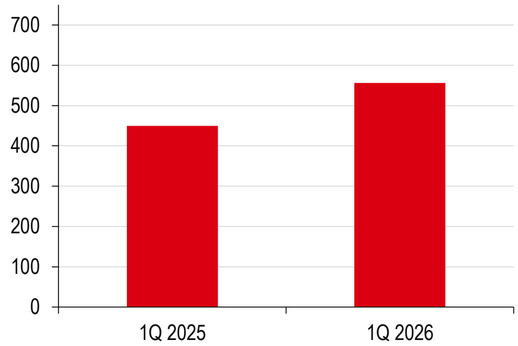
Source: Company data, HSBC

Total mass to orbit (t), 2023-30e



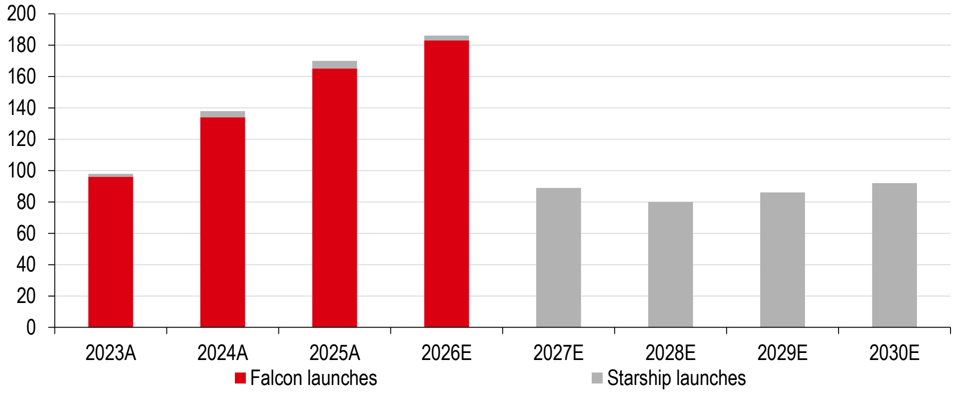
Source: Company data, HSBC estimates

Total mass to orbit (t), 1Q25 vs 1Q26



Source: Company data, HSBC

SpaceX launches by spacecraft, 2023-30e



Source: Company data, HSBC estimates

How does Space generate revenues?

With a TAM of USD370bn, SpaceX's market leadership should drive growth

The company identifies a space-enabled solutions TAM of USD370bn, versus 2025 revenues of USD4.1bn. The company's TAM is based on the 2025 space-enabled solutions market size of USD370bn according to space consulting firm Novaspace, but excludes satellite communication services (including Starlink) as they are included in the connectivity TAM.

McKinsey forecasts the space-based economy to reach USD1.8trn (a range of USD1.4-2.3trn) by 2035 with annual growth of c9% per year, including communications.

In our view, SpaceX is the market leader of the launch market as it offers the lowest launch costs and represents c90% of global mass to orbit. Starship offers meaningfully lower launch costs versus the broader peer group, and we see potential for other businesses to lose market share given that the cost disadvantage may make them obsolete.

SpaceX's price advantage

Company	Rocket	Cost per kg launch (USD)
SpaceX	Falcon 9	3,200
	Falcon Heavy	1,700
Blue Origin (yet to be commercially viable)	New Glenn	1,300-1,600
United Launch Alliance	Vulcan Centaur	9,300
	6 Solids/Upgrade	5,500
Rocket Lab	Electron	30,000
Arianespace	Ariane 62	30,800
	Ariane 64	46,200

Source: Bloomberg intelligence, HSBC

Short-term customer appetite and the commercial viability of Starship are the main constraints, rather than competitor market disruption in the short-term. Starship's lower costs means that SpaceX can lower prices versus the peer group.

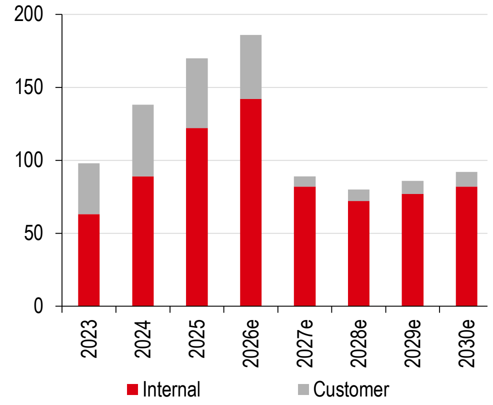
Customers should prioritise reliability and cost when considering launch contracts, meaning that market share gains for SpaceX are likely if Starship reaches commercial viability. As a reminder, in the event of an explosion the cost of the lost cargo sits with the customer, not the launch company (i.e. you pay more for higher reliability), we think.

There are two revenue streams for the Space segment:

1. **Launch services:** Where revenues are charged for the deployment of customer (commercial or government) payload (cargo) to the required orbit. This has mostly been using Falcon 9 and Falcon Heavy spacecraft. These are fixed price contracts that range from one to five years.
2. **Launch and development:** Where revenues are charged for the development of spacecraft alongside the provision of launch and mission services to government agency programmes using Falcon 9, Falcon Heavy, Starship, and Dragon spacecraft. These are fixed price contracts that can range from one to 14 years.

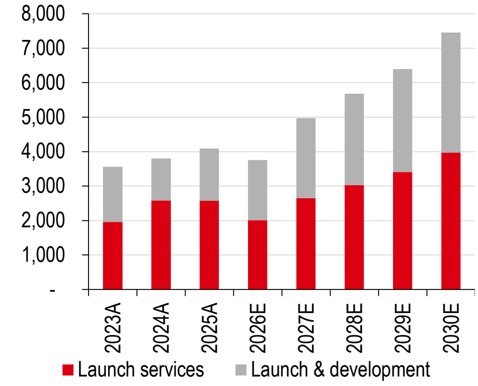
SpaceX does not charge internal revenues for launches related to Starlink missions; however, the launch costs are capitalised through net PPE.

Number of internal vs customer launches, 2023-30e



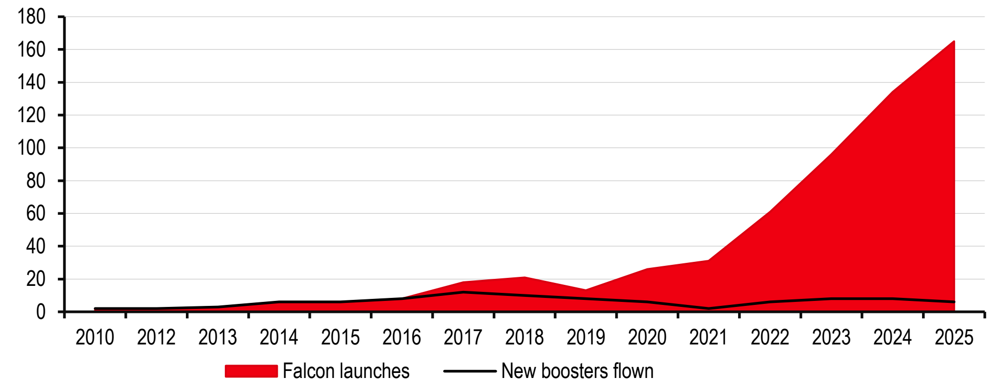
Source: Company data, HSBC estimates

Launch vs launch and development revenues, 2023-30e (USDm)



Source: Company data, HSBC estimates

Total number of launches vs new boosters flown



Source: Company data, HSBC

Fixed price contracts provide margin upside on 'inked-in' contracts

The S-1 prospectus highlights that any cost over-runs with regards to meeting customer contracts must be borne by SpaceX. However, the inverse should also be true, i.e. if a Starship rocket were to be used over a Falcon rocket, the margin upside would be notable for SpaceX. Contract lengths of 1-14 years mean that the earlier the Starship rocket is commercially operational the better for SpaceX.

Gross margin benefits on fixed-price contracts for SpaceX

	unit	Falcon 9	Starship	Future Starship	Long-term ideal
Payload to orbit	t	13	100	200	200
Contract size	t	13	100	200	200
Contract size	**USDm**	**90**	**692**	**1,385**	**1,385**
Cost per kg payload	USD/kg	2,700	1,000	185	50
Launch cost	USDm	35.1	100.0	37.0	10.0
Other COGS	USDm	5.8	5.8	5.8	5.8
COGS	USDm	40.9	105.8	42.8	15.8
Cost of sales	**%**	**45.4%**	**15.3%**	**3.1%**	**1.1%**
Gross margin	**%**	**54.6%**	**84.7%**	**96.9%**	**98.9%**
▲ vs Falcon-9	**bp**		**3,012bp**	**4,231bp**	**4,426bp**

Source: Company data, HSBC estimates

What contracts are currently in place?

- A USD4.16bn deal with the US Space Force under the Space Based Air Moving Target Indicator program. Satellite deployment is expected to be completed by 2028.
- A USD2.29bn deal to build the Space Data Network Backbone. This will provide connectivity for the "Golden Dome" defence contract.
- The UK Ministry of Defence reportedly began using the Starshield network at the start of 2026 for its military operations.
- SpaceX has proposed a direct-to-cell satellite service for government use with an estimated USD100m monthly operational fee and a potential USD500m launch fee.
- SpaceX is under contract to launch Globalstar's HIBLEO-4 replenishment satellites. These missions are for sustaining Globalstar's existing LEO telecommunications infrastructure.
- SpaceX successfully launched 10 satellites for Spire Global on the Transporter-16 mission. These payloads support missions ranging from Earth observation and IoT connectivity to highly accurate measurements of Earth's magnetic field for the National Geospatial-Intelligence Agency (NGA).
- HANCOM InSpace: as part of its commercial services, SpaceX launched the Sejong-3 satellite to support private commercial data applications for South Korean providers.
- 34th Station Resupply: SpaceX is contracted for the 34th resupply mission to the International Space Station (ISS), using the Dragon spacecraft to deliver cargo and scientific experiment equipment.
- Rideshare Collaborations: SpaceX continues to facilitate small-satellite missions through its Transporter series, recently manifesting payloads for companies like Momentus to demonstrate rendezvous and proximity operations (RPO) sensors and power processing units for NASA and the US Air Force.

SpaceX competitors

- **Blue Origin (US)**: The New Glenn remains in its testing stage, successfully completing three test flights, but most recently (May 2026) suffered an explosion on the launch pad during a static fire test. The New Shepard rocket has completed 38 successful flights.
- **United Launch Alliance (ULA, US):** Recent launches have been dominated by Amazon LEO and US Government launches, at a higher cost to customer versus Falcon 9.
  The higher cost base and the upcoming commercial use of Starship means that ULA may struggle to compete. Furthermore, we would expect Amazon contracts to shift to Blue Origin given management links.
- **Rocket Labs (US):** Another launcher of Amazon LEO contracts with similar risk exposure to ULA. Bloomberg Intelligence has Electron rocket costs at c10x of SpaceX.
- **Arianespace (EU):** A major European player, again exposed to Amazon LEO.
- **Indian Space Research Organisation (India):** India deploys its spacecraft from the Satish Dhawan Space Centre. Recent clients include SpaceMobile. Whilst cost per kilogram is difficult to find, we expect higher costs than SpaceX's Falcon 9, let alone Starship.
- **Roscosmos (Russia):** This is the state-owned organisation of Russia launch companies. Given regulatory pushbacks against Russia, we see this as an unlikely competitor to SpaceX.

Corey Chan* (Reg. No. S1700518100001)
Head, A-share Industrials & Renewables Research
HSBC Qianhai Securities Limited
corey.chan@hsbcqh.com.cn
+86 21 5066 2001

** Employed by a non-US affiliate of HSBC Securities (USA) Inc, and is not registered/ qualified pursuant to FINRA regulations*

The competitive landscape in China

We have written on China's LEO viability before (see, *China Aerospace, Rocketing High*, 12 February 2026). We highlight key aspects below.

Where is China in its LEO progress?

Key commercial rocket makers have successfully sent rockets into orbit, with a total of 25 flights in 2025, according to the CNSA. Key commercial rocket makers aim to launch the first flights of reusable rockets in 2026. In 4Q25, CASC's Long March 12A and LandSpace's Zhueque-3 failed in their first-stage booster landing attempts.

China and the US: Key commercial rocket companies

Country	Company	Type	Rocket	Fuel	Current stage
China	LandSpace	Private	Zhuque-2	Liquid	Not reusable; expected pricing of RMB40-50k per kg
China	LandSpace	Private	Zhuque-3	Liquid	Rocket's second stage successfully entered orbit in December 2025
China	Galactic Energy	Private	Ceres-1	Solid	Not reusable; priced at cRMB50k per kg
China	Galactic Energy	Private	Pallas-2	Liquid	Medium to large reusable rocket; first flight targeted for October 2026
China	Space Pioneer	Private	Tianlong-2	Liquid	Not reusable; priced at cRMB75.5k per kg
China	Space Pioneer	Private	Tianlong-3	Liquid	Reusable liquid rocket; first flight targeted for 2026
China	CAS Space	Mixed Ownership	Kinetica-1	Solid	Not reusable; priced at cRMB60-70k per kg
China	CAS Space	Mixed Ownership	Kinetica-2	Liquid	Medium-sized rocket with a designed payload capacity of 12 tonnes
China	Orienspace	Private	Gravity-1	Solid	Not reusable; priced at cRMB30k per kg
China	Orienspace	Private	Gravity-2	Liquid	Reusable rocket under development for largescale networking of medium and large satellites and commercial high-orbit launch missions
China	Deep Blue Aerospace	Private	Nebula-1	Liquid	Working on vertical take-off and vertical landing technology (VTVL) in 2026
China	iSpace	Private	Hyperbola-1	Solid	Not reusable; priced at cRMB106k per kg
China	iSpace	Private	Hyperbola-3	Liquid	Reusable liquid rocket; first flight targeted for 1H26
US	SpaceX	Private	Falcon 9	Liquid	Mature reusable rocket; hundreds of missions in 2025
US	SpaceX	Private	Starship	Liquid	World's largest reusable rocket, still testing second-stage booster reuse technology
US	Blue Origin	Private	New Glenn	Liquid	Successfully recycled first-stage booster in November 2025
US	Rocket Lab	Private	Electron	Liquid	Working on reusable technology
US	Rocket Lab	Private	Neutron	Liquid	Reusable rocket; first flight expected in 1H26

Source: Wind, Tencent News, Company websites, HSBC Qianhai Securities

China: LEO satellite market demand forecasts

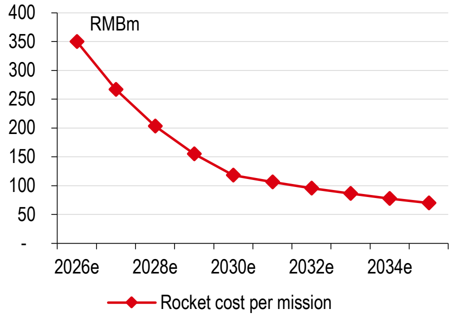
The US Government has restrictions on international client use. Therefore, we expect the Chinese and US/EU launch services to be broadly separate markets.

We estimate that the total number of LEO satellites launched by China will increase to c13k units in 2035e, from 100 units in 2025e, driven by the country's ambitious LEO constellation plans and increasing satellite replacement demand. We believe this could generate a market opportunity of RMB82bn for China's rocket and satellite supply chain by 2035e, of which the largest share goes to the satellite supply chain.

- **Rocket:** We expect the commercial rocket market size to contract to RMB14.8bn in 2035e from RMB20.6bn in 2026e (CAGR of -4%) as we expect fewer rockets will be needed after the adoption of reusable rocket technology. We expect the commercialisation of the first-stage reuse technology by 2030e and the second-stage reuse technology by 2035e in China. This should lower the mission rocket cost by c80% in 2026-35e, based on our estimates. In addition, we expect the number of satellites that each rocket can carry will increase to 60 units in 2035e from 15 units in 2026e, given the use of more medium to large rockets. SpaceX's Starship can carry up to 400 small Starlink satellites at a time, compared with 25-30 units in the Falcon 9 rocket (according to SpaceX).
- **Satellites:** We expect China's LEO satellite market to grow to RMB64.1bn in 2035e, from RMB13.3bn in 2026e (CAGR of 19%), driven by 34% CAGR in the number of satellite launches, partially offset by a declining cost curve. We expect the cost of satellites to drop by 67% in 2026-35e on mass production and design iterations.
- **3D printing:** Major aerospace entities and companies, including NASA, SpaceX, CAS, iSpace, and Space Pioneer, are using 3D printers to build rocket parts as the technology can improve efficiency, reduce weight, and save cost. For instance, using 3D printing in fuel tank production can cut the production time by 72% and the cost by 59%. Space Pioneer used 3D printing to build 90% of its Tianlong-2 engine parts. By doing so, the company was able to shorten the engine production time by 70-80% and cut the cost and weight by 40-50%, according to Wind. The trend of using stainless steel and carbon fibre to build rockets also makes 3D printing a viable technology as these materials are more suitable for 3D printing than aluminium alloys. We expect the size of China's 3D printing market in rocket manufacturing in terms of the value of products manufactured to increase to RMB3.0bn in 2035e, from RMB1.1bn in 2026e (CAGR of 11%), driven by an increasing value share of rocket parts that can be built by 3D printing (to c40% by 2035e from c3% in 2026e).
- **Space solar:** Power systems, including solar cells, account for c11% of a satellite's cost. Currently, the power load of a satellite ranges from a 1kW base to a 20kW base, depending on the function of the satellite (according to Wind). However, the power usage per satellite could increase in the future to power AI datacentres (AIDCs) in space. On 30 January 2026, SpaceX applied for permission to launch one million satellites to power AI (according to the BBC, 1 February 2026). Elon Musk announced plans to build 100GW solar production capacity (according to Reuters, 23 January 2026), following his comments in an earlier interview that he expected 100GW of new AIDCs to be delivered into space. This implies a 100kW power output for a single satellite. We believe this presents opportunities for China's solar supply chain as the country accounted for 80-90% of the global solar product manufacturing in 2024. Perovskite and p-Type HJT are both viable space solar cell technologies, but the cost of the former is nearly 4x that of the latter (source: Wind). Depending on which solar cell technology is being employed, we expect the global market size of space solar to reach RMB11-43bn in 2035e, assuming that 50% or 3.2k units of the total number of global satellites launched in 2035e will be space-based AIDCs.

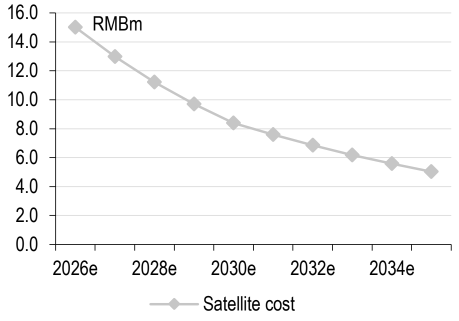
When do we think that the rocket cost will drop to SpaceX levels?

China: We expect the rocket cost to drop 80% in 2026-35e on maturing reusable rocket technology



Source: Wind, HSBC Qianhai Securities estimates

China: We expect the satellite cost to drop 60-70% in 2026-35e on mass production and supply chain integration



Source: Wind, HSBC Qianhai Securities estimates

Connectivity / Starlink

- Starlink could be a key part of the AI and robotics ecosystems; futuristic enterprise use cases could be vital for Starlink's fair value
- Consumer addressable market looks lower than reported due to regulation, technological barriers, and (lack of) government support
- We value Starlink at USD250bn with an EV/Sales-based SOTP

Madhvendra Singh*, CFA
Senior Analyst, TMT
HSBC Bank Middle East Limited, DIFC
madhvendra.singh@hsbc.com
+971 450 93348

Phani Kanumuri
Analyst, TMT
HSBC Securities (USA) Inc.
phani.kanumuri@us.hsbc.com
+1 240 709 8135

Ali Naqvi*
Senior Global TMT Analyst
HSBC Bank Middle East Limited, DIFC
ali.naqvi@hsbc.com
+44 20 7992 2176

** Employed by a non-US affiliate of HSBC Securities (USA) Inc, and is not registered/ qualified pursuant to FINRA regulations*

Starlink: global satellite connectivity for B2C and B2B

Starlink is SpaceX's connectivity business, it delivers broadband connectivity. Its market-leading low-Earth orbit (LEO) satellite constellation provides coverage across two segments: (1) consumer broadband and mobile services, and (2) enterprise and government solutions.

SpaceX cited a total addressable market (TAM) of USD1.6tn for Starlink (cUSD1.4tn for consumer and cUSD200bn for enterprise and government solutions). We believe the serviceable market in consumer broadband and mobile is likely smaller than the stated TAM, but we see stronger upside potential from expansion in enterprise and government demand, especially with expansion within its own and related ecosystem.

We view Starlink primarily as a complement to incumbent telecom operators in the retail connectivity market, helping to bridge the connectivity gap in underserved and remote markets rather than as a frontal threat. However, we see meaningful growth potential in enterprise use cases, especially across logistics and travel, as well as in government and defence. Longer term, Starlink could play a role as a data backbone for physical AI and space datacentres, although the addressable market is difficult to size given these segments are nascent.

Connectivity estimates

USDm, FYE Dec	2023A	2024A	2025A	2026E	2027E	2028E	2029E	2030E
Consumer	2,817	4,830	7,208	10,211	14,615	18,618	23,631	28,796
Enterprise & Government	1,052	2,769	4,179	5,339	7,289	10,204	14,286	20,000
Revenue	**3,869**	**7,599**	**11,387**	**15,551**	**21,903**	**28,822**	**37,917**	**48,797**
Cost of revenue	2,786	4,768	5,921	7,788	10,742	13,847	17,836	22,468
Research and development	381	453	575	953	1,256	1,538	1,871	2,213
Selling, general, and administrative	233	333	468	1,068	1,564	2,058	2,707	3,484
Restructuring charges	-	-	-	-	-	-	-	-
Impairment	-	39	-	-	-	-	-	-
Total costs and expenses	**3,400**	**5,593**	**6,964**	**9,810**	**13,561**	**17,443**	**22,414**	**28,165**
Income (loss) from operations	**469**	**2,006**	**4,423**	**5,741**	**8,342**	**11,379**	**15,502**	**20,632**
D&A	884	1,508	2,376	3,802	5,572	7,620	10,253	13,195
EBITDA	1,353	3,514	6,799	9,543	13,914	18,999	25,755	33,826
Adjustments	249	335	369	546	712	865	1,137	1,464
Segment adjusted EBITDA	**1,602**	**3,849**	**7,168**	**10,089**	**14,626**	**19,864**	**26,893**	**35,290**
Share based compensation	249	296	369	546	712	865	1,137	1,464
Capital expenditures	2,455	3,498	4,178	6,662	9,225	14,256	15,703	17,225

Source: SpaceX, HSBC estimates

Valuation

Given the relative nascency of the Starlink operations and growth potential, especially in the enterprise segment, we use SOTP-based EV/Sales on 2028e revenues for the residential broadband and enterprise segments. In addition, we use 2040e EV/Sales multiple discounted to current for not yet launched residential mobile and futuristic B2B services to derive our valuation of around USD250bn for Starlink. Our SOTP approach comprises:

- 5x 2028 EV/Sales for the Residential broadband segment (two times that of US telecoms.
- 7.5x 2028 EV/Sales for the Enterprise segment (three times that of US telecoms multiple).
- 10x 2040 EV/Sales for the “futuristic ventures” segment. We reflect Starlink’s ambitions regarding autonomous driving, airlines, ships. and IoT market. as well as the data backbone for robots, to derive potential 2040e revenue from these ventures. We then assign a 10x EV/Sales multiple before discounting the EV back to 2026 using a WACC of 8.5%. We assign a 5% probability of success to temper our ambitious revenue forecasts and to take into account possible restrictions from government and regulatory headwinds.
- Similarly for the residential mobile segment, we use a 5x 2040e EV/Sales multiple discounted to 2026.
- To derive our WACC for calculating the net present value for the futuristic ventures and the residential mobile activities, we use a cost of equity of 11.0% (risk free rate of 4.25% and enterprise risk premium of 4.20%, both based on HSBC strategy team estimates; see *Cost of equity 2026*, 13 July 2026, beta of 1, and an execution risk premium of 2.5%), and cost of debt of 6%, rounded to the closest 0.5%.

We apply our innovation premium at the group level to the sum of the EVs from each segment.

Starlink: SOTP-based valuation on EV/Sales

	Revenue – 2028	EV/Sales (x)	EV
Residential broadband	18,618	5.0	93,090
Enterprise	10,204	7.5	76,532
Residential mobile		5.0	24,887
Futuristic venture		10.0	55,407
EV			**249,916**
EV/EBITDA – 2027			**17.1x**
EV/EBITDA – 2028			**12.6x**

Source: HSBC estimates

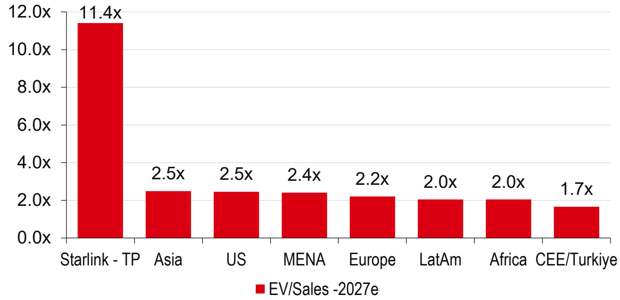
To sense check our valuation, we compare our Starlink valuation and fundamentals with US telcos. We expect Starlink to have 36-47% of US telco EBITDA by 2028. However, our valuation is 71-87% of US telcos EV. The valuation premium for Starlink is derived from faster growth and higher operating cash flow margin, which we expect to be at or above 40% after the operations reach maturity compared with US telco operating cash flow margin of 25% for US telco average (2027)

Our valuation for Starlink is 71-87% of US telco EV, although it forms 19-28% of revenues and 36-47% of EBITDA

Starlink as percentage of US telcos	2028 Revenue	2028 EBITDA	Target EV to current EV
AT&T	21.4	38.4	74.4
VZ	19.3	35.8	70.6
TMUS	28.0	47.3	86.8

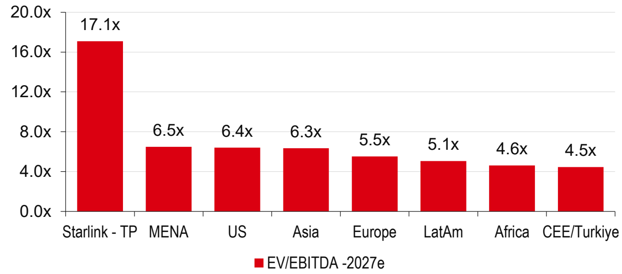
Source: HSBC estimates

Our Starlink fair value EV/Sales is 4-6 times that of current telco multiples



Source: Company data, HSBC estimates

Our Starlink fair value EV/EBITDA multiple is 2.4 to 3.4 times that of current telco multiples



Source: Company data, HSBC estimates

Bear and bull case scenario valuation

Most of our target valuation is driven by future use cases and the terminal value. We assess valuation under both bull and bear case scenarios. Based on the assumptions set out in the table below, we derive an implied fair value of approximately cUSD550bn in the bull case and cUSD100bn in the bear case.

Starlink: Bear, bull, and base case valuation scenarios

Assumptions	Bear case	Base case	Bull case
Residential broadband			
Starlink broadband subscribers as a percentage of addressable homes	2.0	4.2	6.0
ARPU change vs base case (per year, ppt)	-2.0	0.0	2.0
Residential mobile			
Residential mobile subscribers as a percentage of the addressable population	0.5	0.9	2.0
ARPU change vs base case (per year, ppt)	-2.0	0.0	2.0
Enterprise			
Enterprise growth vs base case (per year, ppt)	-5.0	0.0	5.0
Futuristic ventures			
Probability of success %	2.5	5.0	12.5
Multiples			
Residential broadband revenue multiple (x) - 2028e	2.5	5.0	7.5
Residential mobile revenue multiple (x) - 2040e	2.5	5.0	7.5
Enterprise revenue multiple (x) - 2028e	5.0	7.5	10.0
Futuristic revenue multiple (x) - 2040e	7.5	10.0	12.5
Implied EV (current, USDbn)	**100**	**250**	**550**

Source: HSBC estimates

How does Starlink work?

Starlink has c9,600 satellites in function and has the ambition to materially increase the capacity of its broadband and mobile constellations. Traditionally, satellites are placed in geosynchronous orbit (GEO), which is 36,000km from Earth. Given the distance, the throughput is low and latency is high. Starlink operates in the low Earth orbit (LEO), 500-2,000km for higher throughput and lower latency than traditional satellite companies.

Starlink uses Ku (11-16GHz), Ka (17-30GHz), and E-band (71GHz-86GHz) for data transmission. Each Starlink satellite has five advanced Ku-band phased array antennas and three dual-band (ka-band and E-band) antennas. Ku and Ka band are used for data transfer between satellite and ground terminals (user and ground station). E-band is used for backhauling to gateway Earth stations. New and more capable satellites also use V-band (37.5-52.4GHz) for data transfer.

Different categories of satellites providing connectivity

	Category	Latency	Coverage
36,000 km	**GEO** *Geo-synchronous earth orbit*	~600ms	***3-4 satellites*** to provide global coverage (24 hours)
36,000 km To 2,000 km	**MEO** *Medium earth orbit*	~180ms	***10-15 satellites*** to provide global coverage (24h), non-high latitude
2000 km To 160 km	**LEO** *Low earth orbit*	~30ms	***Dozens to thousands*** of satellites to provide global coverage (24h)

Source: Company data

Broadband now and possibly more mobile services tomorrow, in theory
There is an important distinction between Starlink's broadband and mobile propositions. Broadband is mature, and the company's expansion plans focus on increasing B2C customers. By contrast, Starlink's mobile strategy is primarily B2B partnering with mobile network operators (MNOs) to reduce "dead zones" and extend coverage. Reflecting this, Starlink Mobile is classified within the Enterprise segment. Key constraints for Starlink to offer B2C mobile include 1) lower signal strength as it travels from satellite to ground station at high frequencies, and 2) lack of indoor penetration (i.e., difficulty reaching users inside buildings). Our addressable market assumptions (read page 52) assume that Starlink can ultimately address the B2C mobile market, consistent with its stated ambition to launch Gen2 services. But for Starlink to become a full-scale telecom competitor, it may need to secure spectrum rights across multiple jurisdictions, each with distinct regulatory regimes. Starlink has indicated that it intends to continue partnering with mobile operators, which may mitigate disruption risk in the short to medium-term.

The markets covered by Starlink, and where it could expand

Now: Currently, 164 countries have Starlink broadband access. Starlink mobile is available in 12 countries, and 19 countries where it expects services to be available soon. We see less connected users in high-income economies like the US, Germany, the UK, France, Japan, South Korea, and Australia as markets with strong potential.

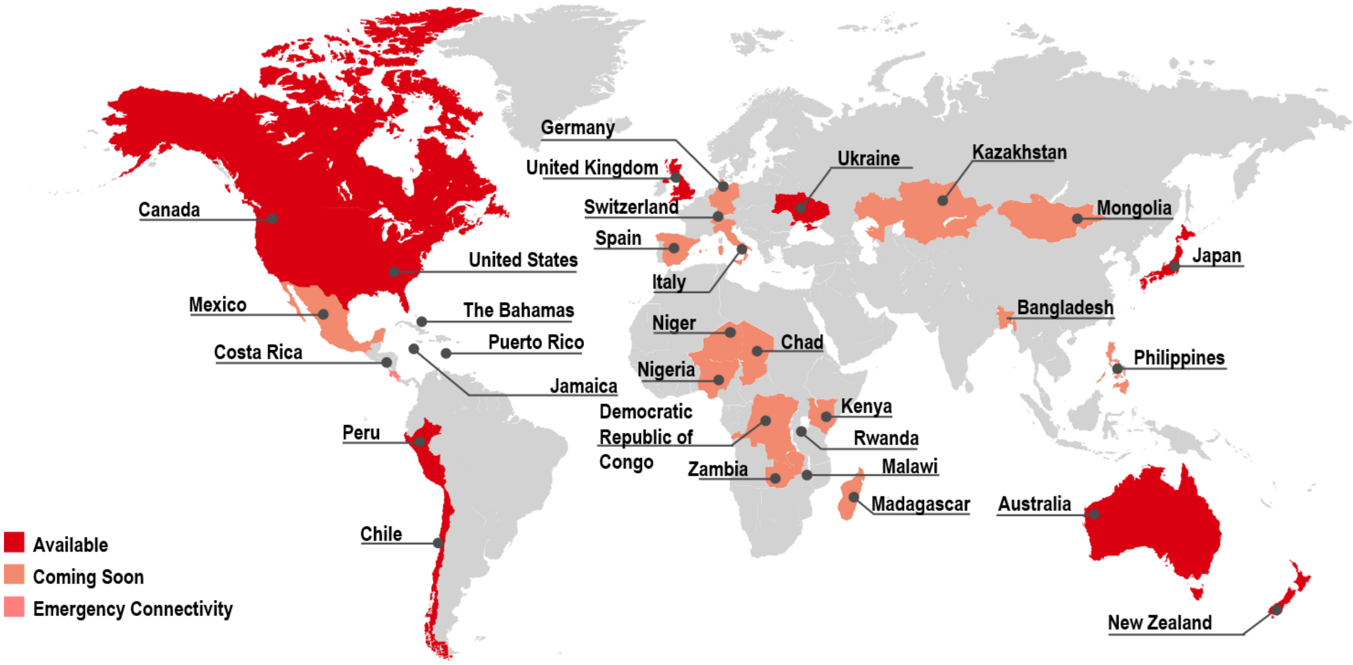
In the future: We use four criteria with minimum value assumptions to identify the markets that have appropriate income (GDP per capita of above USD2,500), some room for growth in internet users (percentage of internet users less than 85%), high percentage of rural users (greater than 20%), and the size of the population (greater than 20m) that could see good adoption of Starlink. The markets that meet the above criteria are Bangladesh, Indonesia, Philippines, Egypt, Angola, Mexico, Ivory Coast, Vietnam, South Africa, Colombia. Algeria, Iraq, and Ukraine.

Can satellites compete with telcos?

Traditionally, there are some key technical reasons why satellite operators struggled to compete with telcos. These include **low indoor coverage, constrained capacity, lower throughput, and higher latency.** Starlink aims to launch its Starlink V3 broadband satellite in 2H26, which has 1Tbps downlink speed vs 96Gbps for V2 Mini (broadband) and a Starship can launch 60 satellites vs less than 30 for V2 Mini (broadband). Overall, Starlink expects one satellite launch to provide 20x more download speed when using V3 broadband.

Still, despite much stronger network capacity, Starlink could face limitations to compete effectively against telecom operators:

Starlink Mobile: Availability



Source: Company data, HSBC

Starlink vs fixed broadband vs mobile: Comparison of speeds

Technology	Cycle	Speed/Latency
60ms Starlink	V1	50-100Mbps; more than 60ms
	V2 Mini	Up to 475Mbps with 225Mbps during peak hours; 25ms to 60ms latency
	V3 (Yet to launch)	Aiming for more than 1Gbps and less than 20ms latency
Fixed broadband	vDSL	Maximum of 350Mbps; 20ms-50ms latency
	Cable	Up to 1Gbps with DOCSIS 4.0 reaching 10Gbps theoretical limit; 10ms-25ms latency
	Fiber	Up to 10Gbps but can be higher in enterprises; The maximum speed achieved was 300Tbps in lab; 1ms-20ms latency
Mobile	2G	15-200kbps on average; 200ms to 1000ms latency
	3G	1-8Mbps on average; 100ms to 300ms latency
	4G	15-100Mbps on average with theoretical maximum of 1Gbps; 20ms to 50ms latency
	5G	100-500Mbps with theoretical maximum of 20Gbps; 1ms-20ms latency
	6G (Not yet fully defined)	Theoretical maximum of 1Tbps; Theoretical latency of less than 1ms

Source: Various sources, HSBC

- **Indoor signal loss:** TMUS in a recent investor conference said that D2D (direct-to-device) does not work well indoors due to power loss from transmission from satellites to mobile devices, which makes it more difficult to pass through roofs/walls. Furthermore, Starlink V3 broadband would orbit closer to the earth (350km) than V2 Mini (broadband) orbit (c550km). While this helps with higher speeds, it also means a lower coverage area and leads to more propulsion needed to remain in orbit (hence a shorter lifespan eventually).
- **Beam size and capacity:** Each beam in Starlink's V2 Mini (broadband) has 160sqkm of radius and each beam can hold a limited number of users (512 as per the company, subject to capacity availability). However, although V3 broadband could lower the capacity constraints in urban markets, it is unlikely to work in high dense areas.

Starlink: Capacity increases on each new generation of satellites but capacity constraints could continue to exist

Satellite	V1	V2
Number of beams	128	1024+
User per beam	128	512
Area per beam (sqkm)	900	160

Source: Company data

How traditional telcos collaborate with Starlink

From the major US telecoms, the messaging has been consistent, in that LEO (low Earth orbit) satellite constellation operators are a completement to the existing cellular voice and data infrastructure and not a competitive threat. Helped by current broadcast technology, as well as established distribution channels, telecom operators have the advantage in providing and distributing voice and data services compatible with existing devices and the track record in providing a reliable service.

While public comments from telecom providers point to satellites as a complementary technology, the potential for Starlink to deploy a major direct-to-device (D2D) strategy should not be ruled out. Future generation protocols (such as 6G) are likely to make D2D data transmission more integral to the communication protocol, especially as Starlink acquires spectrum assets in key markets, such as the US.

Potential US telecom and satellite service joint ventures

AT&T, T-Mobile, and Verizon formed a joint venture (JV) aiming to help end wireless dead zones in the US by pooling spectrum to address coverage gaps. The JV aims to standardise technology standards between operators and help satellite providers reach more customers through a unified platform. While the JV is in early formation with no confirmed mandate, it was communicated as a way to improve access for US consumers. However, it will also encourage and ease competition, at least in the US market, and push other satellite operators to increase capacity. New or smaller entrants will presumably negotiate with the JV and have access to a wider range of US consumers without dealing with telecom operators individually.

Starlink's acquisition of US Spectrum

EchoStar sold its AWS-4 spectrum (40MHz) and H-Block spectrum (10MHz), totalling 50MHz of combined mid-band spectrum, to SpaceX for USD17bn (Echostar press release, 8 September 2025) via a combination of cash and SpaceX stock. Later, the agreement was extended to include 15MHz of unpaired spectrum, with SpaceX acquiring 65MHz of spectrum for a total consideration of up to USD19.6bn. The acquisition of licences provides flexible-use spectrum to deliver direct-to-consumer mobile services, enabling Starlink Mobile to operate on its own frequencies.
In comparison, AT&T acquired 30MHz of nationwide 3.45GHz mid-band spectrum and approximately 20MHz of nationwide 600MHz low-band spectrum for USD23bn. Our view is that we cannot rule out Starlink as a potential future competitor to the traditional telecom operators, especially if technological barriers become lower, but we do not see it as likely in our base case.

Key telco-Starlink partnerships globally

We have identified several partnerships between telecom players and Starlink, as well as other satellite companies, globally. Below we highlight some of the key telco-Starlink partnerships that have been announced by the operators and highlight the company's positioning and potential in some of the major markets.

- In **Zambia**, **MTN** completed field testing of D2C services from Starlink and aims to commercialise it soon.
- **Entel** partnered with Starlink to launch D2C service in **Chile** and **Peru**.
- **Airtel Africa** was the first African MNO to partner with Starlink, initially launching enterprise and backhaul solutions in May 2025, and expanded to D2C **services across 14 markets**.
- In October 2025, **STC group** announced a 10-year agreement with **AST SpaceMobile (ASTS),** a peer of Starlink, with a total contract value that could exceed 9% of 2024 revenues (i.e. more than USD1.8bn).
- **Optus** (in **Australia**) and Starlink are working to expand mobile coverage across Australia using direct-to-mobile satellite technology.
- In March 2025, both **Bharti Airtel and Jio** signed deals to explore reselling of Starlink equipment through their retail outlets – subjected to SpaceX receiving its own authorisations to sell its services in India.
- In May 2024, **PT Telkom's** (in **Indonesia**) subsidiary 'Telkomsat' became an authorised Starlink reseller in Indonesia and has previously collaborated with Starlink to provide backhaul services.
- Starlink launched in **Japan** in 2022, and it provides fixed residential, roam, business, and maritime plans. Moreover, Starlink has been providing telecom operators with mobile backhaul service since 2022.
- Telekom **Malaysia** has partnered with Starlink to support rural and enterprise connectivity.
- Globe Telecom in **Philippines** has partnered with Starlink providing direct-to-cell LTE connection to bridge coverage gaps and backup during extreme weather events.

How telcos view the competition from Starlink and satellite companies

Operator	Impact from satellites
AT&T	AT&T continues to view satellites as an important complementary role in connecting customers…but does not think satellite is a substitute for the speed, reliability, and capability of its assets
T-Mobile US	Sees no effective way for satellites to compete with a terrestrial network as they are limited by the laws of physics.
Verizon	Sees satellites as a complementary service and thinks it is difficult for satellites to compete with terrestrial networks
AMX	AMX stated in its 2026 investor day that it is looking at partnerships with satellite companies. While it is watching satellite companies' potential to turn as competitors closely, it does not see it as a likely scenario
MTN	MTN's strategy with satellite providers varies by geography and corresponding operating conditions. In some markets it is in partnerships for backhaul and in some countries it compares with satellite operators. In Zambia, MTN completed field testing of direct to cell services from Starlink and aims to commercialise it soon. The group is taking a technology-agnostic approach, deploying the solutions that best enable monetisation while delivering reliable service to African households and businesses.
Megacable	Satellites do not compete significantly in Mexico and satellite offerings are targeted at rural areas rather than urban and have a much higher price
Telefonica Brasil	Most of the current broadband net adds for Starlink are in areas that are not connected well and do not see any major competitive concerns
Entel	Partnered with Starlink to launch direct-to-cell service in Chile and Peru. Around 10% of customers porting into Entel said they choose it specifically because of Starlink in 1Q26. Starlink has a strong value not only in upselling plans, but also in attracting new customers.
Airtel Africa	Airtel Africa became the first African MNO to partner with Starlink, initially launching enterprise and backhaul solutions in May 2025 and planning to expand to D2D services across 14 markets. The initial rollout supports basic data for selected applications and text messaging in remote areas, positioned as a failover rather than a replacement for Airtel's existing network. Management stated that it is engaging with regional regulators, and deployment will proceed in line with country-specific regulatory approvals.
Mobily	Mobily has introduced D2D services but states that the immediate domestic impact of LEO D2C services in Saudi Arabia will remain limited as domestic approval for these satellite connections is currently restricted, primarily to the aviation and maritime sectors. Mobily maintains that traditional terrestrial 4G and 5G networks will continue to deliver superior capacity, latency, and cost efficiency for mass-market usage vs emerging LEO alternatives.
STC	In October 2025, STC group announced a 10-year agreement with AST SpaceMobile (ASTS), a peer of Starlink, with a total contract value that could exceed 9% of 2024 revenues (i.e. more than USD1.8bn). As part of the agreement, STC committed a USD175m prepayment. The final contract value is dependent on transaction volumes; STC will receive a percentage of the revenues generated from use of the services. Commercial services are expected to launch in 4Q26, subject to regulatory approvals. Under the agreement, the partnership covers STC's core Saudi market. In addition, ASTS granted STC exclusive rights in Libya, Syria, Lebanon, Djibouti, Tunisia, and Sudan, and preferential rights in Kuwait, Oman, Yemen, Bahrain, Jordan, Iraq, Algeria, Morocco, and Egypt (subject to availability). Rather than focusing solely on D2D consumer applications, STC Group is pursuing comprehensive, multi-orbit satellite strategy aimed at enterprise and advanced communication solutions. It has signed a strategic agreement with Telefónica Global Solutions (TGS), the global business arm of the Spanish telecom. This will enable STC to develop and offer customised connectivity services that span across LEO, medium earth orbit (MEO), and geostationary orbit (GEO) networks, ensuring flexible coverage options for diverse enterprise requirements.

Source: Company data, HSBC

We calculate a lower TAM than communicated by SpaceX

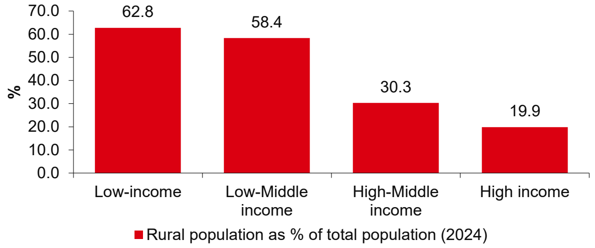
Our analysis suggests a TAM that is 78-92% lower than the TAM implied by Starlink's weighted-average ARPU assumptions. We see geographical exclusion, affordability, relative pricing, technology, and other limiting factors.

Consumer Broadband and Mobile: SpaceX's TAM assumptions too ambitious in our view Starlink estimates the majority of its TAM (USD1.4trn out of USD1.6trn) lies in traditional telecoms segments of consumer broadband (USD660bn) and mobile segments (USD740bn). For the calculation of TAM, Starlink uses the average of ARPUs for broadband and mobile services among four income groups: high, high-middle, low-middle, and low.

We believe the market Starlink can realistically capture – its serviceable obtainable market (SOM) – is materially smaller. We estimate a total SOM in consumer broadband and mobile of USD105-310bn (vs TAM of USD1.4trn provided by the company), with USD40-120bn in consumer broadband (vs TAM of USD660bn company estimates) and USD60-185bn in mobile (vs company estimated TAM of USD740bn).

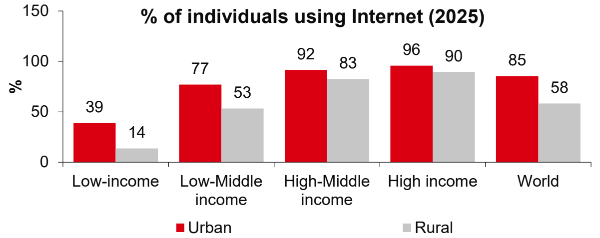
- **Demographics:** Starlink's initial target base is rural and underserved populations (c40% of the global population, as per the prospectus). SpaceX states that Russia and China are excluded from its TAM. However, the mobile TAM appears to use a c8bn population input, which likely includes both countries. In our analysis, we exclude Russia and China from all estimates to ensure consistency. We look at Brazil and Starlink's expansion to further validate our view (see, *Brazil Telecoms: Starlink: Beyond the headline numbers*, 8 April 2026 for more details). Our analysis suggests that Starlink's strong broadband net adds were more pronounced in low population density states and municipalities with smaller numbers of broadband subscribers. Starlink has more than 2% market share in only four of the top 100 municipalities by broadband subscribers (Feb 2026). Its presence in large and key states like Sao Paulo and Rio de Janeiro, and key municipalities is not significant, and we do not expect significant gains in these states in the future.
- **Technology:** LEO satellites operate several hundred kms above the Earth, which inherently affects latency and signal strength. While Starlink highlights improvements in throughput and latency with newer satellite generations, we believe satellite connectivity will structurally struggle to match terrestrial networks in dense urban and suburban areas due to the need to propagate signal between ground stations and satellites, where fibre and 4G/5G typically offer superior economics and performance. Furthermore, terrestrial networks see regular and gradual technology upgrades. Still, the pace of change has been significant for Starlink as it plans to launch the V3 version of satellites in 2H26, nearly seven years after launching its first satellites.

Rural population is more prominent in low-income countries, where Starlink could have affordability concerns



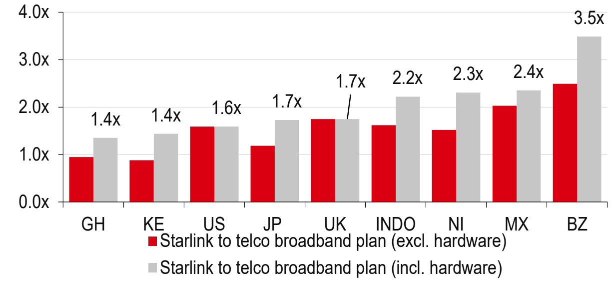
Source: World Bank, HSBC

A big part of the unconnected population – a target audience for Starlink – is in low-income countries, again could have affordability concerns



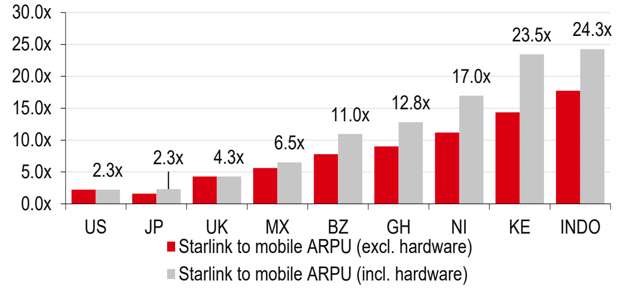
Source: ITU, HSBC

Broadband: Starlink's plans are 1.4-3.5x more expensive than telcos in key global markets



Source: Company data, HSBC
MX - Mexico;BZ-Brazil;KE-Kenya;NI-Nigeria;GH-Ghana;US-US;UK-UK;JP-Japan;INDO-Indonesia

Mobile: Starlink's plans are 2.3-24.3x more expensive than telcos in key global markets



Source: Company data, HSBC

- **Affordability:** Many rural households are concentrated in low- and lower-middle income countries, where affordability is a key constraint. Starlink's TAM methodology applies a weighted-average ARPU across countries, which we believe overstates the revenue opportunity in lower-income rural markets. In our approach, we apply ARPUs by income cohort, assume higher serviceability in rural versus urban areas, and estimate an addressable market range accordingly. Our analysis suggests the obtainable market in rural areas is more than 70% lower than the TAM implied by Starlink's weighted-average ARPU assumptions.
- **Relative pricing vs telcos:** Starlink's pricing is at a premium to many terrestrial broadband and mobile offerings, which could limit mass-market uptake – particularly outside the high-income cohort. Based on our analysis, we estimate that Starlink's broadband pricing is 1.4-3.5x higher on average than comparable terrestrial plans. While not exactly a comparable, Starlink's pricing premium vs mobile ARPUs suggests significantly different pricing is likely needed if Starlink aims to compete on the mobile frontier. In addition, initial capex/upfront installation costs could also be prohibitive.
- **Regulation:** Starlink offers fixed broadband in 164 markets, covering 3.3bn people (41% of the 8bn global population and 51% of the 6.4bn population excluding China and Russia). The TAM could exclude several other markets based on regulatory approvals. Mobile service availability is lower, currently in 12 countries, with a further 19 countries that Starlink expects soon. The communications sector is one of the most heavily regulated sectors globally and regulatory frameworks differ by country; securing licences, landing rights, and spectrum approvals across all markets is likely to be a meaningful hurdle.
- **Government intervention:** Given the importance of telecoms in connectivity and social order, regulators typically tend to tightly control key market participants and tend to favour local operators or local subsidiaries of MNCs with local management.

Consumer broadband + Mobile TAM for Starlink, according to HSBC (USDbn, rounded to the nearest USD5bn)

		% Urban TAM that can be serviced					
		0.0	**5.0**	**10.0**	**15.0**	**20.0**	**25.0**
% Rural TAM that can be serviced	**0**	0	40	80	120	165	205
	20	60	105	145	185	225	265
	40	125	165	205	245	285	330
	60	185	225	270	310	350	390
	80	250	290	330	370	410	450
	100	310	350	390	435	475	515

Source: HSBC estimates

Consumer broadband TAM for Starlink, according to HSBC (USDbn, rounded to the nearest USD5bn)

		% Urban TAM that can be serviced					
		0.0	**5.0**	**10.0**	**15.0**	**20.0**	**25.0**
% Rural TAM that can be serviced	**0**	0	15	35	50	70	85
	20	25	40	60	75	90	110
	40	50	65	80	100	115	130
	60	70	90	105	120	140	155
	80	95	110	130	145	165	180
	100	120	135	155	170	185	205

Source: HSBC estimates

Mobile TAM for Starlink, according to HSBC (USDbn, rounded to the nearest USD5bn)

		% Urban TAM that can be serviced					
		0.0	**5.0**	**10.0**	**15.0**	**20.0**	**25.0**
% Rural TAM that can be serviced	**0**	0	25	45	70	95	120
	20	35	60	85	110	130	155
	40	75	100	120	145	170	190
	60	110	135	160	185	205	230
	80	150	175	195	220	245	265
	100	185	210	235	255	280	305

Source: HSBC estimates

Enterprise addressable market: assessing the potential

Starlink reported an Enterprise TAM of USD205bn out of which USD200bn for fixed broadband market to SMEs and USD5bn related to defence and government. Starlink also sees potential for future use cases in; 1) Enterprise Mobility for aviation, maritime, and vehicle fleets, 2) Enterprise back-up and failover connectivity, 3) expanded government application, 4) Smart device connectivity, and 5) in-orbit transport.

We attempt to estimate a market size for the future use cases, particularly connected cars driven by increased autonomous driving. We estimate a USD650bn addressable market, most of which comprises connected cars and IoT. However, the competition in these segments is high and several operators and satellite networks alone should not be able to service these segments. Aviation and Maritime have better growth prospects and lower competition, but the market size is relatively low. Physical AI and data backbone for space datacentres are exciting development areas but given the lack of proven cases, we think it is premature to value the market opportunity.

The key emerging areas are as follows:

Connected vehicles: With increasing focus on autonomous driving, we see strong potential for growth in data consumption in cars and other vehicles. We assume 12.5% of vehicles globally will have high data consumption on the back of more autonomous driving and this leads to a TAM of USD563bn. We note that a significant part of current data uploaded for training AI is done over home internet. We do not expect Starlink to capture a majority of this demand, although we see it as a backup and potentially an alternate channel for data transmission and is likely to capture some part of the growth.

Land Mobility partnerships: John Deere, the California Fire Department, and passenger rail operators such as Brightline (Florida), and Italo Treno.

Enterprise Mobility excl. connected cars: We see strong potential TAM for the mobility segment with focus on providing internet in airlines, cargo ships, and on highways. We use the total number of aeroplanes and ships, excluding China and Russia. Starlink's plans have a combination of; 1) price per vehicle (ship, aircraft, etc.) and 2) additional charges for any usage over allocated quota. We use the price of unlimited plans for these mobility vehicles to calculate TAM.

Airline partnerships (implemented or committed): United Airlines, Southwest Airlines, Qatar Airways, Lufthansa Group, British Airways, Alaska Airlines, Frontier Airlines, Wizz Air, Volaris, Jet SMART, Cebu Pacific, American Airlines and Hawaiian Airlines.

Cruise operator partnerships: Carnival Corporation, Royal Caribbean Group, MSC Cruises, and Norwegian Cruise Line Holdings.

IoT/Smart devices: As per GSMA, 2-3bn IoT devices could be addressable by satellites by 2030 and could represent a USD10bn revenue opportunity. To calculate TAM, we use wide-area IoT estimates by Ericsson by 2031 and use ARPU of USD4, nearly half of global mobile ARPU.

Enterprise back-up and failover connectivity: While companies may prefer the terrestrial networks for connectivity, satellite connectivity could provide an additional layer of security with Starlink providing back-up and failover continuity.

In-orbit connectivity: Starlink allows third parties to purchase its space laser hardware and connect third-party satellites to Starlink network, allowing others to offload data to a ground station anywhere on Earth, while bypassing the need to build their own relay architecture or ground stations. Moreover, Starlink appears to be the only realistic option to act as a data backbone, when and if it launches space datacentres.

Fixed B2B connectivity market: Starlink serves a broad fixed-site customer base across industries such as retail and financial services that require high availability for critical operations as well as reliable connectivity in remote or hard-to-serve locations.

Government organisations: For government customers, Starlink provides connectivity services for public services, social impact, humanitarian efforts, and disaster response in even the most remote environments. Separately with Starshield, Starlink developed a secure,

Enterprise: Emerging verticals could have significant potential – HSBC estimates

Starlink TAM (emerging enterprise verticals)	Starlink position	USDbn
Airlines		3
Maritime		12
Vehicles		563
IoT		73
Total		**650**
Key verticals to watch but not in our forecasts		
Physical AI		
Data backbone for space datacentres		

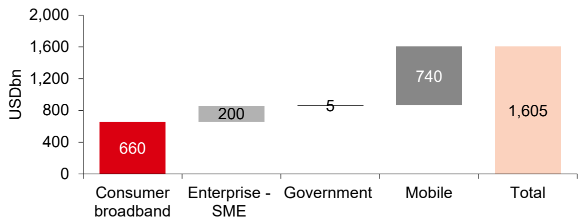
Source: HSBC estimates, multiple sources; Note: dark Green indicates strong potential to be market leader, and light green indicates significant competition dedicated satellite network designed specifically for US Government customers and national security applications. In May, US Space Force selected SpaceX for two major contracts, one of which is to build a space data network backbone worth USD2.3bn.

Starlink TAM: SpaceX vs HSBC estimates
As explained earlier, we have identified factors that limit Starlink's addressable market in the Consumer segment including geographical exclusion, affordability, relative pricing, and technology. However, although we value futuristic bets as we acknowledge that B2B represents an opportunity, we assign a 5% probability to achieve our 2040 revenue forecasts in our valuation.

The **upside risks** to our TAM estimates include Technology upgrades that enable Starlink to compete more effectively in urban areas. Mobile players could compete to partner with Starlink, giving Starlink favourable terms in negotiation, especially smaller operators who lack geographical and population coverage and could use the help with limited capex and costs, while Starlink expands beyond connectivity into newer services.

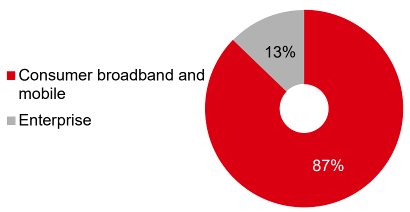
The **downside risks** include lower pricing power than expected, regulatory restrictions, and Starlink failing to acquire spectrum to compete with mobile operators, impacting growth momentum. Local regulations, especially around decrypting data and mandatory use of international gateways/switches in controls of the central regulators/authorities could be another hurdle/risk.

Starlink: TAM of USD1.6trn according to SpaceX…



Source: Company data, HSBC

…and the Consumer segment has the largest TAM opportunity



Source: Company data, HSBC

Starlink: Our estimates for TAM are lower in consumer broadband and mobile markets

TAM (USDbn)	Company reported TAM	Realistic addressable market
Consumer broadband	660	40-120
Mobile	740	60-185
Consumer broadband + Mobile	**1,400**	**105-310**
Futuristic areas	NA	650

Financials

Revenue growth: Starlink has shown strong growth in subscribers; 4.5x by 1Q26 vs 2023. We expect 38% subscriber CAGR in 2026-30e. Our 2030 subscriber base forms 2.5% of total global households, excluding China and Russia, or 4.6% of 2025 global fixed broadband subscribers (excl. China and Russia). However, we expect ARPU to decline by 9% CAGR in 2025-30e as it expands internationally to more low-income markets. Overall, we expect 32%, 37%, and 34% consumer (includes both broadband and mobile), enterprise, and total revenue CAGR in 2025-30e, respectively.

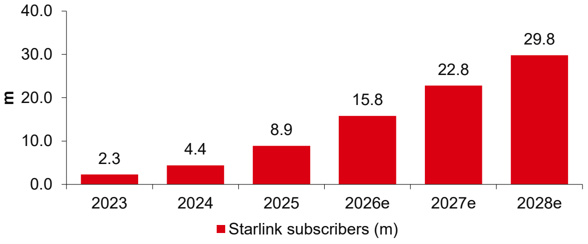
EBITDA and capex margins: We expect EBITDA margin to increase to 72% by 2030 vs 64% in 1Q26, as cost of revenue and R&D costs decline as a percentage of sales, but we expect SG&A costs to increase as a percentage of sales as it aims to expand its distribution network to expand its subscriber base. We also expect capex intensity to remain high at 35-45% in 2026-30e vs 37% in 2025 as Starlink aims to increase its satellite count to increase its transmission capacity.

Connectivity segment estimates

USDm, FYE Dec	2023A	2024A	2025A	2026E	2027E	2028E	2029E	2030E
Consumer	2,817	4,830	7,208	10,211	14,615	18,618	23,631	28,796
Enterprise & Government	1,052	2,769	4,179	5,339	7,289	10,204	14,286	20,000
Revenue	**3,869**	**7,599**	**11,387**	**15,551**	**21,903**	**28,822**	**37,917**	**48,797**
Cost of revenue	2,786	4,768	5,921	7,788	10,742	13,847	17,836	22,468
Research and development	381	453	575	953	1,256	1,538	1,871	2,213
Selling, general, and administrative	233	333	468	1,068	1,564	2,058	2,707	3,484
Restructuring charges	-	-	-	-	-	-	-	-
Impairment	-	39	-	-	-	-	-	-
Total costs and expenses	**3,400**	**5,593**	**6,964**	**9,810**	**13,561**	**17,443**	**22,414**	**28,165**
Income (loss) from operations	469	2,006	4,423	5,741	8,342	11,379	15,502	20,632
D&A	884	1,508	2,376	3,802	5,572	7,620	10,253	13,195
EBITDA	1,353	3,514	6,799	9,543	13,914	18,999	25,755	33,826
Adjustments	249	335	369	546	712	865	1,137	1,464
Segment adjusted EBITDA	**1,602**	**3,849**	**7,168**	**10,089**	**14,626**	**19,864**	**26,893**	**35,290**
Share based compensation	249	296	369	546	712	865	1,137	1,464
Capital expenditures	2,455	3,498	4,178	6,662	9,225	14,256	15,703	17,225

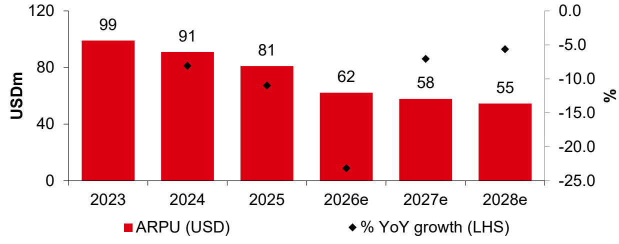
Source: SpaceX, HSBC estimates

We expect strong growth in Starlink subscribers as it expands more aggressively geographically



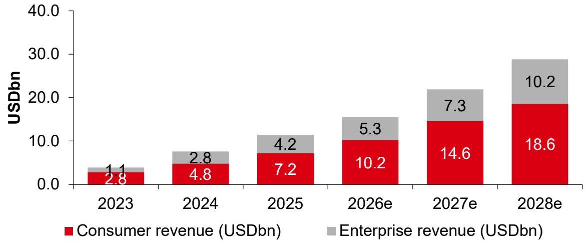
Source: Company data, HSBC estimates

ARPU is likely to decline as Starlink expands internationally into low-income regions



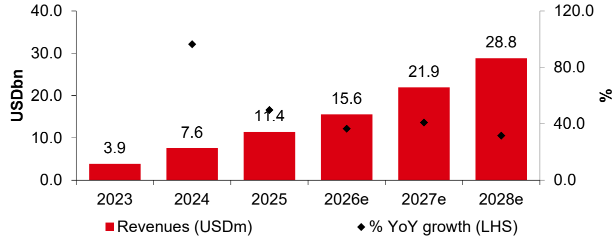
Source: Company data, HSBC estimates

We expect enterprise as a proportion of revenues to increase over time



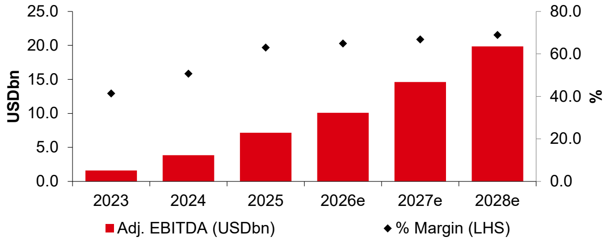
Source: Company data, HSBC estimates

We expect 37% revenue CAGR in 2025-28e for Starlink



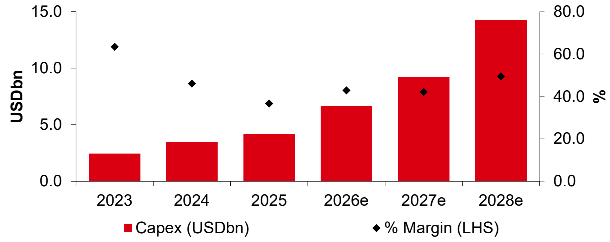
Source: Company data, HSBC estimates

EBITDA margin could expand to more than 70% for Starlink



Source: Company data, HSBC estimates

We expect capex margin to remain high



Source: Company data, HSBC estimates

AI

- We value the AI segment at an EV of USD150bn comprising X (USD6bn), Grok (USD91bn), NeoCloud Deals (USD29bn), and Cursor (USD24bn)
- We assign zero value for Terafab and orbital datacentres as the former is in early stages of development and the necessity for the latter seems small, at this point
- We think the AI segment disclosure is limited; we estimate a number of key inputs

AI: X, Grok, and compute leasing

Abhishek Shukla*, CFA
Senior Analyst, Technology
HSBC Bank Middle East Limited, DIFC
abhishek2.shukla@hsbc.com
+971 4 5093343

** Employed by a non-US affiliate of HSBC Securities (USA) Inc, and is not registered/ qualified pursuant to FINRA regulations*

SpaceX's AI segment is formed of X (formerly Twitter), Grok (the large language model; LLM), and the compute leasing business, which currently has deals with Alphabet and Anthropic.

SpaceX cited a TAM of USD26.5trn for the segment, comprising USD2.4trn in AI infrastructure, USD0.76trn in consumer subscriptions, USD0.6trn in advertising, and USD22.7trn in enterprise applications. We believe the serviceable market is smaller than the stated TAM and see both Terafab (a collaboration with Tesla and Intel to manufacture chips) and orbital compute opportunities long-term options rather than short-term realities.

AI segment: Long-term estimates

USDm, FYE Dec*	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Revenue	**2,961**	**2,620**	**3,201**	**18,901**	**22,909**	**27,770**	**36,083**	**50,015**
X	2,961	2,616	3,089	3,118	3,669	3,973	4,302	4,658
Grok	0	4	112	1,707	6,596	12,239	18,171	24,974
Compute offtake by Anthropic	0	0	0	8,125	0	0	0	0
Compute offtake by Google	0	0	0	3,680	2,760	0	0	0
Compute offtake by others	0	0	0	0	4,098	4,928	5,766	10,569
Cursor	0	0	0	2,271	5,331	6,083	7,204	8,640
Non-GAAP operating profit	**39**	**-1,348**	**-5,868**	**-3,840**	**-8,674**	**-9,019**	**-7,209**	**-4,235**
X	39	-291	-716	126	492	722	887	1,023
Grok	0	-1,057	-5,152	-10,707	-12,102	-11,907	-11,190	-10,361
Compute offtake by Anthropic	0	0	0	4,761	0	0	0	0
Compute offtake by Google	0	0	0	2,760	2,070	0	0	0
Compute offtake by others	0	0	0	0	779	936	1,095	2,008
Cursor	0	0	0	-780	-368	683	1,357	1,919
Non-GAAP operating margin	**1.3%**	**-51.5%**	**-183.3%**	**-20.3%**	**-37.9%**	**-32.5%**	**-20.0%**	**-8.5%**
X	1.3%	-11.1%	-23.2%	4.0%	13.4%	18.2%	20.6%	22.0%
Grok	NA	-28924.1%	-4618.8%	-627.3%	-183.5%	-97.3%	-61.6%	-41.5%
Compute offtake by Anthropic	NA	NA	NA	58.6%	NA	NA	NA	NA
Compute offtake by Google	NA	NA	NA	75.0%	75.0%	NA	NA	NA
Compute offtake by others	NA	NA	NA	NA	19.0%	19.0%	19.0%	19.0%
Cursor	NA	NA	NA	-34.4%	-6.9%	11.2%	18.8%	22.2%
Capex	**463**	**5,633**	**12,727**	**57,723**	**40,000**	**40,000**	**40,000**	**40,000**

Source: Company data, HSBC estimates *Split between X and Grok is estimated by HSBC even for historical periods

Valuation summary

AI segment

- **X social media platform:** We value X at USD6.2bn using 2x 2026e EV/Sales. This is in-line with second tier social media peers such as SNAP and PINS.
- **Grok:** We value Grok at USD90.7bn, using 3.63x 2030e EV/Sales, in-line with the OpenAI post-money valuation of USD852bn from March 2026 (source: OpenAI).
- **Compute infrastructure:** SpaceX rents its compute infrastructure to AI competitors. We value the Anthropic contract at USD4.5bn using eight months EBIT and the Alphabet contract at USD4.5bn using six months EBIT. The short timelines limit the contribution to our valuation. In addition, we assume that further deals are signed in the future. We value these future deals at USD20,081m, which is 10x 2030e non-GAAP EBIT.
- **Cursor: SpaceX has agreed to acquire this agentic AI company for USD60bn with SpaceX shares.** We cannot reconcile such a high price and rather value it at USD24.5bn, using 2.83x 2030e EV/Sales, in-line with Anthropic's latest financing round with a valuation of USD965bn from May 2026 (source: Anthropic). Closing of the acquisition could occur in Q3 2026.
- **Terafab & Orbital compute:** We value these both at zero value. We believe that the company remains far from recreating the complex hardware supply chain and will face hefty costs. We do not assume costs either.

AI segment: Sum-of-the-parts valuation (USDm): Base case valuation

Segment	Valuation	Methodology
X	6,236	2x 2026e revenue, in-line with tier 2 social media peers such as SNAP and PINS
Grok	90,656	3.63x 2030e revenue, in-line with OpenAI most
Compute offtake by Anthropic	4,535	Good deal, but likely to be short lived. Six months' EBIT discounted to today
Compute offtake by Google	4,472	Good deal, but likely to be short lived. Eight months' EBIT discounted to today
Compute offtake by others	20,081	10x 2030e non-GAAP EBIT
Cursor	24,450	Gross value of USD24,450m is 2.83x 2030e revenue (in-line with Anthropic). USD2.7bn acquired cash and USD60bn future acquisition cost reflected in diluted share count.
Terafab	-	Still at a framework agreement stage
Orbital AI	-	Not needed in the foreseeable future
Total	**150,430**	

Source: Company data, HSBC estimates

Bull and bear case scenarios

For our bear case (USD96.6bn) we make the following adjustments:

- We trim our X central case multiple (which is 2x 2026e sales) by 20%.
- We discount our Grok valuation by 50% to reflect lower operating margins versus OpenAI.
- Future compute offtake valuation is trimmed by 20% to reflect uncertainties.
- We lower our Cursor valuation by 20% to reflect lower market share versus Anthropic.

For our bull case (USD452.0bn) we make the following adjustments:

- We increase our X multiple by 20%.
- We base our Grok valuation on Palantir EV/Sales, an uplift of 4.2x.
- Future compute offtake valuation is lifted by 20% to reflect higher certainties.
- We lift our Cursor valuation by 20% to reflect higher growth opportunities.

AI segment valuation: Bear, base, and bull case valuation scenarios (USDm)

Segment	Bear case	Comment	Base case	Comment	Bull case	Comment
X	5,197	20% discount to base case	6,236	In-line with second tier social media EV/Sales	7,483	20% premium to base case
Grok	45,328	50% discount to OAI EV/Sales, given lower margins	90,656	In-line with 2030e OAI EV/Sales	382,105	In-line with PLTR 2030e EV/Sales
Compute offtake by Anthropic	4,535	In-line with base case	4,535	Based on 8 months' EBIT	4,535	In-line with base case
Compute offtake by Google	4,472	In-line with base case	4,472	Based on 6 months' EBIT	4,472	In-line with base case
Compute offtake by others	16,734	20% discount to base case	20,081	10x 2030e EV/EBIT	24,097	20% premium to base case
Cursor	20,375	20% discount to base case	24,450	In-line with Anthropic 2030e EV/Sales	29,340	20% premium to base case
Terafab	-	In-line with base case	0	Still at vision stage	-	In-line with base case
Orbital AI	-	In-line with base case	0	Not needed in the foreseeable future	-	In-line with base case
Total	**96,641**		**150,430**		**452,032**	

Source: Company data, HSBC estimates

Grok: Interesting, but a highly loss-making venture

Grok has two difficult challenges to overcome in our opinion:

- **Sizable capex requirements:** Both Grok and OpenAI target the consumer AI segment where they could pose a threat to Google search. But Google has a highly cash generative core search business and can afford to invest hundreds of billions of dollars per year on AI while continuing to offer Gemini and Google AI summaries/Overviews for free to users. OpenAI does not have a cash generative business, but it has nonetheless committed to spending USD687bn in compute capacity over 2026-30e, according to our estimates. If Grok wants to win sizeable market share it will need to match the above two players in capex, which could require multiple rounds of USD100bn+ equity issuances, in our opinion. Our forecasts assume USD58bn of capex in 2026 after USD13bn in 2025, and USD40bn per annum in 2027-30.
- **Uncertain revenue outlook:** The AI chatbot market is highly competitive with low switching costs, and there are signs that the availability of multiple "free" AI offerings from Google is putting pressure on competitor subscription pricing. Furthermore, neither OpenAI nor Grok are currently earning meaningful advertising revenues. These factors could entail a significant cash burn for both corporates.

We expect Grok to earn revenues of USD24,974m and register non-GAAP operating losses of USD10,361m in 2030e, and we see downside risks to our forecasts baring a dramatic change in the industry landscape and/or Grok's execution.

Grok: Long-term assumptions

USDm, FYE Dec*	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Grok users and subscribers (m): end of period								
Monthly active users	**-**	**-**	**89.0**	**210.7**	**322.9**	**432.1**	**556.4**	**708.7**
Unpaid users	-	(0.0)	88.1	199.9	301.1	401.0	515.7	656.5
Paid subscribers	-	0.0	0.9	10.8	21.8	31.1	40.8	52.2
Paid subscribers as a % of total			1.0%	5.1%	6.7%	7.2%	7.3%	7.4%
ARPU (USD/month)								
Advertising ARPU	-	-	-	0.023	0.474	0.876	1.147	1.332
Subscription ARPU	0	30	30	21	20	19	19	19
Revenue	**0**	**4**	**112**	**1,707**	**6,596**	**12,239**	**18,171**	**24,974**
Advertising	0	0	0	40	1,424	3,688	6,310	9,365
Subscription	0	4	110	1,493	3,852	6,103	8,227	10,614
Estimated enterprise AI	0	0	2	174	1,319	2,448	3,634	4,995
Non-GAAP operating profit	**0**	**-1,057**	**-5,152**	**-10,707**	**-12,102**	**-11,907**	**-11,190**	**-10,361**
Margin		-28924.1%	-4618.8%	-627.3%	-183.5%	-97.3%	-61.6%	-41.5%

Source: Company data, HSBC estimates *Even historical financials are estimated by HSBC

Valuation

We value Grok at USD90,656m based on a 2030e EV/Sales multiple for OpenAI. Below we also check different methods but returning lower valuations:

- **Current Anthropic ARR multiple:** Anthropic raised USD65bn at a pre-money valuation of USD900bn (USD965bn post-money) in May 2026 in a private placement. The company had reached an annualised revenue run rate (ARR) of USD47bn in May 2026, suggesting a market cap/latest ARR of 19.1x. We estimate Grok's current ARR at about USD600m with worse profitability (HSBC estimated non-GAAP operating margin of -1,786% in 1Q26) suggesting an optimistic case valuation of USD11bn using the same EV/Sales. Anthropic expects to reach a positive non-GAAP operating profit in 2Q26.
- **Current OpenAI ARR multiple:** OpenAI raised USD120bn at a pre-money valuation of USD730bn (USD852bn post-money) in March 2026 in a private placement. At the time of the equity raise, OpenAI's ARR was about USD25bn, suggesting a market cap/ARR of 29.2x.

OpenAI is focussed on the consumer segment as is Grok, suggesting OpenAI may be a better peer than Anthropic. However, OpenAI's 1Q26 non-GAAP operating margin of -122% is better than Grok's 1Q26 non-GAAP operating margin of -c1,800% (this is not a typo).

- **2030 EV/Sales for OpenAI:** we expect OpenAI to earn 2030e revenues of USD201bn, suggesting current EV/2030e sales of 3.6x. As both OpenAI and Grok are focussed on consumer AI, we have valued Grok at USD90,656m, which is 3.6x its 2030e revenue of USD24,974m.
- **2030 EV/Sales for Anthropic:** Alternatively, we expect Anthropic to earn revenue of USD318bn in 2030e, suggesting current EV/2030e sales 2.83x. On this metric, Grok would be worth USD70.7bn, which is 2.83x our 2030e Grok revenue estimate of USD24,974m.

Why not to compare with Palantir multiples?
We note some comparisons in the media (for example The Globe and Mail on 2 June 2026 compares the two while explaining why the comparison may be a bad idea) with Palantir's EV/Sales multiples. Palantir trades at 15.3x 2030e HSBCe EV/Sales. If we value Grok at the same multiple, it could be valued at USD382bn. We consider such comparison unsuitable though given the vast difference in operating profitability (we expect Palantir to earn a 2030e non-GAAP operating margin of 59.5% (2026e 60.8%) vs -41.5% for Grok).

Grok subscription revenue: potentially downgrading customers amidst competition
We estimate Grok generated revenues of USD138m in 1Q26. The company had 1.9 million paid subscribers at the end of 1Q26 and we estimate ARPU of USD30/month.

- Historically, Grok had two subscription tiers. SuperGrok was priced at USD30/month while SuperGrok heavy was priced at USD300/month. We assume the vast majority of paying customers opted for SuperGrok and estimate a historical composite ARPU of USD30/month.
- On 25 March 2026 xAI launched SuperGrok Lite at USD10/month. It appears it was in response to similar moves by OpenAI/ChatGPT. OpenAI historically sold its Plus subscription at USD20/month. In August 2025, OpenAI launched a USD5/month "Go" version for India. In January 2026, "Go" was extended to the rest of the world for USD8/month. According to The Information, OpenAI expects more than 80% of Plus subscribers to downgrade to Go by the end of 2026. We assume that Grok's ARPU will settle at USD19/month vs USD13/month for OpenAI ChatGPT.

Grok: Short-term estimates

USDm, FYE Dec*	1Q25A	2Q25A	3Q25A	4Q25A	1Q26A	2Q26E	3Q26E	4Q26E
Grok users and subscribers (m): End of period								
Monthly active users	17.6	34.4	64.0	89.0	117.0	148.7	180.2	210.7
Unpaid users	17.6	34.2	63.6	88.1	115.1	144.0	172.4	199.9
Paid subscribers	0.0	0.2	0.4	0.9	1.9	4.7	7.8	10.8
Paid subscribers as a % of total	0.3%	0.5%	0.7%	1.0%	1.6%	3.2%	4.3%	5.1%
ARPU (USD/month)								
Advertising ARPU	-	-	-	-	-	-	-	0.072
Subscription ARPU	30	30	30	30	30	27	23	20
Revenue	**2**	**11**	**32**	**66**	**138**	**308**	**500**	**761**
Advertising	0	0	0	0	0	0	0	40
Subscription	2	11	31	65	136	303	469	585
Estimated enterprise AI	**0**	**0**	**0**	**1**	**2**	**5**	**31**	**135**
Non-GAAP operating profit	**-877**	**-1,115**	**-1,405**	**-1,755**	**-2,468**	**-2,628**	**-2,748**	**-2,863**
Margin	-38136.4%	-9916.6%	-4439.8%	-2644.8%	-1786.0%	-853.3%	-550.0%	-376.2%

Source: Company data, HSBC estimates *Even historical financials are estimated by HSBC

Is OpenAI's user base c8x the size of Grok?
ChatGPT had about 920m **weekly** active users (WEUs) in February 2026. Grok had about 117m **monthly** active users (MAUs) in March 2026, and therefore it cannot be said that ChatGPT's user base is 7.9x the Grok user base. According to Sensortower, Grok's weekly active users are about 59% of its monthly active users as sporadic use of Grok is more common. Consequently, Grok's 117m monthly active users are equivalent to about 70m weekly active users. On this basis, OpenAI's user base is nearly 13x Grok's user base.

By December 2030e, we estimate that Grok will have 709m monthly active users, including 425m weekly active users vs slightly below 2bn weekly active users at ChatGPT. To narrow the gap, Grok would need be willing and able to marshal levels of compute capacity in-line with OpenAI's plans, which we consider unlikely as:

- We expect OpenAI to spend USD687bn on computing costs over 2026-30e. We do not see Grok being able to spend an equivalent amount without requiring multiple rounds of USD100bn+ equity or debt issuances. We estimate total capex for the AI segment at USD218bn over the same period
- If Grok continues to focus on the consumer AI segment, it will need to compete with Google. Google has significant cash flow from its core business and faces existential threat to its search business from the likes of OpenAI/Grok if they manage to gain a strong foothold in the market. It is likely that Google will continue to invest aggressively on AI compute and products to keep nascent competition at bay.

We estimate that 7.4% of Grok MAUs will be subscribers by 2030e
We assume that Grok's paid subscriber penetration will increase from 1.6% of MAUs in March 2026 to 7.4% of MAUs (or 12.3% of its WAUs) by 2030e, in-line with our estimates for ChatGPT over the same timeframe.

In the table below, we show a snapshot of the consumer AI driven chatbot industry. Please note that we show the average number of users through the year and not year-end figures.

AI driven chatbots industry landscape

USDm, CY	2025E	2026E	2027E	2028E	2029E	2030E
Chatbot & other consumer applications	**12,601**	**23,944**	**37,497**	**52,409**	**70,548**	**93,460**
Anthropic	315	1,886	4,026	6,815	11,086	17,828
OpenAI	8,757	13,810	18,493	23,139	28,462	34,314
Google	760	1,501	2,906	4,865	7,309	10,218
xAI	110	1,493	3,852	6,103	8,227	10,614
Other	2,659	5,253	8,219	11,488	15,464	20,486
Market shares	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**
Anthropic	2.5%	7.9%	10.7%	13.0%	15.7%	19.1%
OpenAI	69.5%	57.7%	49.3%	44.2%	40.3%	36.7%
Google	6.0%	6.3%	7.8%	9.3%	10.4%	10.9%
xAI	0.9%	6.2%	10.3%	11.6%	11.7%	11.4%
Other	21.1%	21.9%	21.9%	21.9%	21.9%	21.9%
Number of users (avg for the year)						
Anthropic	NA	NA	NA	NA	NA	NA
OpenAI (weekly)	627	969	1,182	1,391	1,614	1,843
Gemini (weekly*)	297	552	685	805	935	1,067
xAI (weekly*)	26	92	164	229	299	382
Other	NA	NA	NA	NA	NA	NA
Number of paying users (avg for the year)						
Anthropic	NA	NA	NA	NA	NA	NA
OpenAI	34	79	119	149	184	221
Gemini	3	9	19	31	47	66
xAI	0	5	17	27	36	47
Other	NA	NA	NA	NA	NA	NA
ARPU (USD/month)						
Anthropic	NA	NA	NA	NA	NA	NA
OpenAI	22	14	13	13	13	13
Gemini	22	14	13	13	13	13
xAI	30	23	19	19	19	19
Other	NA	NA	NA	NA	NA	NA

Source: Company data, Menlo Ventures, HSBC estimates

Grok advertising revenue

Grok remains an ad-free product, but management has in the recent past shared that it plans to introduce ads into Grok's responses (FT, 7 August 2025), following in the footsteps of OpenAI.

We believe AI chatbots (ChatGPT/Google Gemini/Grok) benefit from high-intent conversational ad inventory, which offers contextual data that should be superior to traditional search keywords or identity graphs. Supporting this bullish take, Alphabet in its 4Q25 earnings call shared that its AI-powered conversational features (AI Overviews and AI mode) are monetising at rates comparable to traditional search despite the early-stage nature of the ad channel.

We expect Grok to earn advertising revenues of USD9.4bn by 2030e (2026e USD0.04bn) with 656.5m non-paying ad-supported MAUs by December 2030e (373m WEUs). Ad supported MAUs should average 589m in 2030e, according to our estimates. We expect ARPU of USD1.3/month per MAU (USD2.33/month per WEU) in 2030e. We model comparable ad ARPUs for OpenAI, Google Gemini, and Grok.

Advertising ARPU for various companies

USDm, CY	2022A	2023A	2024A	2025E	2026E	2027E	2028E	2029E	2030E
Google search									
Daily active users (m)*	1,703	1,836	1,928	2,060	2,181	2,283	2,373	NA	NA
y-o-y		7.8%	5.0%	6.9%	5.9%	4.7%	3.9%	NA	NA
ARPU (monthly, USD)	7.9	7.9	8.6	9.1	9.9	10.2	10.6	NA	NA
y-o-y		0.0%	7.8%	6.1%	8.5%	4.0%	3.9%	NA	NA
Search advertising revenue	162,450	175,033	198,084	224,532	257,881	280,680	302,933	NA	NA
Meta									
Daily active people#	2,960	3,190	3,350	3,580	3,790	3,968	4,123	4,264	4,397
y-o-y		7.8%	5.0%	6.9%	5.9%	4.7%	3.9%	3.4%	3.1%
ARPU (monthly, USD)	3.2	3.4	4.0	4.6	5.3	6.1	6.7	7.3	7.8
y-o-y		7.7%	15.9%	14.3%	16.9%	14.2%	10.7%	8.3%	6.9%
Advertising revenue	113,641	131,948	160,632	196,174	242,722	290,218	333,733	373,944	412,308
Snap									
Daily active users (m)	354	400	437	470	485	504	524	539	555
y-o-y		12.9%	9.3%	7.5%	3.2%	3.9%	4.0%	2.9%	3.0%
ARPU (monthly, USD)	0.9	0.8	0.9	0.9	1.0	1.1	1.1	1.2	1.3
y-o-y		-11.5%	6.5%	2.9%	8.0%	7.6%	7.2%	7.5%	7.1%
Advertising revenue	3,961	3,960	4,611	5,104	5,686	6,355	7,083	7,834	8,637
Total revenue	4,603	4,602	5,359	5,931	6,608	7,386	8,232	9,105	10,037
Pinterest									
Monthly active users	440	477	532	592	640	683	720	758	799
Daily active users (m) **	264	286	319	355	384	410	432	455	479
y-o-y					8.2%	6.6%	5.5%	5.3%	5.3%
ARPU (monthly, USD)	0.9	0.9	1.0	1.0	1.0	1.1	1.1	1.2	1.2
y-o-y						5.4%	4.2%	4.5%	3.2%
Advertising revenue	2,801	3,051	3,644	4,217	4,726	5,308	5,837	6,419	6,979
OpenAI									
Weekly active users (non-paying)	NA	NA	NA	599	953	1,174	1,381	1,602	1,828
y-o-y					59.2%	23.2%	17.6%	16.0%	14.1%
ARPU (monthly, USD)					0.12	0.80	1.52	2.00	2.33
y-o-y						540.2%	90.6%	32.1%	16.3%
Advertising revenue				-	1,421	11,203	25,103	38,466	51,062
xAI									
Monthly active users (non-paying)	NA	NA	NA	43	148	256	355	461	589
Weekly active users (non-paying)	NA	NA	NA	26	87	147	202	262	335
y-o-y					236.9%	68.7%	37.8%	29.9%	27.7%
ARPU (monthly, USD)					0.04	0.81	1.52	2.00	2.33
y-o-y						2005.8%	87.9%	31.7%	16.2%
Advertising revenue				-	40	1,424	3,688	6,310	9,365

Source: Company data, HSBC estimates. Note: *1Q25 daily active users stood at 2bn, according to Google. Other years are estimated by HSBC. #Distorted by the inclusion of WhatsApp, which does not generate meaningful ad revenue. **Assuming daily active users are 60% of monthly active users. Typically, major platforms, such as Meta, have daily active uses that are c80% of monthly active users, while smaller platforms, such as Snap, have the ratio at 50%. NA – Not applicable/available

However, we estimate lower 2030e ad ARPU for AI chatbots than that for established platforms such as Google search or Meta's social media offerings today. We think modest monetisation assumptions are justified, despite chatbots' promising ad potential, mainly as chatbots trail Big Tech platforms in daily engagement (averaging c14 daily minutes per MAU for ChatGPT and likely lower for Grok).

Why should Grok's margins be lower than OpenAI's?

OpenAI has higher revenue/compute capacity and economies of scale, suggesting Grok's margin should be worse than OpenAI's

OpenAI reported revenue of USD5,800m in 1Q26, an ARR of USD25bn in March 2026, and a non-GAAP operating margin of -122% in 1Q26. We would expect a weaker margin for Grok than for OpenAI due to lower scale in a high fixed cost industry (cost of training models).

Our 2030e non-GAAP operating margin assumption for Grok is -41.5%. We explain below why our estimates may be optimistic:

- **Economies of scale – OpenAI is larger today than what we forecast for Grok in 2030e:** We expect Grok to report 2030e revenue of USD24,974m, which is less than the ARR OpenAI had in 1Q26. If economies of scale were the only determinant of margins, Grok in 2030e should report non-GAAP operating margin lower than OpenAI in 1Q26 (-122%).

- **OpenAI is more compute* efficient than Grok:** At the end of December 2025, OpenAI had reached an ARR of about USD21.4bn and it had total (rented) compute capacity of 1.9GW, suggesting an ARR/compute ratio of USD11.26bn/GW. At the end of March 2026, Grok had an ARR of about USD550m and total compute capacity of 1GW, suggesting an ARR/compute ratio of USD0.55bn/GW. Even if we assume that some of the capacity was unused, the ARR/compute ratio will remain below that of OpenAI's. It is also worth considering why Grok has spare compute when almost all the other LLM providers are securing more capacity. If Grok earns orders of magnitude lower revenue per GW of compute capacity*, it is difficult to see why its margins would not be below those of OpenAI.

  **Please also refer to the section on AI segment capex "Not all GW's are made equal" to understand why the GW used by OpenAI might not be fully equivalent to the GW used by SpaceX, even though the above point still holds.*

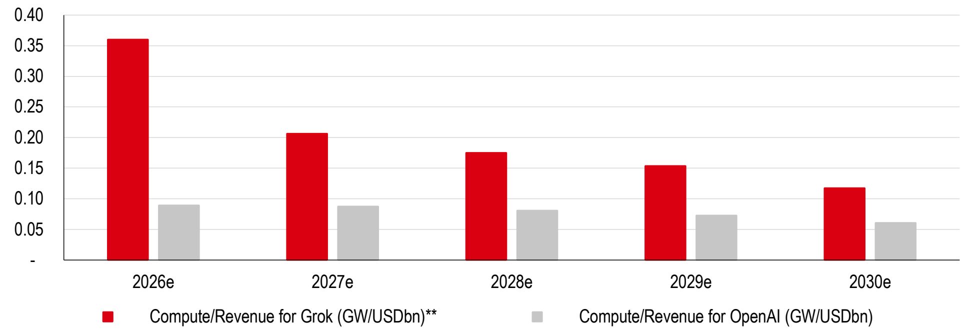
Grok: Margin profile

USDm, FYE Dec*	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26e	3Q26e	4Q26e
Grok users and subscribers (m): End of period								
Monthly active users	17.6	34.4	64.0	89.0	117.0	148.7	180.2	210.7
Unpaid users	17.6	34.2	63.6	88.1	115.1	144.0	172.4	199.9
Paid subscribers	0.0	0.2	0.4	0.9	1.9	4.7	7.8	10.8
Paid subscribers as a % of total	0.3%	0.5%	0.7%	1.0%	1.6%	3.2%	4.3%	5.1%
ARPU (USD/month)								
Advertising ARPU	-	-	-	-	-	-	-	0.072
Subscription ARPU	30	30	30	30	30	27	23	20
Revenue	**2**	**11**	**32**	**66**	**138**	**308**	**500**	**761**
Advertising	0	0	0	0	0	0	0	40

Source: Company data, HSBC estimates *2Q, 3Q, 4Q25 estimated by HSBC

The chart below compares compute capacity per USDbn of annualised revenue for Grok and OpenAI. OpenAI is more compute efficient than Grok as per our estimates and as per historical trends.

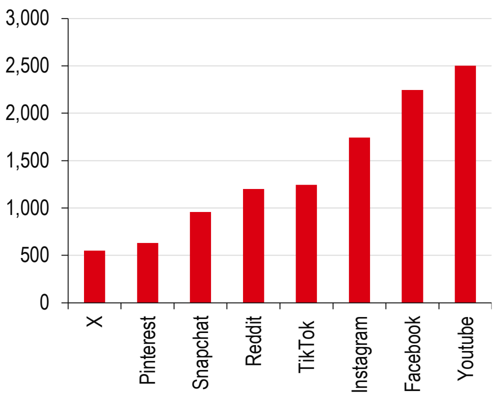
Compute required per USD1bn of revenue (GW/USDbn)



Source: Company data, HSBC estimates *Grok and Cursor revenue used as denominator

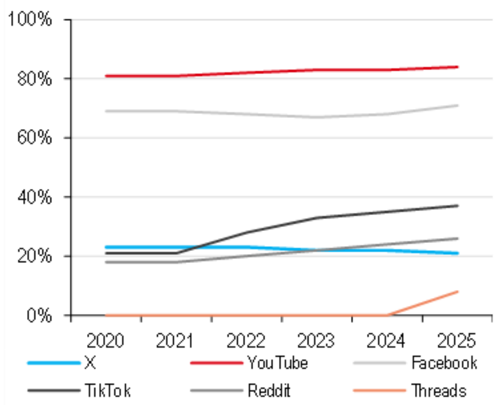
X: A popular media but difficult to monetise

Global monthly active users (m)



Source: 2026 eMarketer, SpaceX prospectus, HSBC

US adults who say they use social media platforms



Source: Pew research survey conducted on June 2025

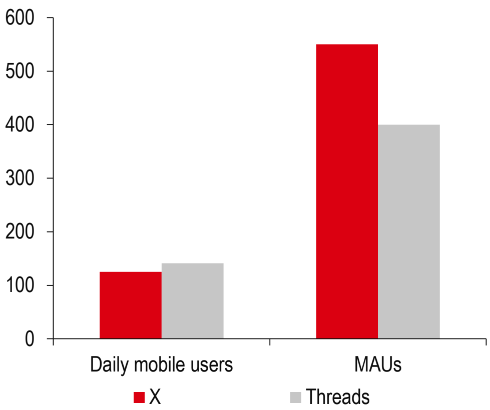
Mohammed Khallouf*, CFA
Global Tech Platforms Analyst
HSBC Bank Middle East Limited, DIFC
mohammed.khallouf@hsbc.com
+971 4 509 3337

** Employed by a non-US affiliate of HSBC Securities (USA) Inc, and is not registered/ qualified pursuant to FINRA regulations*

Social media service X (formerly known as Twitter) today has 550m MAUs, placing it amongst the 10 largest (ex-China) online platforms globally, and with 21% of US adults reporting that they used it in 2025.

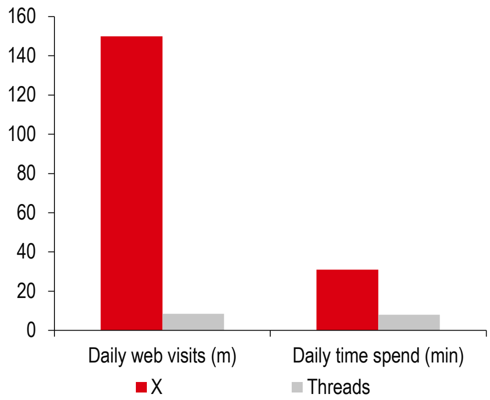
Competitive environment shifting but X remains firmly in the lead

User base (m)



Source: Similarweb Jan 2026, HSBC

Engagement metrics



Source: Sensortower 25Q2, HSBC

X is the dominant player in its microblogging segment, but it has been facing growing competition from Meta Platforms, which launched its rival platform Threads in 2023, with 3P data indicating that Threads surpassed X in daily mobile users in early 2026 (although X continues to have a larger MAU base).

However, X retains a substantial lead in daily website traffic, and we note that a sizable portion of Threads users are funnelled from Instagram thanks to Threads' native integration. The later acts to inflate headline numbers but does not necessarily translate into real engagement, as reflected in the sizable difference in average time spent by users (X at over 30min per day vs 8min for Threads). Threads serves as an extension of Meta's social media platforms while X remains the mainstream global hub for gathering information and opinions.

Monetisation remains a work in progress

Despite its reach, X has historically lagged in monetisation with a marginal share of social media industry revenues (at c1% in the US market in FY26e according to eMarketer) given a global ad ARPU we estimate at USD0.3 (monthly per MAU) in 2025.

X users tend to lack purchase intent

X's ad monetisation is held back by the nature of the platform, as users tend to lack purchase intent/commercial data signals, with engagement heavily swayed to current affairs and political posts. Furthermore, many larger brand advertisers pared back spending on the platform given the switch to a "light" approach to content moderation following its 2022 acquisition.

New ad tech introduced

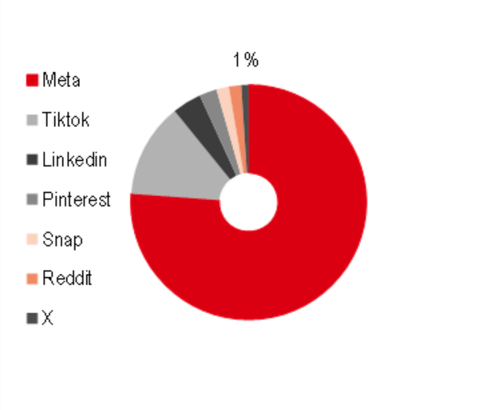
X's ad revenues returned to growth in 2025 (+7% y-o-y) but contracted in 1Q26 (-20% y-o-y), which management in the prospectus attributes to an ad platform overhaul that impacted ad sales during the rebuild. The redesign of the ad manager focuses on AI-powered ad ranking and retrieval systems to bring it in-line with Big Tech offerings. We think steps to leverage X's in-house AI advantage are positive and should further benefit from increased in-app conversational usage of Grok, with around one-fifth of X MAUs using the functionality, boosting the quality of the contextual user data on hand for ad targeting.

Headway for greater subscriber penetration

Aside from ads, X has a small subscriber base comprising 4.4m X Premium and Premium+ tier paid subscriptions, which we estimate accounted for USD211m of revenues in 1QQ26. Subscribers can avail of blue checkmarks to authenticate their accounts, see fewer to no ads, and enjoy increased usage limits and faster responses to in-app queries put to Grok. We note that X's subscriber penetration at 0.8% of MAUs trails that of best-in-class peer Snap (which has 25m Snapchat+ subscribers for a 2.6% conversion rate) despite the advantage of a built-in AI assistant, highlighting headway for greater subscriber penetration, in our view.

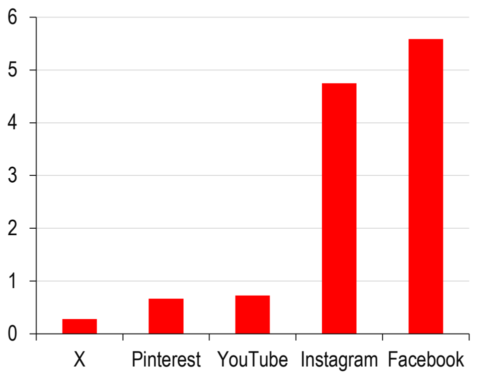
X plans to diversify its offering into "communications, media, payments, banking, and commerce". Fintech features appear to be first in-line with the "X Money" product, which aims to enable payments and other financial services. We are somewhat more sceptical of the prospects for "Super apps" in western markets due to existing competition and a cultural preference we believe for specialised apps over the Asian all-in-one model. Furthermore, according to Reuters (29 April 2026) the platform has yet to receive payment licenses from several states, highlighting the regulatory hurdles when it comes to financial offerings.

US social media market revenue share (%)



Source: 2026 eMarketer, HSBC

ARPU, monthly per MAU (USD)



Source: Company data, eMarketer, HSBC. Monthly ad revenue per MAU

HSBC's valuation reflects operating challenges

Social media valuation comparison

Corporate	Ticker	EV/Sales (2026e)	Sales CAGR (2025-27e)	EBITDA* margin (2027e)
Alphabet	GOOGL.O	7.9	20%	44%
Meta	META.O	6.6	23%	48%
Pinterest	PINS.K	1.7	14%	30%
Reddit	RDDT.N	6.3	41%	46%
Snap	SNAP.K	1.0	12%	22%

Source: Eikon LSEG 22 July 2026, as reported by corporate including any adjustments

We value X in-line with SNAP and PINS at USD6.2bn

The table above shows the relative valuation of various social media platforms. We think X's marginal market share of industry revenues, low profitability, and uncertain growth outlook suggest a valuation in-line with tier-2 social media peers SNAP and PINS rather than Big Tech. We value X at USD6,236m, which is 2x our 2026e revenue estimate of USD3,089m.

Cursor: excellent strategy but at a high price, missing details

The proposed Cursor acquisition would allow Grok to focus on lucrative enterprise AI

We are cautious on the financial outlook for Grok's consumer facing business. In this context, we believe Grok's proposed acquisition of Cursor, an AI powered code editor, can help Grok pivot towards enterprise AI, which enjoys better financial dynamics, in our opinion.

We value Cursor at USD24,450m, which is 2.83x 2030e revenue of USD8,640m, a multiple in-line with Anthropic's most recent equity raise. Our target valuation of USD24,450m is 12.7x 2030e non-GAAP operating profit of USD1,919m. In addition, we add USD2.7bn for the cash that Cursor had on its balance sheet at group level, to derive a target valuation of USD27,150m. The table below shows non-GAAP EV/EBIT for major technology companies under our coverage. Our implied 12.7x 2030e non-GAAP EV/EBIT valuation is towards the high-end of what we typically see in the sector.

Non-GAAP EV/EBIT in the technology sector

EV/EBIT (x)	2021	2022	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Oracle	21.8	21.7	19.9	18.1	17.4	15.0	12.9	10.2	8.1	6.4
AAPL	46.2	46.1	41.8	36.1	31.9	27.4	25.4	23.4	21.4	NA
Adobe	14.8	13.2	11.2	10.0	8.8	7.6	6.7	5.9	NA	NA
Autodesk	34.2	25.9	23.2	20.7	16.7	13.4	11.2	9.5	NA	NA
Amazon	68.2	78.2	42.9	29.4	26.0	21.7	17.7	14.9	12.6	10.5
Salesforce	33.5	23.7	15.3	12.8	11.8	10.6	8.7	7.3	6.0	4.8
CRWV	NA	-672.0	23,748.1	66.0	78.2	98.1	29.8	18.0	15.3	13.9
Fair Isaac	65.3	49.0	42.1	35.7	28.6	18.3	18.0	18.5	18.9	NA
GOOG	1.2	47.2	40.3	31.2	27.0	20.2	16.8	14.2	NA	NA
IBM	25.3	22.1	20.3	20.0	16.3	16.2	14.7	12.8	11.7	10.7
Intuit	21.1	17.9	14.5	12.9	10.4	8.2	6.4	5.0	NA	NA
META	38.3	59.0	35.2	23.6	19.6	19.5	16.6	13.8	11.3	9.3
Microsoft	37.4	34.9	29.5	25.0	20.5	17.4	14.2	11.7	9.8	8.2
ServiceNow	67.6	53.1	39.5	30.1	23.7	18.5	14.1	10.3	7.1	4.8
Nvidia	40.6	57.2	138.7	58.8	36.6	18.9	12.1	NA	NA	NA
Palantir	645.6	694.8	475.9	283.6	147.8	70.9	45.9	38.6	31.8	26.7
SAP	29.9	25.1	24.1	19.5	15.1	13.5	11.3	9.7	8.5	7.3
Snowflake	-2,488.7	868.6	408.3	410.4	202.4	118.4	65.3	44.7	NA	NA
Twilio	12,239.9	-7,355.0	62.9	44.5	32.3	26.6	22.8	NA	NA	NA
Zoom	13.3	13.9	11.7	11.1	10.1	8.7	7.9	NA	NA	NA

Source: LSEG DataStream, company data, HSBC estimates

AI-driven coding tools revenue on a surge – annualised run rate

USDm	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26
AI driven coding tools	**3,058**	**3,584**	**4,200**	**5,244**	**6,581**	**8,535**	**11,425**	**15,091**	**19,436**
Claude Code	300	420	588	1,000	1,600	2,560	4,341	5,916	7,897
OpenAI: Codex	67	121	217	391	625	1,000	1,470	2,054	2,735
GitHub CoPilot	950	997	1,047	1,072	1,125	1,181	1,333	1,691	2,091
Cursor	774	929	1,000	1,300	1,600	2,000	2,257	2,863	3,539
Other coding	967	1,117	1,347	1,482	1,630	1,793	2,023	2,567	3,174

Source: Company data, Menlo Ventures, HSBC estimates. Note: Numbers in red are based on some media reports or estimated on the basis of some numbers reported by the company

This suggests a net negative value of USD32,850m (USD27,150m - USD60,000m), for SpaceX shareholders in our view. If SpaceX fails to acquire Cursor, SpaceX will need to pay a break fee of USD1.5bn and a USD8.54bn deferred service fee under the Compute agreement. We assume that SpaceX will go ahead with the acquisition.

SpaceX's S-1 filing has provided little details about Cursor's financials (USD3.1bn of total assets, including USD2.7bn of cash and USD0.6bn total liabilities), and nor have we seen any official press release from Cursor. The following statement taken verbatim from MarketWise.com is indicative of details that are available from media sources. But these are not audited numbers and give no details about profitability. About Cursor: *"Its annualised revenue reached USD100m by January 2025, USD500m by June 2025, and USD1bn by November 2025. In February, annualised revenue topped USD2bn."*

AI-based coding tools landscape

- Microsoft's GitHub allows developers to store, manage, track, and share their code. GitHub is the world's largest repository for software development. It provides tools for collaboration, such as version control, code reviews, and project management, enabling multiple people to work on the same project simultaneously without overwriting each other's work.
- GitHub CoPilot is an AI-driven tool that helps programmers write code and solve a problem described in natural language. GitHub CoPilot uses foundation LLMs, including OAI's GPT models, Anthropic's Claude, xAI's Grok, and Google's Gemini. Microsoft revealed in its 4QCY25 results conference call that GitHub CoPilot has 4.7m users, up 75% y-o-y. GitHub CoPilot offers several plans, including Pro (USD10/month), Business (USD19/month), and Enterprise (custom, often higher than Business). We estimate a December 2025e ARR of USD1,072m, assuming an ARPU of USD19/month. We expect ARPU to rise rapidly as Microsoft migrates to usage-based pricing. Microsoft also recently launched its own reasoning and coding LLM models, which is likely to further intensify competition in the segment. While Microsoft is a challenger in the LLM side of the business, Microsoft GitHub has a large footprint, which Microsoft should be able to leverage to sell its LLM offering.
- Anthropic's Claude Code was fully launched to the public in May 2025. ARR increased to USD1bn by December 2025 and USD2.5bn by February 2026.

AI-driven coding tools– HSBC long-term estimates

USDm, CY	2025E	2026E	2027E	2028E	2029E	2030E
Number of professional programmers (m)	**25**	**24**	**24**	**23**	**23**	**22**
% using AI coding tools	16%	79%	99%	100%	100%	100%
Number of programmers using AI coding tools (m)	**4**	**19**	**24**	**23**	**23**	**22**
ARPU (USD/month)	49	90	131	157	188	224
AI Coding tools ARR	**2,378**	**21,079**	**37,079**	**43,777**	**51,076**	**59,545**
Claude Code	234	9,020	18,078	21,785	25,123	28,679
OpenAI: Codex	66	3,054	5,741	6,862	8,034	9,375
GitHub CoPilot	850	2,146	3,149	3,593	4,255	5,104
Cursor	581	3,607	5,331	6,083	7,204	8,640
Other coding	646	3,251	4,780	5,455	6,460	7,747

Source: Company data, HSBC estimates

- OAI's Codex showed a surge in user adoption after it launched GPT-5.3 Codex in early February 2026. The total number of users more than tripled to 1.6m in early March 2026 versus early 2026. According to Yahoo Finance, the ARR reached USD1bn in early 2026 and has increased significantly since then. ARR of USD1bn with 1.6m users would suggest an ARPU of about USD52/month.

Uncertainty around long-term ARPU assumptions

We are confident that by the end of 2027e (if not sooner), the vast majority of professional software programmers will be using AI driven coding tools. Software professionals are obviously tech savvy and among the early adopters of such technology. Increase in the user base as a revenue driver is unlikely to last beyond 2027e, in our opinion. In fact, if a number of programmers are laid off due to AI driven productivity gains, the user base could even shrink. Besides users, there is uncertainty around ARPU. Nvidia CEO Jensen Huang said in a LinkedIn interview recently that Nvidia could offer its software engineers tokens worth half their salary (an engineer earning USD500K per annum may spend USD250K per annum on AI tokens). While Nvidia might be an extreme example, we are hearing anecdotes from across the industry of token costs being substantially higher than the users previously estimated. Currently, most AI coding tools providers are talking about ARPU of USD10-200/month and our forecasts are rising within this range. But evolution of ARPU is uncertain in our view.

Putting coding tool sector-wide revenue in the global macro context

Our estimated 2030e global AI driven coding tool revenue is 9% of our estimated 2030e global IT services and software programming employee costs. We estimate global IT services and software revenue will reach around USD4.3trn by 2030e (2025 USD2.9trn). R&D expense, especially for large software/technology companies, tend to be about 15% of revenue (CY2026e estimates – Oracle 10.6%, Microsoft 10.4%, SAP 20.2%, ServiceNow 16.2%, Intuit 12.8%, Meta 33.4%, Adobe 13.4%, Salesforce 11.6%). This suggests total R&D (mostly programming) cost of USD643bn in 2030e. We estimate that AI driven coding tools will have global revenue of USD59.5bn in 2030e, which is 9.3% of USD643bn.

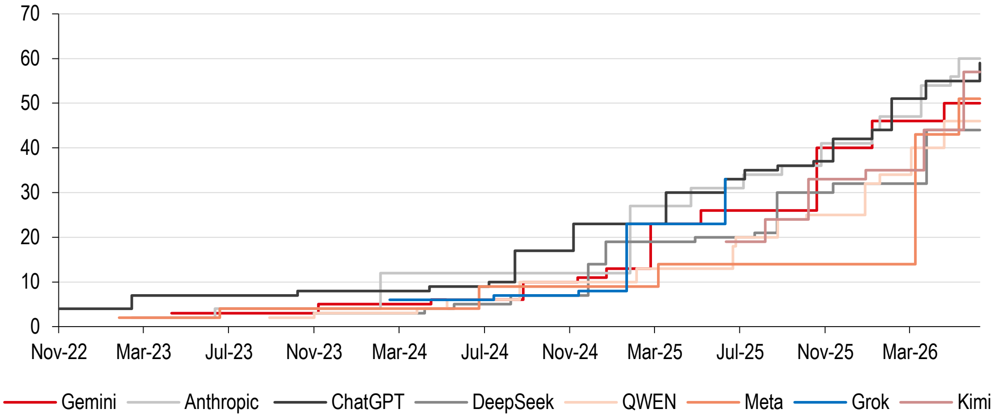
Margin uncertainty for Cursor is Grok's opportunity

Cursor does not have its own LLM model. Cursor users are able to choose the underlying LLM (Anthropic/OpenAI, etc). Cursor needs to pay the LLM provider, a cost that is passed on to the end customer. We should think of Cursor as a software driven tool that leverages other people's AI offerings. This creates a major risk for Cursor as its margins could be at the mercy of LLM providers. So long as the LLM foundation model market is fragmented and performance of various models comparable, Cursor would have good bargaining power as it has access to the end consumer and underlying LLM used can be changed. However, if the end users perceive one LLM (say Anthropic's) is better at coding than others, the LLM provider will have the bargaining power and could squeeze Cursor.

In this context, SpaceX with its in-house LLM models can reduce the cost of third party LLM access and help boost Cursor margins. The performance of various LLM models is converging, suggesting that, eventually, it should be possible for Cursor to "backward integrate" and use in-house LLM models.

The chart below shows the relative "intelligence" of various LLM providers. There is clearly a trend of convergence such that in theory, it should be possible to replace Anthropic or OpenAI models with in-house Grok models.

Comparative "intelligence" of various LLM models



Source: Company data, HSBC estimates

How about "in practice"?

In practice, Anthropic has consistently gained market share at the expense of others. We define foundation model subsegment as the business of providing LLM foundation models to various deployment agents such as Cursor. Foundation models are used for purposes other than coding too.

We estimate that Anthropic, OpenAI, and Gemini control 90%+ of the foundation model market

Anthropic's usage share of the ex-China foundation model market increased from 24% in December 2024 to 32% in June 2025, and to 44% in November 2025, according to Menlo Ventures. We estimate the market share increased to 54% in May 2026. We estimate that Anthropic (54%), OpenAI (18%), and Google Gemini (22%) have cornered more than 90% of the foundation model market. The above observation suggests that the performance in various "intelligence tests", as shown in our charts, might not be a good indicator of real world performance. When and whether Grok is able to convince Cursor users to switch to Grok LLM from Anthropic/Claude is to be seen and cannot be presumed on the basis of intelligence scores alone.

What margins do we model for Cursor?

We estimate 2026e non-GAAP operating margin of -34.4%, improving to +22.2% by 2030e.

We have no actual data points on historical margins for the company. But we have seen some media articles (several months old) suggesting that the gross margins were negative for the company. Margins were adversely affected by many individual users who were paying USD20/month subscription charges but incurring higher token costs that Cursor had to pay to Anthropic/OpenAI, etc. Cursor had better margin with enterprise users as the billing was based on both number of seats and token usage. By 2030e, we estimate a gross margin of about 40%. OpenAI (which is admittedly consumer focussed and not enterprise focussed) had a gross margin of 42% (ignoring Microsoft revenue share) in 1H25. We expect Cursor's 2030e revenue to be in-line with OpenAI's ARR reached in 1H25.

We estimate R&D costs/revenue of 13.2% by 2030e (2026e 25.8%). Long-term R&D costs/revenue is in-line with what we see for major software companies.

We estimate SG&A/revenue of 4% as Cursor has a lean team and marketing is likely to be word of mouth, in our opinion.

Compute deal with Anthropic: excellent, but could be short lived

The deal with Anthropic is highly favourable to SpaceX as the former is paying nearly twice the market rent for compute capacity, as per our estimates; we expect Anthropic to terminate the agreement within six months as its traditional compute providers ramp up capacity

We consider the deal highly value accretive for SpaceX as it implies asset turnover of 54.8% (vs 30% typical in the sector) and ROIC of 32%, well above cost of capital and 8-12% earned by other AI focussed cloud infrastructure providers. It appears to us that Anthropic faced a short-term capacity squeeze as its revenue increased from ARR of USD9bn in December 2025 to USD47bn in May 2026 and was forced to pay a premium to SpaceX. However, as Anthropic's traditional cloud service providers (Amazon, Google, and increasingly Microsoft) ramp-up capacity, we would expect Anthropic to terminate the over-priced compute capacity offtake within six months, in-line with agreement terms.

In May 2026, SpaceX entered into Cloud Services Agreements with Anthropic to provide cloud-based compute capacity including c325,000 NVIDIA GPUs, backed by hyperscale-class CPUs, exabyte-scale storage, and high-speed networking and interconnects purpose-built for AI workloads. Pursuant to these agreements, Anthropic will pay SpaceX USD1.25bn per month starting in July 2026. The payment in May and June 2026 will be lower as the capacity to be rented to Anthropic is being ramped up. After the initial three-month period, the agreement may be terminated by either party upon 90 days' notice.

Assets worth USD27bn to generate USD15bn per year of revenue
Anthropic/SpaceX have not disclosed the type of 325,000 Nvidia GPUs being leased.
We assume that the latest available (in large quantity) GB300 NVL 72 racks are being leased. If an earlier type of GPU, say H100, is being leased, the asset value will be lower and asset turnover and ROIC for SpaceX higher.

- We assume 4,514 GB300 NVL72 racks, which will contain 325,000 B300 GPUs. We estimate that these racks cost about USD3m per rack (including USD2.6m Nvidia content), resulting in total GPU cost of USD13.5bn.
- We estimate that components other than GB300 racks may add another 30% to the cost of electronics, resulting in further USD4.1bn of costs.
- 325,000 B300 GPUs are likely to require a datacentre with compute power of about 650MW (assuming 1,400W for B300 GPU and a further 600W for other electronics). Building the datacentre could require USD9.75bn of capex assuming USD15m per MW of construction costs. True construction costs for SpaceX could be significantly lower.
- Putting the three together, we estimate total assets to be leased at USD27bn.

USD1.25bn per month rent would imply annual rent of USD15bn, which suggests asset turnover of 55%. When we look at peers, who typically lease datacentres for longer duration, asset turnover is generally around 30% or lower. We see no reason why Anthropic would continue paying twice as much rent for compute capacity as the market price.

Asset turnover among cloud infrastructure providers: USD100 of assets earn USD30 per year of revenue

USDm, FYE Dec	2024A	2025A	2026E	2027E	2028E	2029E	2030E
Asset turnover: Gross assets, ex wip and including ROU assets							
Oracle: OCI	35.6%	31.9%	26.9%	27.5%	30.4%	31.9%	33.2%
Amazon: AWS	85.8%	66.7%	51.4%	41.4%	37.0%	35.7%	36.4%
Google Cloud	NA	NA	NA	NA	NA	NA	NA
Microsoft	41.6%	39.3%	37.9%	36.2%	34.4%	33.4%	33.2%
CoreWeave	NA	22.8%	23.5%	25.8%	26.8%	26.8%	26.7%
Non-GAAP operating margin							
Oracle	44.2%	42.9%	41.9%	39.2%	36.9%	35.6%	35.0%
Amazon: AWS	45.1%	43.1%	44.4%	44.7%	45.0%	44.4%	43.9%
Google Cloud	14.1%	23.7%	30.6%	29.9%	29.9%	NA	NA
Microsoft	45.0%	46.7%	46.9%	46.3%	46.0%	45.9%	45.6%
CoreWeave	18.6%	13.0%	6.9%	14.2%	16.4%	15.5%	15.1%
ROIC							
Oracle: OCI	10.3%	9.3%	7.8%	8.0%	8.8%	9.3%	9.7%
Amazon: AWS	38.7%	28.8%	22.8%	18.5%	16.7%	15.9%	16.0%
Google Cloud	NA	NA	NA	NA	NA	NA	NA
Microsoft	16.4%	15.7%	15.2%	14.5%	13.8%	13.4%	13.3%
CoreWeave	25.2%	8.2%	3.7%	7.3%	9.5%	10.4%	11.2%

Source: Company data, HSBC estimates

Why do we model 59% non-GAAP operating margin on the deal?

- We expect depreciation costs of 30.4% of revenue vs more than 50% for peers. The core asset is GPUs, which is depreciated over six years. Therefore, USD100 of assets will face annualised depreciation of USD16.6. If USD100 of assets generate cUSD55 revenue per year, as explained above, depreciation will be 16.6/54.8 = 30.4% of revenue.
- Sales and marketing costs should be close to zero as it is likely a CEO-to-CEO deal, in our view, without a large salesforce to pay for.
- Electricity costs are about 6% of revenue for CoreWeave, a GPU/AI focussed cloud infrastructure provider. As Anthropic seems to be paying twice as much rent as the market rate, electricity costs might be only 3% of revenue in the current deal.
- As xAI owns its datacentres, it will not need to pay separate rent for the datacentre. The depreciation for a self-owned datacentre is included above (in fact overestimated as datacentres are depreciated over 15+ years).
- Other costs are 15-16% of revenue for CoreWeave. We estimate that these would be about 8% of revenue in the Anthropic deal.
- Overall, we expect operating costs of 41.4% of revenue (30.4% depreciation, 3% electricity, 8% other costs), suggesting a non-GAAP operating margin of 58.6%.

A non-GAAP operating margin of 58.6% with 54.8% asset turnover suggests ROIC of more than 32%.

Pivot to a neo-cloud business model could be a good move

As explained earlier, we are somewhat ambivalent on the prospects for Grok's consumer focussed business. We think the company is making a good (albeit expensive) pivot to enterprise AI (via the Cursor acquisition) and leasing the rest of the compute capacity like neo-clouds lease. However, there is no definitive signal from management that they intend to deemphasize consumer AI chatbots.

Compute deal with Google: even better deal; but short lived too

We consider the deal more value accretive to Grok as it implies asset turnover of 119% and ROIC of 89%, far above cost of capital and 8-12% earned by other AI focussed cloud infrastructure providers.

Google agreed in May 2026 to lease 110,000 Nvidia GPUs along with related other components from SpaceX. Google will pay USD920m per month to SpaceX starting October 2026. A lower rent will be payable in September 2026 as SpaceX ramps up capacity. After December 2026, the deal can be terminated by either party with 90 days' notice.

Assets worth USD9.26bn to generate annual revenue of USD11bn
Anthropic/SpaceX have not disclosed the type of 110,000 Nvidia GPUs being leased.
We assume that the latest available (in large quantity) GB300 NVL 72 racks are being leased.
If an earlier type of GPU, say H100, is being leased, the asset value will be lower and asset turnover and ROIC for SpaceX higher.

- We assume 1,528 GB300 NVL72 racks, which will contain 110,000 B300 GPUs.
  We estimate that these racks cost about USD3m per rack, resulting in total GPU cost of USD4.583bn.
- We estimate that components other than GB300 racks may add another 30% to the cost of electronics, resulting in further USD1.375bn of costs.
- 110,000 B300 GPUs are likely to require a datacentre with compute power of about 220MW (assuming 1,400W for B300 GPUs and a further 600W for other electronics). Building the datacentre could require USD3.3bn of capex assuming USD15m per MW of construction costs. True construction costs for SpaceX could be lower.
- Putting the three together, we estimate total assets to be leased of USD9.26bn.

USD920m per month rent would imply annual rent of USD11bn, which suggests asset turnover of 119%. If annual rent is more than the cost of acquiring an asset, which has c6 year life, it is highly value accretive for the lessor. We assess a non-GAAP operating margin of about 75% for SpaceX and ROIC of 89.5%.

We again find the deal terms puzzlingly lucrative for SpaceX. We see no reason why Google would in the long-term rent compute capacity at 3x the market rate and then re-lease it to customers at a lower rate as part of its GCP offering. We assume the deal will be terminated by 1Q27, in-line with the termination clause in the agreement.

We note that Google owns a c5% stake in SpaceX, which could be worth significantly more than the value of the above order.

We expect further neo-cloud deals in the future

We assume that SpaceX will continue to lease its spare capacity to other customers. However, we assume financial details that are in synch with peers and the industry. Typically, cloud infrastructure businesses earn c10% ROIC and c30% asset turnover. We assume the same.
We estimate long-term non-GAAP operating margin of 19%.

- We estimate 16% for CoreWeave although CoreWeave's management has guided for long-term non-GAAP operating margin of 25%. SpaceX may earn higher margin due to higher backward integration (self-owned datacentres) and a track record of operational excellence and cost reduction.

- Oracle's management has guided for long-term gross margin of 30-40% on its AI compute business (35% at the midpoint). If we assume a further 8% costs for R&D, SG&A, etc, Oracle could earn long-term non-GAAP operating margin of 27%. Oracle's long-term margins are likely to be higher than SpaceX's neo-cloud segment as we expect Oracle to have larger scale and Oracle will have synergies with the non-GPU cloud segment (which has higher margins) and Oracle's enterprise software offerings.
- SpaceX's non-GAAP operating margins cannot be compared with the margins of the likes of AWS (2026e 44.4%) as the latter have many orders of magnitude higher scale (2026e revenue USD161,654m) and substantial CPU-based general purpose cloud.

We value the segment at USD20,081m, which is 10x 2030e non-GAAP operating profit. In comparison, Microsoft, Oracle, and CoreWeave trade at 8.4x, 8.7x, and 15.3x 2030e non-GAAP EV/Operating profit.

SpaceX: Neo-cloud segment*

USDm, FYE Dec*	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Cloud capacity leased to third parties (GW): Period end	-	-	-	-	0.34	0.40	0.50	1.13
Cloud capacity leased to third parties (GW): Period avg					0.30	0.37	0.43	0.78
Revenue (USDm)	**0**	**0**	**0**	**0**	**4,098**	**4,928**	**5,766**	**10,569**
Revenue per GW (USDm/GW)					13,500	13,500	13,500	13,500
Non-GAAP operating profit	**0**	**0**	**0**	**0**	**779**	**936**	**1,095**	**2,008**
Margin					19.0%	19.0%	19.0%	19.0%

Source: Company data, HSBC estimates *Excluding deals with Google and Anthropic

Frank Lee*
Global Head of Tech Hardware & Semi Research
The Hongkong and Shanghai Banking Corporation Limited
frank.lee@hsbc.com.hk
+852 2996 6916

Pulkit Aggarwal*
Associate
Bangalore

** Employed by a non-US affiliate of HSBC Securities (USA) Inc, and is not registered/ qualified pursuant to FINRA regulations*

Assigning zero value to Terafab

Terafab is in "vision stage" with no tangible progress on design or build out

According to SpaceX's S-1 filing, Terafab aims to be the world's largest chip manufacturing facility, with the goal of achieving one terawatt of annual compute production capacity. SpaceX plans to develop fully vertically integrated end-to-end capabilities spanning the design of lithography masks, fabrication of logic and memory chips, and advanced packaging, all in a vertically integrated closed-loop single plant, to rapidly iterate to improve chip design and performance. SpaceX plans to design chips that are optimised for the space environment.

SpaceX has agreed with Tesla on a general framework for the future development of Terafab. Intel joined the project in April 2026 and is expected to contribute its expertise in designing, fabricating, and packaging ultra high-performance chips to help Terafab scale.

We see Terafab as an early-stage vision. None of the three (potential) partners have shown an ability to build GPUs that are better than Nvidia's GPUs. We explain later in this section why making space-based GPU better/more economical than Earth-based GPUs is harder even if the cost of launching into space were zero.

A Tesla, Intel, and SpaceX collaboration

Terafab is a joint venture between SpaceX, Tesla, and xAI to develop a large, vertically integrated semiconductor fabrication facility (fab), announced in March 2026. With neither of these companies having any experience in semiconductor designing, manufacturing, or packaging, Intel has since joined as a manufacturing partner. Other than operating foundry, Intel is expected to contribute across the entire chip development cycle, specifically in design, fabrication, and advanced packaging of ultra-high-performance chips.

What SpaceX aims to achieve through Terafab

The purpose of the Terafab project is to address the shortfall in AI compute capacity by bringing the entire chip-making supply chain (design and lithography, to chip fabrication, packaging, and testing) under one roof with the ultimate goal of achieving 1TW in compute output.

Core objectives:

- **Scale:** The facility aims to produce over one terawatt (1TW) of compute capacity annually. This scale is expected to meet the demand for Tesla's autonomous vehicles and Optimus robots, SpaceX's AI-compute satellites, and xAI operations.
- **Vertical integration:** The semi industry is highly fragmented, requiring companies to ship wafers across the globe for different stages of production. By building a vertically integrated fab with logic, memory, and packaging all under one roof, Terafab aims to circumvent the logistic delays and allow rapid iteration.

Cost and time needed to scale

SpaceX has proposed an initial investment of USD55bn to build Terafab (Reuters, 6 May). The Terafab project would involve multi-phase chip fabrication, an advanced computing complex, and the total investment could rise to USD119bn if additional phases are completed.

We assume that it could take Terafab at least a decade to achieve the 1TW scale

While there is no official timeline, we can use the USD165bn investment in Arizona, US, by TSMC as a reference. TSMC announced its first phase of investment in May 2020 with plans to enter volume production in 2024. TSMC has subsequently announced multiple phases of investments adding both chip fabrication alongside advanced packaging facilities. The Fab 4, which will house the N2 and A16 technology, will start volume production by the end of the decade. The timeline for volume production in Fab 5 and 6, which will utilise more advanced nodes, is still undecided but goes beyond 2030. Therefore, we believe it is reasonable to assume that it could take Terafab at least a decade to achieve the 1TW scale.

1TW of annual compute output from a "vertically integrated" facility – is it achievable?

SpaceX places the Terafab initiative as a central part of achieving enough supply for its future chip requirements, particularly as it develops orbital AI at scale, and design chips that are optimised for the space environment. With deployment of orbital AI compute satellites set to begin as early as 2028 according to SpaceX, we believe the Terafab will not be operational in time due to several reasons, as described:

- **Economic and financial viability:** The cost of developing a world-class leading-edge fabrication facility runs into tens of billions of dollars. Terafab aims to integrate designing, logic, and memory fabrication and advanced packaging all under the same roof, which will require significant capex and R&D spending. TSMC has announced USD165bn investment in the US over a decade after, and it expects 30% of its advanced capacity to be manufactured in the US. Terafab's current budget is USD119bn. The current major GW based compute deployment deals span 50-60GW by the end of the decade, for which the entire foundry industry combined is unable to deliver the hardware and is in the process of expanding their capacity. Therefore, the internal demand might not be large enough to justify the capex and R&D needed for an integrated fab with 1TW of annual output.
- **Single-source Intel dependency:** Intel is expected to contribute its expertise in designing, fabricating, and packaging ultra high-performance chips to help Terafab scale. Intel's foundry business is currently navigating a multi-year transition as Terafab remains its only committed external customer for its foundry services. Being locked in an agreement with solely Intel might present a risk of underperforming yields as its 14A node, which Terafab is expected to utilise, remains untested.

- **Memory bottleneck:** Memory fabrication is a highly specialised discipline led by companies like Samsung, SK Hynix, and Micron. Attempting to manufacture memory in-house, with no experience or partners, alongside logic processors will add immense layers of complexity to the process, potentially leading to lower yields. If Terafab fails in this task, it could be forced to buy commodity memory from the open market. With the unprecedented price increases in memory, the entire cost structure might be severely impacted.
- **Semi equipment bottleneck:** The global semi fabrication industry is heavily reliant on a specialised supply chain, most notably the extreme ultraviolet (EUV) lithography machines manufactured almost exclusively by ASML. Terafab would require a number of these EUV machines to scale to its ambition of an output of 1TW. However, ASML's output of these machines is limited by its own supply chain complexities and could be a major bottleneck.
- **Energy constraints:** Energy would be a major constraint for Terafab as it aims to achieve 1TW of AI compute capacity. Energy is already a bottleneck for datacentres, with a scale of compute much lower than Terafab's 1TW target. On-site power generation would require additional capex impacting the financial viability of the project.
- **Talent deficit:** Recruiting engineering talent to build a fab of this scale in order to compete with the industry leaders in logic and memory fabrication would not be easy given the mismatch between supply and demand. The challenges in securing engineering and technical talent was vocalised by TSMC during its Arizona fab expansion where it had to send more than 1,000 Taiwanese engineers to Arizona to support operations. Since Terafab does not have an existing pool of talent with the skills needed to set up a semi foundry, poaching talent from incumbents would be a difficult and expensive task.

We assign zero value to datacentres in space

We see no economic rationale for setting up largescale datacentres in space within the next decade, even though SpaceX expects to begin deploying its orbital AI compute satellites as early as 2028. We are conscious that Google's Project Suncatcher is working on "equipping solar-powered satellite constellations with TPUs and free-space optical links to one day scale machine learning compute in space" (Press release 4 November 2025). We also believe that by the time (if ever) space-based datacentres become viable, there would be entities other than SpaceX that have mastered the science of reusable space rockets and SpaceX will likely earn launchers' value in a competitive market and not a de facto monopoly profit in a large market. For example, we note Blue Origin, Rocket Lab, China Aerospace Science and Technology Corporation, European Space Agency, Arianespace, and Indian Space Research Organisation are trying to develop reusable rockets.

What are the key arguments for putting datacentres in space?

- **Cheaper electricity:** Satellites carrying datacentres in dawn-dusk sun synchronous orbits could receive sunlight 24/7 all year. Consequently, these datacentres could run with minimal battery backup and would have cheaper electricity costs than Earth-based datacentres. Solar panels based in these orbits could generate 5x as much electricity per day as Earth-based solar panels because space-based panels enjoy 24-hour sunlight and solar radiation is not attenuated by the environment.
- **More resource availability:** Proponents expect the electricity demand for AI datacentres to rise so much that Earth's resources and a realistic electricity generation capacity will be overwhelmed. SpaceX aims to launch 100GW of compute capacity per year into space.

- **Fewer regulatory and operational hurdles in space:** Securing electricity grid connections, particularly in the US, has become time consuming. There are "not in my backyard" protest groups and many people are concerned that datacentres are crowding out other users of water and electricity.
- **Space-based datacentres would be labour efficient:** Because space-based datacentres will not have any humans working in them, they would likely be more labour efficient and could have lower general and administrative costs.

What is our stance on these specific arguments?

Utility cost is only 5-6% of revenue

Focussing on electricity is opposite of the 80:20 rule: The table below shows the cost structure of a GPU focussed (i.e. AI focussed) cloud (CoreWeave). Utility cost is 5-6% of revenue. The main cost is capital (expressed via depreciation). Depreciation could be c50% of revenue. Putting datacentres in sun synchronous orbit would likely reduce the utility cost while risking or inflating most other costs. It is the opposite of what we understand from the 80:20 rule.

CoreWeave: Margin profile

FYE Dec, USD 000	2023A	2024A	2025A	2026E	2027E	2028E	2029E	2030E
Revenue	**228,943**	**1,915,426**	**5,131,096**	**12,540,198**	**24,592,231**	**36,455,125**	**46,055,872**	**53,156,420**
Rent	46,980	336,980	902,980	2,559,121	4,475,786	6,270,281	7,921,610	9,142,904
As a % of revenue	20.5%	17.6%	17.6%	20.4%	18.2%	17.2%	17.2%	17.2%
Power/utility	12,312	88,312	291,312	746,939	1,352,573	2,005,032	2,533,073	2,923,603
As a % of revenue	5.4%	4.6%	5.7%	6.0%	5.5%	5.5%	5.5%	5.5%
Depreciation	103,210	863,413	2,454,457	6,388,148	11,696,320	17,475,957	22,360,807	26,031,766
As a % of revenue	45.1%	45.1%	47.8%	50.9%	47.6%	47.9%	48.6%	49.0%
SG&A, R&D, etc	65,738	270,876	815,180	1,986,956	3,577,058	4,720,939	6,079,375	7,016,647
As a % of revenue	28.7%	14.1%	15.9%	15.8%	14.5%	13.0%	13.2%	13.2%
Non-GAAP Operating profit	**703**	**355,845**	**667,167**	**859,034**	**3,490,494**	**5,982,916**	**7,161,007**	**8,041,499**
Margin	0.3%	18.6%	13.0%	6.9%	14.2%	16.4%	15.5%	15.1%

Source: Company data, HSBC estimates

Gas powered power plants are easier alternatives: A 1GW terrestrial AI datacentre requires capex of USD50bn. We estimate that a 1GW captive gas power plant requires additional capex of USD1.3bn. Captive gas power plant adds less than 3% to the capex cost of the project. Admittedly, timelines for heavy duty gas turbines are currently stretched but our analysis indicates that the US should be able to increase its power capacity to accommodate incremental power demand from AI datacentres. Moreover, by the time the technology for space-based datacentres is perfected and becomes economical, many of the bottlenecks on earth may have been solved.

US electricity shortages are manageable: In the table below, we show Visible Alpha consensus estimates for Nvidia's datacentre segment. From this, we estimate incremental AI compute that uses Nvidia capacity by assuming USD30bn of Nvidia content per GW of AI compute (in-line with our estimates and comments from Nvidia). We estimate that Nvidia accounts for c55% of global AI capacity additions by GW. Thus, we estimate global AI compute additions. We assume that 60% of global AI compute capacity is being installed in the US. This gives us an indication of incremental AI compute power demand in the US.

We start with power capacity estimates from the US Energy Information Administration (US EIA). The US EIA expects overall installed power capacity in the US to rise to 1,427GW by 2030, from 1,222GW in 2025. However, not all power plants are made equal. Coal and gas-based power plants could easily operate at 80-90% capacity utilisation/plant load factor (PLF). Solar (24-26%) and wind (32-35%) tend to operate at much lower PLFs. Therefore, we use a weighted power capacity in the US where non-hydro renewable power is weighted at 30% while other power capacity is weighted 90% of nameplate capacity.

US power capacity is highly underutilised on most days and hours of the year and the capacity is built to cater to peak demand. Peak demand was 759GW in 2025. We compare peak load demand to weighted capacity to understand true headroom in the system.

Our analysis indicates that US peak demand will remain at 87-88% of weighted capacity up to 2027e. However, it will rise to 98% of weighted capacity by 2030e. It is possible that more projects will be announced over the coming years in view of the upcoming shortages and the true tightness in the market will be lower. We also note that the EIA base case assumes coal power plant capacity will be reduced by 70W by 2030e, probably due to environmental concerns. It is highly likely that some or even all of these plants' lives could be extended for a few more years if a genuine power shortage scenario were to be imminent. If no coal power plants are shut, peak power demand will be 92% of weighted capacity in 2030e, suggesting a manageable situation.

AI compute and US power supply demand situation

USDm, CY	2025	2026E	2027E	2028E	2029E	2030E
AI compute demand as per NVDA consensus						
NVDA datacentre revenue (USDm)	193,737	362,091	513,965	641,095	718,917	770,738
NVDA content per 1GW of AI compute (USDm/GW)	30,000	30,000	30,000	30,000	30,000	30,000
AI compute fuelled by NVDA (GW)	**6**	**12**	**17**	**21**	**24**	**26**
NVDA's market share in AI compute by GW	56%	51%	57%	55%	55%	55%
Overall AI compute incremental demand (GW)	**12**	**24**	**30**	**39**	**44**	**47**
Overall AI compute incremental cumulative demand (GW)	-	24	54	93	136	183
% installed in the US		60%	60%	60%	60%	60%
US AI compute incremental cumulative demand		**14**	**32**	**56**	**82**	**110**
Nameplate power capacity (GW)						
USA	**1,222**	**1,292**	**1,343**	**1,379**	**1,396**	**1,427**
Other than coal, solar, wind & hydro (90% weight)	664	690	706	720	745	780
Coal (90% weight)	159	156	153	138	114	88
Renewable	399	447	485	521	537	559
Hydro (90% weight)	25	25	25	25	25	25
Solar & wind (weighted at 30%)	374	422	460	496	512	534
Weighted power capacity (GW)						
USA: base case	875	910	933	943	949	964
USA: If coal not shut down	875	912	938	962	989	1,027
Peak electricity demand (GWh)						
USA	759	788	822	861	903	948
Rest of USA (2% CAGR)	759	774	790	805	822	838
Incremental AI demand	-	14	32	56	82	110
Peak electricity demand/weighted power capacity						
USA: base case	87%	87%	88%	91%	95%	98%
USA: If coal not shut down	87%	86%	88%	89%	91%	92%

Source: US EIA, Company data, Visible Alpha, HSBC estimates

Likewise, we do not believe space is needed to avoid a water crunch on earth (space-based datacentres will use radiation cooling without using water). A 1MW datacentre requires around 10,000 to 25,000 litres of water per day (source: IEA). Therefore, 1GW could require 25m litres per day (at max) or 9,125m litres per year. The cost of generating water from sea water desalination is about USD0.4 to USD1.0 per 1,000 litres (for example, Sorek B desalination plant in Israel is contracted to supply desalinated water at about USD0.41 per 1,000 litre). Therefore, 9,125m litres of water could cost USD9m per year. Capex for 1GW of AI compute is USD50bn (according to Nvidia & HSBC estimates). Annual rent received is USD15-17.5bn (as per HSBC estimates). Therefore, the cost of water is very small compared with the capex or annual rent.

100GW per annum incremental compute demand is unrealistic: As explained above, if we use Nvidia consensus estimates, AI compute demand is likely to be about 47GW in 2030e. Only a small percentage (if not 0%) would be deployed in space. If multiple companies are able to master reusable space rockets, it is unlikely that SpaceX will corner 100% of the space AI datacentre market.

Lower labour costs in space may not be the right argument: The argument that space-based datacentres will have no labour costs, and therefore an advantage, is unlikely. It is possible to operate an Earth-based datacentre without employees, and it will run as well or badly as a space-based datacentre. There is a reason why companies hire employees for Earth-based datacentres.

- Every year, 3-6% of GPUs tend to fail, which need to be replaced. In the first three years of their lives, GPUs tend to be under guarantee from Nvidia. These GPUs will not be replaced in a space-based datacentre and other equipment that is meant to work along with GPUs will also be idle.
- Cooling systems require maintenance. Without staff, filters clog, pumps break, and temperatures rise until the servers automatically shut down to prevent permanent damage.
- Other parts of the electronic system fail. It needs only one part of a large system to fail to make the entire system unusable.

Therefore, space-based datacentres face serious limitations. We should assume that companies hire humans to work on Earth-based datacentres only because they add more value than they cost.

What the space datacentre bulls do not talk about

Sun-synchronous orbit will have high latency: Sun-synchronous orbits are almost perpendicular to the Earth's equator. A quick look at the map will show that such a satellite would spend less than 10% of its time above the US or Europe. Most of the time will be spent over oceans, polar regions, or poorer or sparsely populated countries, which are not the main source of AI demand. The South Pole is about 14,000km from California (to pick one spot in the US) while the North pole is about 6,000km. Europe is about 6,000km from New York while Japan is about 5,500km from California. It would make more sense to build geographically dispersed datacentres within the US and to some extent in regions such as Europe and Asia Pacific. Peak inference demand in the US, Japan, and Europe should come at different times in the day ensuring high utilisation throughout the day. Cloud infrastructure providers generally try to build inference focussed datacentres close to end demand for lower latency. In this context, Sun-synchronous satellites are poor choices for inference workloads, in our opinion.

Latency does matter a lot: Locations that are farther away from major usage centres enjoy lower rents. The table below provides relative rents for datacentres in various regions within the US. Rents in Northern Virginia are significantly higher than in Dallas, even though power is more easily available in Dallas. The reason why North Virginia commands a premium is that it is a global hub for hyperscalers. It sits at the centre of global internet traffic and has easy access to transatlantic subsea cable in Virginia Beach.

2H 2025 wholesale datacentres: Primary market fundamentals

Market	Inventory (MW)	Increase (MW y/y)	Available MW	Vacancy rate	2025 net absorption (MW)	Rental rates (kW/month)
Northern Virginia	4,040	1,110	22	1%	1,102	213
Atlanta	1,459	459	29	2%	456	175
Dallas-Ft. Worth	1,067	476	26	2%	471	158
Chicago	905	264	11	2%	264	205
Phoenix	807	205	11	1%	206	190
Silicon Valley	489	21	23	5%	(1)	228
Hillsboro, OR	475	-	1	0%	2	200
New York Tri-State	190	-	14	7%	(1)	203
Total	**9,433**	**2,534**	**135**	**1%**	**2,498**	**198**

Source: CBRE, HSBC

A 1GW datacentre rented in Northern Virginia would incur rent of USD2,550m/year, on average, according to CBRE data. In Dallas the rent would be USD1,890m/year. As a 1GW datacentre earns annual rent of about USD15,000m (HSBC estimates, depending upon location, lease term, and negotiated arrangement), the difference in annual rent is about 4.5% of revenue, even though both locations are within the US. As a reminder, electricity is 5-6% of revenue (e.g. 6.0% of revenue for CoreWeave in 2026e, according to HSBC estimates). For datacentres in Sun-synchronous orbits, where they are far away from the US, the reduction in desirability due to higher latency would likely more than offset any electricity cost advantage.

Training models better on ultra large clusters on Earth: Training large language models is more cost effective on mega clusters that are tightly connected within the same datacentre. SpaceX plans to deploy satellites with AI compute capacity of 100KW/tonne. This suggests even with a 200 tonne Starship launch, one launch will not have more than 20MW of compute capacity, likely split among multiple satellites. These compute clusters will be small compared with hundreds or even thousands of megawatt compute clusters on Earth and will not be a good choice for training large language models, in our view.

LEO satellites have lower life than terrestrial datacentres: Typically, low Earth orbit satellites tend to have a life of 5-7 years (vs 15+ years for terrestrial datacentres). Even in LEO, satellites face some drag (space is a relative term). A 100KW orbital datacentre may require radiators with a surface area of about 160sqm to cool the heat generated by GPUs. According to our Asia Energy Transition equity research team, 300-400KW (say 350KW on average) of solar array are needed to support 100KW IT load on a satellite. A 350KW solar panel in space may span around 1,400sqm. Such massive surface area is likely to dramatically increase drag and either reduce satellite life or require much more “keep in orbit” propellent.

Space-based datacentres may cost twice as much as Earth-based datacentres, even ignoring the launch cost: According to Semi Analysis, the capex cost of electronics will be similar on Earth and in space. However, the cost of datacentres housing the electronics in space would be almost 8x the cost of a comparable datacentre on Earth. If we ignore launch costs, the datacentre in space would be 3.8x as costly. Space-based datacentres will need propulsion hardware to keep the satellite in orbit during its operation and to deorbit once the satellite has reached the end of its life. The cost of the propulsion system per MW can be as much as the cost of the entire datacentre on Earth. Radiators and solar panels will add significantly to the capex requirement in space, although solar panels will result in lower electricity costs later. Shielding electronic equipment from space radiation and communication with Earth and other satellites are likely to add significant costs to space-based datacentres.

Can we have some numbers please?

In the table below we compare the costs in three scenarios:

- Earth-based 1GW datacentre with self-owned datacentre.
- Space-based Sun-synchronous datacentre with current cost of launch included.
- Space-based Sun-synchronous datacentre when cost of launch is ignored.

We consider only three costs – depreciation over the expected life (which is different between space and Earth), cost of capital at 10% (although space should arguably be higher), and cost of electricity. We ignore labour costs on Earth-based datacentres as explained before. Earth-based datacentres could operate without staff, but the option to deploy people can add value to Earth-based datacentres but cannot be a source of lower value.

We conclude that annualised cost of operating a space-based datacentre is about USD41bn/GW per year (vs USD12.98bn/GW for Earth-based datacentres). Even if the cost of launch were to be ignored (as these decline over time), space-based datacentres still have higher cost of USD25.6bn/GW.

Conclusion: space-based datacentres are unlikely to become a mainstream phenomenon in the foreseeable future.

Comparative economics of space-based datacentres vs terrestrial datacentres

USDm	Capex	Lifespan (years)	Annual depreciation	Annual capital cost	Annual costs (USDm/GW)
Terrestrial datacentre					
Datacentre	12,500	15.0	833	1,250	2,083
Electronics	37,500	6.0	6,250	3,750	10,000
Annual electricity costs					900
Total costs					**12,983**
Orbital datacentre: Including cost of launch					
Datacentre	100,000	5.0	20,000	10,000	30,000
Electronics	37,500	5.0	7,500	3,750	11,250
Annual electricity costs					0
Total costs					**41,250**
Orbital datacentre: Excluding cost of launch					
Datacentre	47,875	5.0	9,575	4,788	14,363
Electronics	37,500	5.0	7,500	3,750	11,250
Annual electricity costs					0
Total costs					**25,613**

Source: Company data, SemiAnalysis, HSBC estimates

AI Segment – Financials

We expect AI segment non-GAAP operating profit to improve from USD-5,868m in 2025 to USD-4,235m in 2030e. 2026e and 2027e non-GAAP operating profit is helped by significant contribution from Anthropic and Google compute offtake deals, which we believe are likely to be short-term. 2027e and 2029e non-GAAP operating profit shows deterioration vs 2026e as generous compute offtake deals taper off and Grok expands with negative margins. Capex trajectory is highly uncertain. If SpaceX wishes to succeed as an LLM provider it will need to invest more than USD100bn per annum in capex, in our opinion. If capex is materially higher than our estimates, AI segment revenue and losses through 2030e would likely be higher.

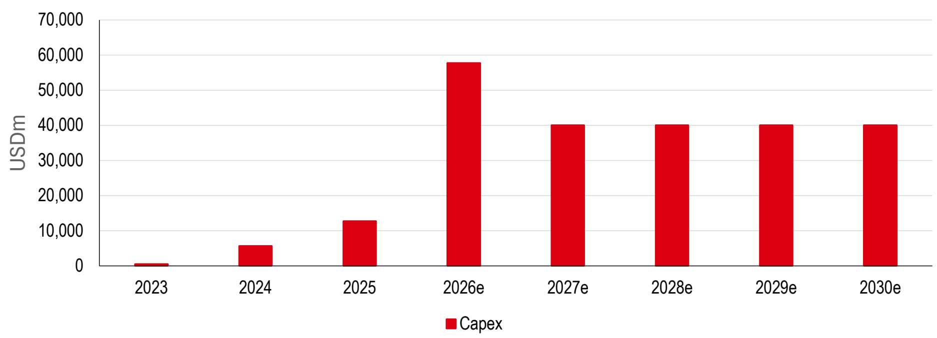
AI segment: Long-term estimates

USDm, FYE Dec*	2023A	2024A	2025A	2026E	2027E	2028E	2029E	2030E
Revenue	**2,961**	**2,620**	**3,201**	**18,901**	**22,909**	**27,770**	**36,083**	**50,015**
X	2,961	2,616	3,089	3,118	3,669	3,973	4,302	4,658
Grok	0	4	112	1,707	6,596	12,239	18,171	24,974
Compute offtake by Anthropic	0	0	0	8,125	0	0	0	0
Compute offtake by Google	0	0	0	3,680	2,760	0	0	0
Compute offtake by others	0	0	0	0	4,098	4,928	5,766	10,569
Cursor	0	0	0	2,271	5,331	6,083	7,204	8,640
Non-GAAP operating profit	**39**	**-1,348**	**-5,868**	**-3,840**	**-8,674**	**-9,019**	**-7,209**	**-4,235**
X	39	-291	-716	126	492	722	887	1,023
Grok	0	-1,057	-5,152	-10,707	-12,102	-11,907	-11,190	-10,361
Compute offtake by Anthropic	0	0	0	4,761	0	0	0	0
Compute offtake by Google	0	0	0	2,760	2,070	0	0	0
Compute offtake by others	0	0	0	0	779	936	1,095	2,008
Cursor	0	0	0	-780	-368	683	1,357	1,919
Non-GAAP operating margin	**1.3%**	**-51.5%**	**-183.3%**	**-20.3%**	**-37.9%**	**-32.5%**	**-20.0%**	**-8.5%**
X	1.3%	-11.1%	-23.2%	4.0%	13.4%	18.2%	20.6%	22.0%
Grok	NA	-28924.1%	-4618.8%	-627.3%	-183.5%	-97.3%	-61.6%	-41.5%
Compute offtake by Anthropic	NA	NA	NA	58.6%	NA	NA	NA	NA
Compute offtake by Google	NA	NA	NA	75.0%	75.0%	NA	NA	NA
Compute offtake by others	NA	NA	NA	NA	19.0%	19.0%	19.0%	19.0%
Cursor	NA	NA	NA	-34.4%	-6.9%	11.2%	18.8%	22.2%
Capex	**463**	**5,633**	**12,727**	**57,723**	**40,000**	**40,000**	**40,000**	**40,000**

Source: Company data, HSBC estimates. *Split between X and Grok is estimated by HSBC even for historical periods

Why do we expect a "peculiar" capex profile?

AI segment: Capex profile



Source: Company data, HSBC estimates

Much of the capex for the segment is meant for GPU focussed (i.e. AI focussed) compute capacity. Before we get into our forecasts for capex, we need to understand a nuance about SpaceX's historical capex.

Not all GWs are made equal

According to SpaceX, it had deployed 1GW of IT compute capacity by 1Q26. Generally, 1GW of Nvidia GPU-based compute capacity requires USD50bn of capex including USD10-15bn for the datacentre and USD35-40bn for the IT equipment. However, SpaceX's AI segment had cumulative capex since inception of USD26,546m as at 1Q26. Granular consolidated balance sheet details reflect total servers and networking gross PP&E of USD23,850m at 1Q26. Even if we accept that SpaceX may have constructed datacentres at a lower cost than peers, it is difficult to see how IT equipment capex could be so much lower than expected.

While we do not have definitive answer to the disconnect, we believe it may suggest that part of the compute capacity is not based on Nvidia GPUs. Some of the ASIC-based compute setups can have significantly lower cost per GW (lower priced as the performance does not match Nvidia performance).

The prospectus refers 100,000 units of H100, 110,000 GB200, and 110,000 GB300 GPUs but these would represent no more than 560MW of compute capacity out of the total of 1GW. It is possible that the rest of the compute is ASIC. The HSBC Equity Research team covering Nvidia and Broadcom suggested that Broadcom has some customers that it has not named and one of these could be SpaceX.

So how do we model capex per GW?

We assume that all the compute will be Nvidia-based and assume USD50bn of capex for 1GW of compute. We made this simplifying assumption to ensure better comparability with peers.

Why compare with OpenAI compute?

We have no bottom-up model for AI compute requirement. The market is fast moving and translating compute capacity for tokens generated to revenue earned is unclear and volatile.

OpenAI has provided significant details on its long-term revenue trajectory and compute cost outlook. Please refer to our report (*B2B up, B2C down: our new AI TAM forecasts to 2030* dated 9 June 2026) for more details. We have used OpenAI revenue and compute requirements to tether our compute requirement estimates for SpaceX. Once we get compute requirement in NVDA equivalent GW, we convert required compute into required gross PP&E assuming USD50bn capex for 1GW of Nvidia equivalent compute capacity.

In 2030e, SpaceX could be twice as compute intensive as OpenAI due to lower scale

In the long-term (2030e), we assume that SpaceX will be twice as compute intensive as OpenAI. We estimate that SpaceX will need 0.12GW of compute for USD1bn of annual revenue vs 0.06GW for OpenAI. This reflects historical trends and the fact that we expect OpenAI to have significantly larger economies of scale.

SpaceX: AI segment capex model

USDm, FYE Dec*	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Total installed compute (GW)	**-**	**0.30**	**0.80**	**2.00**	**2.80**	**3.60**	**4.39**	**5.07**
Compute offtake by Anthropic	-	-	-	0.65	-	-	-	-
Compute offtake by Google	-	-	-	0.28	-	-	-	-
Compute offtake by others	-	-	-	-	0.34	0.40	0.50	1.13
Compute required for self	**-**	**-**	**0.60**	**1.43**	**2.46**	**3.20**	**3.89**	**3.94**
Compute/Revenue for Grok (GW/USDbn)**		-	5.38	0.36	0.21	0.17	0.15	0.12
Compute/Revenue for OpenAI (GW/USDbn)	-	-	0.09	0.09	0.09	0.08	0.07	0.06
OpenAI Exit ARR (USDm)	0	0	21400	55249	92926	130384	174596	226081
OpenAI Exit compute capacity (GW)	-	-	1.90	5.00	8.00	10.00	12.00	12.00
Gross PP&E	463	6,096	18,823	76,546	116,546	156,546	196,083	229,987
Gross PP&E per GW of compute (USDm/GW)		20,320	23,529	38,273	41,624	43,485	44,658	45,373
Capex	463	5,633	12,727	57,723	40,000	40,000	40,000	40,000

Source: Company data, HSBC estimates. *Split between X and Grok is estimated by HSBC even for historical periods. **Using revenue of Grok and Cursor as denominator

Future markets – the unvalued unknown

In addition to the markets captured in the USD28trn TAM, SpaceX expects to create new categories of demand over time, fuelled by declining launch costs, more capable satellites, and the growth of largescale computing. As the opportunities are in the nascent stage with uncertain pace and ultimate size, they are excluded from quantifiable TAM calculations. However, SpaceX believes they represent trillions of dollars' worth of economic value.

- **Long-haul point-to-point terrestrial travel:** Starship's exceptional speed, reliability, and cost efficiency has the potential to reshape terrestrial commercial transport by reducing international long-haul journeys to under 30 minutes and enabling travel to almost any point on Earth in an hour or less, according to the company. Despite facing technical, economic, and regulatory hurdles, SpaceX believes it is well-positioned to capture share in terrestrial logistics and transportation.
- **Space tourism:** With the advancement of space technology and orbital flight infrastructure, SpaceX believes that access to space has the potential to create a new category of commercial human tourism, which has historically been limited to professional astronauts.
- **Passenger and cargo transport to the Moon and Mars:** Improvements in reusable launch systems and deep-space transport infrastructure has the potential to enable new forms of interplanetary logistics, including passenger and cargo transport to the Moon and Mars.
- **Energy production and manufacturing on the Moon and Mars:** A sustained human and industrial presence on the Moon and Mars would depend on reliable, largescale energy. Potential approaches include solar power systems that rely on thin atmospheres, strong solar exposure, and advanced energy technologies. SpaceX believes that progress in planetary infrastructure could eventually enable manufacturing using locally available resources.
- **In-orbit manufacturing:** Orbital infrastructure has the potential to enable high-value production, free from constraints such as gravity, thereby unlocking improvements in precision and efficiency. Microgravity conditions may support advances in pharmaceuticals, advanced materials, and semiconductors. In-orbit facilities could also draw on abundant, uninterrupted solar energy, supporting energy-intensive operations with improved sustainability.
- **Asteroid mining:** Asteroids may contain significant quantities of valuable resources (platinum-group metals, rare earth elements, nickel, cobalt, iron). With advances in reusable launch systems, autonomous robotics, and in-situ processing, SpaceX expects asteroid resources to become more accessible, eventually enabling a new category of commercial space resource extraction.

Financial statement review

- Our in-house forensic accountant reviews the financial statements in SpaceX's SEC S-1 registration statement
- SpaceX's accounts are complex and include significant management judgements, which may be subjective
- Although we have no significant concerns about the numbers, we advise caution given the amount of judgement applied to the relatively early stage of the company's development

Dylan Whitfield*, FCA
Head, Accounting Valuations & Forensic Accounting
HSBC Bank plc
dylan.b.whitfield@hsbc.com
+44 20 3359 5903

** Employed by a non-US affiliate of HSBC Securities (USA) Inc, and is not registered/ qualified pursuant to FINRA regulations*

Forensic accounting review – context

We discuss the governance and ownership structure of the company in the next section. However, with that context and given the relatively early-stage of development of a substantial portion of the company's future aspirations (acknowledging its progress to date), as part of this initiation we have sought to understand the key aspects of SpaceX's accounting risk with our in-house forensic accountant. We set out our observations below.

Overall impression

Whilst we do not have any significant immediate concerns about the company's accounting, we advise caution and that particular care is taken when considering the financial information and the company's provided financial disclosures. Inevitably, the company's financial information relies on subjective judgements, estimation, and management's view of the future aspired operations of the company. In addition, the early stage of development of substantial portions (although not all) of the business and the unconventional approach of the company's management should be considered when thinking about the financial reporting of the company and the control environment, which ultimately produces the information and disclosures.

Individual observations

We have reviewed SpaceX's S-1 registration statement (dated 1 June 2026) and noted a number of items that we think investors should be specifically aware of and/or monitor.
We describe our observations below. For the avoidance of doubt, the areas we highlight do not imply that we think there is any incorrect accounting, rather these are areas that we think that investors should be aware of as part of their investment decision-making approach.

Observations by accounting area

- **Recognition of revenue subject to material estimates.** The company has contracts in the Space and Connectivity segments that recognise revenue with reference to total cost at completion estimates and cost incurred to date (as one might expect). As per the audit report;

"…the Company records revenue based upon costs (such as materials and labour hours) incurred to date relative to the total estimated cost at completion. Developing the estimated total cost at completion for each performance obligation requires the use of significant management judgment, including assumptions…[1]".

[1] Audit report contained within the Form S-1 (page F-3)

Although we do not call into question the accounting, long-term contract accounting is, by definition, more complex than straightforward point-in-time sales, therefore we think that investors should be aware of the issue and be comfortable with the materiality of the amounts involved.

- **Accounting for launch vehicles.** How the company accounts for launch vehicles depends on whether the component is reusable or single use. Reusable sections are treated as a property, plant, and equipment (PPE) and single use sections are classified as inventory. This accounting makes logical sense, however, does add complexity when analysing the balance sheet and when considering capex. In addition, the company states that it capitalises refurbishments and renewals that are (put simply) not simply maintenance and repair and makes reference to the capitalisation of certain interest costs (although it is not clear from the S-1 document whether this includes interest costs related to launch vehicles). We note that depreciation is charged on the PPE costs based on number of launches a reusable component makes. Although logical, this does bring a certain amount of subjective estimation into the amount of cost taken to the income statement for each launch.
  In addition, the company considers "adjusted EBITDA" as an appropriate performance metric; however, we note that exclusion of the deprecation related to launch vehicles ignores substantial costs related to the Space segment.

- **Useful economic life of PPE assets.** As at 31 December 2025, the company held USD42,602m of PPE assets on its balance sheet, around 46% of the total assets of the company. The accounting for PPE is, by its nature, subjective as the company must estimate the likely asset life of the assets it holds. These estimates are set out on page F-14 of the S-1 form. As is commonplace, the company provides ranges for the various categories of assets it owns (e.g. datacentre infrastructure is given a 20–25-year asset life). Without granular detail of the assets (which companies are not required to present) it is not possible to conclude whether the chosen asset lives are appropriate; however, we think they appear to be towards the longer-end of what we would normally expect.

- **Support for long-term asset valuations.** The company holds long-term assets (e.g. goodwill, other intangibles assets, and deferred tax), all of which will ultimately require future profits and cash flows to support and maintain their valuation and, therefore, for the assets not be considered impaired. A company that is loss-making has an additional challenge in that losses can be considered one of many indicators of elevated impairment risk. However, we think that it is likely that the company will have prepared some form of future cash flow forecast to support the valuation as part of its year- and period-end financial reporting procedures, which satisfied the auditors that sufficient future economic value was likely to support the valuations. We also take comfort that in previous periods substantial impairments were taken by the company (e.g. USD3,775m related to the impairment of the "Twitter" brand).

- **Cash.** As is permitted, the company includes "highly liquid investments" in its cash balances (which is common practice). However, the company also holds USD377m of restricted cash, which in our view should not be considered when undertaking any analysis of cash flows (for the avoidance of doubt, we do not imply that the accounting is incorrect). Furthermore, if one considers combined operating and investing cash flows (as this company is developing substantial portions of its business), the company is utilising cash rather than generating cash. Whilst this is expected, we think that it is crucial to understand when management expect the consumption of cash to stabilise and the company to generate cash.

- **Share-based payments.** As is common practice, the company removes the accounting entries for share-based payments in its metrics. For the year ended 31 December 2025, this amounted to USD1,947m of cost excluded. As we have set out in *The forensic lens, Share-based payments: Perspective on the impenetrable*, 29 March 2021, we think that, although non-cash, these represent real expenses and therefore should not be removed from any metric that analyses the earnings performance of the business.
- **Alternative performance metric.** As stated above, the company uses "adjusted EBITDA" as an alternative performance metric. We think that this excludes the real costs of depreciation and amortisation, and the substantial interest expense and share-based compensation. Consequently, as we would with all companies, we advise caution when utilising this metric and be clear what the purpose of one's analysis is, if it is based upon this metric.
- **Accumulated deficit.** The company had an accumulated deficit of USD37,035m as at 31 December 2025. Whilst the company remains in net positive shareholders' funds, we note that in order to pay a dividend, a company must have sufficient cash and sufficient distributable reserves. What is permitted to be distributable depends on the legal definition (rather than accounting) rules in any given jurisdiction. Therefore, although a net negative accumulated deficit is not always an issue, the magnitude of the accounting reserves deficit may preclude any distributions by the company in the short-term.
- **Other observations**
  - The company is involved with material litigations and provided USD530m for litigation losses "…that are probable and reasonably estimable…". Whilst this is the correct accounting treatment (i.e. the company has not provided for the "worst case scenario") investors should be cognisant that litigations can be unpredictable and that provision could move materially if the cases are not ruled in the company's favour.
  - The company has long-term finance lease debt of around USD1,200m. It is common practice to exclude lease debt from overall indebtedness; however, accounting standards consider this balance akin to debt. We are aware of divergent views amongst investors as to whether or not lease debt should be considered debt. Therefore, we draw investor attention to this balance in order that they can take their preferred approach.

Learnings from Tesla

- Tesla's insourcing reflects desire to challenge existing auto practices and a lack of established supply chain; SpaceX could follow that model
- Where mature supply chains exist (battery) or Chinese suppliers offer cost leverage (Optimus hardware), Tesla is relying on partners
- SpaceX's capital needs to guide its insourcing strategy; for batteries, chips, and more it could rely on partners or collaborate with Tesla

Tesla's vertical integration by default, not design

Michael Tyndall*, CFA
Senior Global Autos Analyst
HSBC Bank plc
michael.tyndall@hsbc.com
+44 20 3359 6301

Pushkar Tendokar*
Global Autos Analyst
HSBC Securities and Capital Markets (India) Private Limited
pushkarnarendratendolkar@hsbc.co.in
+91 80 4555 2752

** Employed by a non-US affiliate of HSBC Securities (USA) Inc, and is not registered/ qualified pursuant to FINRA regulations*

Tesla presents itself as a disruptor of the long-held car assembly practices…
Traditionally, carmakers have relied heavily on automotive suppliers, both for hardware and automotive software. Tesla was among the first to break this trend, insourcing a sizeable portion of its supply chain, in part because the group wanted to challenge existing processes (mega-casting) or because the existing supply chain was not mature (software development). Building in-house expertise also allowed the group to innovate further. Software development (and application of AI) is driving projects, such as FSD, robotaxis, and humanoid robots, which distinguish it from incumbent OEM peers.

…and looking to extend those principles to other business, especially humanoids
Tesla's prominent vertical integration may have been one of the reasons for product delays, but it has overcome its initial "manufacturing hell"[2] and consistently expanded capacity, which currently stands close to 2.5m units, quadrupling from 600-700k around six years ago (see overleaf below). At the same time, it has managed to grow its Energy Storage Systems (ESS) business, although in this business it has played more of an integrator roll, focusing on software, while buying-in most of its cell needs. Tesla is looking to start production of robots by the end of 2026, where it will again need elevated levels of vertical integration because, in its own words, the supply chain does not currently exist. Both businesses have cross-functional utility – the R&D and AI compute (VLM) developed for the automotive business (especially robotaxis) have complementary uses in robotics.

2 'Tesla Survived Manufacturing Hell--Now Comes The Hard Part', Forbes, 20 December 2018

Despite delays, Tesla was able to ramp up vehicle production and energy storage and now targets mass-scale humanoid production

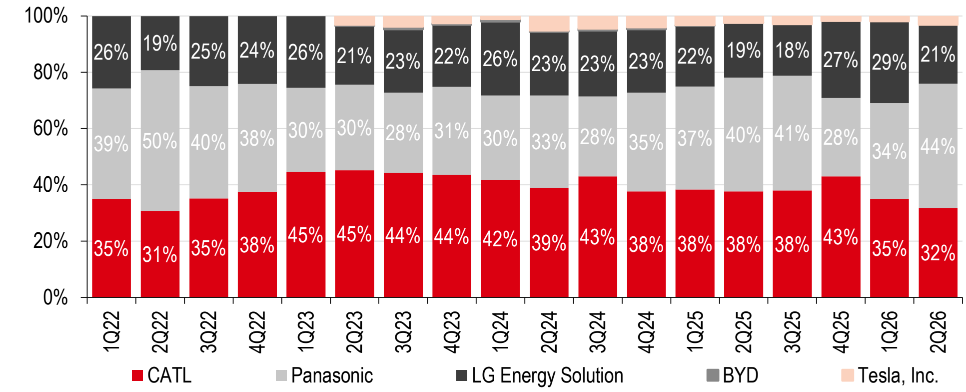
Region	Product	Current status	Capacity YE2019	YE2020	YE2021	YE2022	YE2023	YE2024	YE2025	1Q2026
Automotive										
California	Model 3/Model Y	Production	400k	500k	500k	550k	550k	>550k	>550k	>550k
	Model S/Model X	Production, end-2Q26e	90k	100k	100k	100k	100k	100k	100k	
Shanghai	Model 3/Model Y	Production	150k	450k	>450k	>750k	>950k	>950k	>950k	>950k
Berlin	Model Y	Production				>250k	375k	>375k	>375k	>375k
Texas	Model Y	Production				>250k	>250k	>250k	>250k	>250k
	Cybertruck	Production					>125k	>125k	>125k	>125k
	Cybercab	Pilot Production								
Nevada	Tesla Semi	Pilot Production								
TBD	Roadster	Design development								
Energy generation and storage										
California	Megapack	Production							40 GWh	40 GWh
Nevada	Powerwall	Production							>6 GWh	>6 GWh
Shanghai	Megapack	Production							40 GWh	20 GWh
Texas	Megapack	Construction								
Robotics										
California	Optimus	Construction								*1m*
Texas	Optimus	Construction								*10m*

Source: Company data; note that the humanoid capacity listed in the table above is not the current capacity but the capacity that the line/plant is being designed for

However, battery cell manufacturing is still largely outsourced by Tesla

Where the supply chain has matured, for example in the case of battery manufacturing, Tesla has chosen to rely on a supplier base. As the chart below shows, Tesla's battery needs are distributed across multiple Asian suppliers, whilst only a small portion is catered by inhouse manufacturing. Tesla is ramping up its new battery and material factories, but the scale of such in-house developments is small, and, in our view, Tesla is likely to continue to rely on external supply as its primary source of batteries in the medium-term.

For battery manufacturing, Tesla has relied on suppliers, and SpaceX, after having started with in-house batteries, also appears to be pursuing third party solutions



Source: Rho Motion, HSBC

The path taken by SpaceX is slightly different. It initially relied on in-house batteries but in November 2024 it was reported that (source: KED Global) SpaceX would use LG batteries for "primary and backup power supply, as well as energy storage systems". Thus, having started a different path than Tesla, SpaceX appears to be pursuing third parties as the source of its batteries. Given the scale of its capital needs, we do not think it would like to keep investing in battery manufacturing as long as reliable supply is available.

Humanoids shows more in-sourcing, but likely still reliant on Chinese supply chain
At its Q1 call, the CEO said that "*because it is a completely new supply chain, there's really nothing from the existing supply chains that exist in Optimus*", suggesting deeper vertical integration for humanoids than for vehicles. This is partially true, as seen through the company's recent announcements/actions like discontinuing Model S/X production at Fremont for setting up a 1m p.a. line for Optimus. Tesla also expects its in-house designed AI5/AI6 to power the Optimus robots. This indicates high degree of vertical integration. But for the hardware, press reports (source: SCMP, Feb 2026) suggest that "*Chinese suppliers form the backbone of Tesla's humanoid robot initiative*". Considering that a China-based supply chain has a demonstrated record of low costs and mass production, SpaceX may also follow a similar in-sourcing model for its hardware requirements. The caveat is export control restrictions and rare Earth metals in the case of motors for humanoids.

Semiconductor chips – Terafab to prepare for potential supply shortage in the future
Tesla has built a Cortex (supercomputer cluster) in Nevada that uses more than 100k NVIDIA H100-equivalent chips, powering the training for robotaxis/FSD and Optimus, and is in the process of more than doubling the capacity with another Cortex (Cortex 2).

Limited battery manufacturing but expanding AI compute with plans to build a Terafab to secure chip supply in the long-term

Region	Product	Current status	1Q 2026 capacity
AI training compute			
Texas	Cortex 1	Production	>100k H100e
	Cortex 2	Early Ramp	>130 H100e
Battery manufacturing			
Nevada	LFP	Early Ramp	7 GWh
Texas	4680	Production	40 GWh
	Cathode Materials	Early Ramp	10 GWh
	Lithium Refining	Early Ramp	30 GWh

Source: Company data, HSBC

Tesla is designing its own chips but currently the chips are manufactured by its partners, such as TSMC and Samsung. The CEO has said that even in the best-case scenario the chip production from these suppliers may not be enough. Tesla has therefore said that it would build a Terafab. At its Q1 2026 call, Tesla said that would require cUSD3bn spend and it would be capable of producing a few thousand wafers per month. It also indicated SpaceX taking up the initial phase of the scaled up Terafab. The USD3bn appears to be only the initial spend as SpaceX's S-1 filing states that it expects *"Terafab to be the world's largest chip manufacturing facility"* and cites *"long-term goal of producing one terawatt of compute each year"*.

Governance

- Board size and independence, and dual class share structure is comparable with other US IPOs
- However, concentrated ownership structure means the only person who can remove the CEO is Elon Musk himself
- Elon Musk LTIP heavily geared to share price performance and attaining long-term multiplanetary targets

Corporate governance

Yaryna Kobel*
Corporate Governance Analyst
HSBC Bank plc
yaryna.kobel@hsbc.com
+44 20 3359 6152

Paul Rossington*
Senior Global Tech Platforms Analyst
HSBC Bank plc
paul.rossington@hsbcib.com
+44 20 7991 6734

** Employed by a non-US affiliate of HSBC Securities (USA) Inc, and is not registered/ qualified pursuant to FINRA regulations*

Board size and independence

The board will consist of eight directors; this is in-line with market practices. Indeed, on average, the board size of an IPO company is six members in the EU and the UK, and seven or nine (in controlled companies) in the US[3]. Large boards help a company perform successfully in the short-term after an IPO (measured by Tobin's Q). They are also believed to avoid making risky decisions during turbulent times, such as an IPO[4].

Upon the completion of the offering, the board will include Elon Musk, Gwynne Shotwell, Antonio J. Gracias, Donald Harrison, and Luke Nosek (as the initial Class B Directors) and Ira Ehrenpreis, Randy Glein, and Steve Jurvetson (as the initial Common Stock Directors). Ira Ehrenpreis, Randy Glein, Donald Harrison, Steve Jurvetson, and Luke Nosek are considered by the company independent within the meaning of the listing standards of Nasdaq and Nasdaq Texas. We view a higher number of independent directors as positive, as it significantly and positively affects the stock performance (measured by the price-to-book ratio) achieved 12, 24, and 36 months after the IPO[5]; and largely independent boards are associated with a higher likelihood of survival post-IPO[6].

Investors should be aware that the company intends to utilise a controlled company exemption and does not expect to have a compensation and nominating committee that is composed entirely of independent directors. Nonetheless, the group will establish a compensation and nominating committee to oversee.

---

3 Davis Polk, IPO Governance Survey: Corporate Governance Practices in U.S. Initial Public Offerings, September 2024; F Bertoni, Corporate governance in European IPOs, the Oxford Handbook of IPOs, 2018
4 M Brogi, V Lagasio, V Pesic. Can governance help in making an IPO "successful"? New evidence from Europe, Journal of International Financial Management & Accounting 31(3), March 2020
5 M Brogi, V Lagasio, V Pesic. Can governance help in making an IPO "successful"? New evidence from Europe, Journal of International Financial Management & Accounting 31(3), March 2020
6 N Chancharat, C Krishnamurti, Board Structure and Survival of New Economy IPO Firms, Corporate governance: an international review, volume 20, issue 2, March 2012

Board structure

Executive officers and managing director

CEO, CTO and Chairman – Elon Musk
Has served as CEO, CTO, and Chairman of the SpaceX board since May 2002. He is also the Technoking and CEO and on the board of directors of Tesla since October 2008. In addition, he was also CTO and on the board of directors of X beginning October 2022, CEO and on the board of directors of xAI, beginning March 2023, and following the merger of the X and xAI in March 2025, became President, Treasurer, and CEO and on the board of directors of the combined entity prior to its acquisition by SpaceX in February 2026. Separately, Elon Musk is the founder and CEO of Neuralink Corp. and The Boring Company, an infrastructure company. Elon Musk co-founded PayPal acquired by eBay in October 2002, Zip2 Corporation acquired by Compaq in March 1999. He previously served on the board of Endeavor Group Holdings, Inc. from April 2021 to June 2022.

President and COO – Gwynne Shotwell
Has served as President and COO since 2008 and has been a member of the board of directors since 2009. She was previously VP Business Development from 2002 to 2008. Prior to joining SpaceX she held positions with Microcosm, Inc. and The Aerospace Corporation.Separately, Ms. Shotwell serves on the board of directors of Polaris, Inc., a manufacturer of powersports vehicles, and on the Northwestern University's Board of Trustees.

CFO – Bret Johnsen
Has served as CFO since 2011 and leads the global finance organization responsible for its long-term financial strategy, internal financial operations, interactions with the financial community, and the financial aspects of SpaceX's growth initiatives. Prior to joining SpaceX he served as CFO at Mindspeed Technologies, Inc. from 2008 to 2011, and prior to that spent 9 years at Broadcom Inc. between 1999 and 2008. Separately Bret Johnsen serves as a Trustee of the University of Southern California.

Non-management directors

Ira Ehrenpreis
Has served on the board since February 2026. Ira Ehrenpreis is a founder and managing member of DBL Partners venture capital formed in 2015, and was previously a partner at Technology Partners venture capital. He also serves on the board of directors of Tesla, is Chairman of the VCNetwork, Chair of the National Association of Corporate Directors Northern California and the Co-Chair of the Stanford Precourt Institute for Energy Advisory Council.

Randy Glein
Has served on the board since February 2026 and previously served as a board observer since 2009. He is co-founder and managing partner of DFJ Growth venture capital and currently serves on the board of directors of several private technology companies. He has previously served on the board of directors of Anaplan, Inc. and Tremor Video, Inc. Prior to joining DFJ Growth, he served as CFO of FeedBurner (acquired by Google in 2007) and VP of Tribune Company and its corporate investment group, Tribune Ventures.

Antonio J. Gracias
Has served on the board since October 2010. Antonio Gracias is a founder, CEO and CIO of Valor Management LLC private equity since 2001, and prior to that was a founder and Managing Member of MG Capital private equity from 1995 to 2000. He has also served on the board of Neuralink Corp. and The Boring Company since May 2026. Previously he served as a director of Tesla from 2007 to 2021, a director of Harmony Biosciences Holdings, Inc. from 2017 to 2026, Marathon Pharmaceuticals, LLC from November 2013 to 2017, and SolarCity Corporation from 2012 to 2016.

Donald Harrison
Has served on our board since February 2015 and has served as President, Global Partnerships and Corporate Development at Google LLC since 2017, prior to which he held the same role in a VP capacity from 2020, before which he was Google LLC Deputy General Counsel from 2005 to 2012. Separately he is on the board of directors of Reliance Jio.

Steve Jurvetson
Has served on the board since March 2009. Mr. Jurvetson is a co-founder of Future Ventures venture capital which he founded in 2019, and previously co-founded and served as Managing Director of Draper Fisher Jurvetson venture capital from 1995 to 2017. Prior to this he worked as an R&D engineer at Hewlett Packard, in product marketing at Apple Inc. and NeXT, and a management consultant at Bain & Co. He currently serves as a director of The Metals Company, a deep-sea mining exploration company. He previously served as a director of Tesla from 2009 to 2020, NeoPhotonics Corp. from 2004 to 2011, Planet Labs from 2011 to 2017 and D-Wave from 2003 to 2020. Before co-founding Future Ventures and Draper Fisher.

Luke Nosek
Has served on our board since July 2008. Mr. Nosek co-founded Gigafund venture capital firm in 2017, and has been Managing Partner since inception. Prior to this he co-founded Founders Fund venture capital in 2006, and served as General Partner until 2017. Prior to that, Mr. Nosek co-founded and served as Vice President of Business Development, Vice President of Marketing, and Vice President of Strategy of PayPal from 1998 to 2002. He also serves as a member of the board of directors of various private companies including Last Energy, Emerald Cloud Lab and ResearchGate. He has previously served as a board member of DeepMind prior to its acquisition by Google.

Source: Company filings

Dual-class share structure…

Following the completion of the IPO there will be two classes of common stock: Class A common stock and Class B common stock. Each share of Class A common stock will entitle its holder to one vote per share; each share of Class B common stock will entitle its holder to 10 votes per share. Class A shareholders and Class B shareholders will vote together as a single class on all matters to be voted on by shareholders, except Class B shareholders will be entitled to elect the majority of the board of directors (and to certain other votes – see below). The table below outlines the ownership structure of all outstanding Class A common and Class B common stock respectively before the offering.

Shares beneficially owned before the offering

Named executive officers and directors	Class A common stock	%	Class B common stock	%	Combined voting power %
Elon Musk	849,494,440	12.30%	5,569,053,075	93.60%	85.10%
Gwynne Shotwell	5,460,400	*	7,113,550	*	*
Bret Johnsen	9,583,690	*	—	*	*
Ira Ehrenpreis	809,050	*	564,650	*	*
Randy Glein	277,800	*	—	*	*
Antonio J. Gracias	503,414,530	7.30%	—	*	*
Donald Harrison	—	*	—	*	*
Steve Jurvetson	—	*	—	*	*
Luke Nosek	32,987,360	*	—	*	*
Total	**1,402,027,270**	**20.2%**	**5,576,731,275**	**93.70%**	**86.00%**

Source: Company data, HSBC. *Represents beneficial ownership or voting power of less than 1%.

Upon completion of the offering, Elon Musk will hold approximately 91.6% of all outstanding Class B common stock, and 82.4% of the total voting power of the firm's outstanding common stock and entitled to elect most of the board. Two or more classes of common stock outstanding after an IPO are practiced in the US; 15% of recent controlled-company IPOs had shares with unequal voting rights[7]. Separately, compensation awards made in respect of Class C common stock will be converted into awards in respect of Class A common stock on a one-for-one basis.

…means the only person who can remove the CEO is Elon Musk

What this means is that, if Elon Musk can only be removed from the board and other leadership positions by the affirmative vote of the holders of a majority of the outstanding shares of Class B common stock, then only Elon Musk can remove the CEO (himself); typically, this responsibility is reserved for the board of directors. We think investors should consider this.

Functional and gender diversity

The summary of director bios for all board members states the experience and expertise they bring to the boardroom. We think clarity in this area is key, as research suggests that only task-related knowledge, abilities, and experiences are beneficial for corporate decision-making[8] and functionally diverse boards are more likely to bring innovative solutions[9].

Out of eight board members, only one is a woman (13%). Currently, according to the OECD, 34% of board members in the US are women.

7 Davis Polk, IPO Governance Survey: Corporate Governance Practices in U.S. Initial Public Offerings, September 2024
8 T L Simons et al., Making use of difference: Diversity, debate, and decision comprehensiveness in top management teams, Academy of Management Journal 42(6), 1999
9 K A Bantel et al., Top management and innovations in banking: Does the composition of the top team make a difference?, Special Issue: Strategic Leaders and Leadership, Volume 10, Issue 1, 1989

Ethics, business conduct, and culture

The company states that in connection with the offering, the board will adopt a code of business conduct and ethics applicable to all employees, directors, and officers. Indeed, an IPO firm's culture and business ethics are important to its future performance; IPOs of strong culture firms are associated with more positive price revisions and higher initial returns, e.g., greater under-pricing[10]. Overall, firms with strong ethical culture outperform their peers by approximately 40 percentage points across various measures, including employee loyalty, innovation, adaptability and business growth[11]. And companies led by CEOs with culture among the top three drivers of their business had a three-year revenue CAGR more than double that of other companies: 9.1% compared with 4.4%[12].

With regard to corporate governance, we think investors should consider:

- How a lower degree of independence on the compensation and nominating committee could impact the effectiveness of oversight of the areas under the committee's responsibilities.
- Whether independent directors are likely to provide independent challenge and support to management.
- The controls that may be needed by the company (e.g., a maximum duration of this structure, or voting matters limited to a director and/or blocking a change of control) to ensure that investors can exercise influence over key corporate matters.
- Whether the board/committee has performed board skill mapping aligned with the strategic objectives of the company and wider diversity considerations.
- Whether future diversity needs have been identified and incorporated into succession plans.
- The methods and sources of information used by the board to monitor culture.

10 D Cumming et al., How does corporate culture affect IPO price formation?, Journal of Banking & Finance, volume 163, June 2024
11 LRN, Benchmark of Ethical Culture, 2021; Heidrick & Struggles, Aligning Culture with the Bottom Line: How Companies Can Accelerate Progress, 2021
12 Heidrick & Struggles, Aligning Culture with the Bottom Line: How Companies Can Accelerate Progress, 2021

Executive compensation

SpaceX plans to establish the compensation and nominating committee that will oversee executive compensation; we view this development as positive. Currently, compensation for executives includes: base salary, long-term incentive plans (LTIPs), e.g., stock options and RSUs, retirement benefits, and employee stock purchase plans. See below the 2025 summary compensation for executive directors, noting that Elon Musk draws a nominal salary only.

2025 summary compensation

Named executive officers	Year	Salary (USD)	Option awards (USD)	Stock awards (USD)	All other compensation (USD)	Total compensation (USD)
Elon Musk	2025	54,080	—	—	—	54,080
Gwynne Shotwell	2025	1,080,127 (4)	82,969,515	1,727,160	30,095	85,806,897
Bret Johnsen	2025	825,000	9,013,002	—	—	9,838,002

Source: Company data, HSBC

The company states in the S-1 filing that Elon Musk generally does not participate in any LTIPs. However, as part of efforts to incentivise him to achieve the firm's long-term business objectives, the board granted him a performance-based award of restricted shares of Class B common stock in January 2026. The restricted shares vest upon:

- The company's achievement of specified market capitalisation milestones (up to USD7.5trn) across 15 equal tranches, and
- The company's establishment of a permanent human colony on Mars, with at least one million inhabitants.

In connection with the xAI merger, the company assumed a performance stock award originally granted to Mr Musk by xAI in November 2025. It vests upon both:

- The achievement of specified market capitalisation milestones across 12 equal tranches (up to USD6.565trn, and
- The company's completion of non-Earth-based datacentres capable of delivering 100 terawatts of compute per year.

The company provides information on targets and methodology/adjustments made.

Clawback

The company plans to adopt a compensation clawback policy; we think investors should consider, when available, what this policy entails, as we discuss in *Executive pay: [claw] back to the future*, 11 December 2025.

We think investors should consider:

- Whether the performance metrics in the LTIPs are aligned with the future needs of the business and investor expectations.
- Whether the board/remuneration committee ensures the pay incentives do not encourage undesirable behaviour.
- Whether the company's clawback rules cover all time-based and performance-based awards.
- Whether the trigger events for clawback reflect the firm's governance framework, key risks, and culture.

For more information on how to assess corporate governance, see *How to assess practices in IPOs*, 12 February 2026

Estimates

Income statement summary

USDm	2024A	2025A	2026E	2027E	2028E
Revenue	**14,015**	**18,674**	**38,208**	**49,783**	**62,272**
yoy growth	34.9%	33.2%	104.6%	30.3%	25.1%
Cost of revenue	7,996	9,451	18,178	25,981	31,858
R&D	3,464	8,643	15,556	17,318	18,538
SG&A	1,813	2,644	4,672	7,950	10,391
Restructuring charges	213	487	(11)	-	-
Impairment	63	38	-	-	-
Total costs and expenses	13,549	21,263	38,394	51,248	60,787
EBIT	**466**	**(2,589)**	**(186)**	**(1,465)**	**1,484**
EBITDA	**4,290**	**4,112**	**15,426**	**22,728**	**33,671**
Adj. EBITDA	**5,350**	**6,584**	**18,890**	**28,099**	**39,022**
Interest expense	(1,580)	(1,945)	(2,458)	(2,196)	(1,464)
Interest income	371	492	1,294	1,314	505
Other income - net	985	(177)	(1,876)	-	-
Net other income	(224)	(1,630)	(3,040)	(883)	(958)
PBT	242	(4,219)	(3,226)	(2,348)	526
Tax	(549)	718	184	(399)	89
Net income (loss)	791	(4,937)	(3,409)	(1,949)	436
Other	(773)	-	(671)	-	-
Net income	**18**	**(4,937)**	**(4,080)**	**(1,949)**	**436**
Diluted net income	21	(4,937)	(4,080)	(1,949)	436
EPS (USDc)	-	-	-	-	-
Basic	**0.6**	**(168.7)**	**(120.7)**	**(14.9)**	**3.3**
Diluted	**0.2**	**(168.7)**	**(120.7)**	**(14.9)**	**3.3**
Weighted average shares					
Basic	2,848	2,926	10,775	13,072	13,072
Diluted	9,956	2,926	10,775	13,072	13,072
Dividends	-	-	-	-	-

Source: HSBC estimates

Cash flow statement summary

USDm	2024A	2025A	2026E	2027E	2028E
Net income (loss)	**791**	**(4,937)**	**(3,409)**	**(1,949)**	**436**
Adjustments					
D&A	3,824	6,701	15,612	24,194	32,187
SBC	784	1,947	3,475	5,370	5,351
Intangible asset impairment	-	-	-	-	-
Deferred income taxes	(675)	626	178	(399)	89
Unrealized gains on digital assets	(955)	112	344	-	-
Fixed asset impairment	135	88	5	-	-
Debt extinguishment	-	-	1,526	-	-
CFO pre W/C	**4,103**	**4,696**	**17,723**	**27,216**	**38,064**
	-	-	-	-	-
-					
Inventory	(309)	(413)	(4,824)	(363)	(1,660)
Prepaid expenses	(328)	(673)	(74)	-	-
Accounts payable	472	709	4,172	760	3,472
Deferred revenue	1,876	1,929	1,119	-	-
Operating lease liabilities, net	(37)	(56)	(5)	-	-
Other liabilities	346	1,136	464	-	-
Net working capital	1,673	2,089	(3,015)	1,024	434
Cash flows from operating activities	**5,776**	**6,785**	**14,708**	**28,240**	**38,498**
Capex	(11,163)	(20,737)	(68,587)	(53,847)	(59,109)
Capitalized interest	-	(169)	(7)	-	-
Proceeds from product rebates	-	118	1,195	-	-
Purchases of marketable securities	(3,542)	(611)	(7,801)	-	-
Maturities of marketable securities	3,712	548	-	-	-
Sales of marketable securities	193	1,457	-	-	-
Investments in affiliates	-	(86)	-	-	-
Other investing activities, net	4	(95)	(4)	-	-
Net cash used in investing activities	**(10,796)**	**(19,575)**	**(75,204)**	**(53,847)**	**(59,109)**
Cash flows from financing activities	-	-	-	-	-
Principal payments on finance leases	(154)	(295)	(82)	-	-
Debt proceeds	-	16,055	47,694	(20,025)	-
Payment of debt issuance costs	-	(66)	(23)	-	-
Debt repayments	(77)	(6,858)	(18,295)	-	-
Debt extinguishment premium	-	-	(1,153)	-	-
Share issuance	13,101	18,807	92,682	-	-
Employee equity awards plans	224	328	111	-	-
Payments for share repurchases	(1,021)	(1,125)	(4,346)	-	-
Taxes paid for equity awards	(243)	(496)	(500)	(870)	(909)
Net cash provided by financing activities	**11,830**	**26,350**	**116,088**	**(20,895)**	**(909)**
Fx impact	1	63	36	-	-
Change in cash	6,811	13,623	55,627	(46,502)	(21,521)
Cash at start	4,690	11,501	25,124	80,751	34,249
Cash at end	11,501	25,124	80,751	34,249	12,728
of which restricted	116	377	756	756	756
Non restricted cash	**11,385**	**24,747**	**79,995**	**33,493**	**11,972**
FCF	**(5,387)**	**(14,247)**	**(53,962)**	**(25,607)**	**(20,612)**

Source: HSBC estimates

Balance sheet summary

USDm	2024A	2025A	2026E	2027E	2028E
Current assets					
Cash and cash equivalents	11,385	24,747	79,995	33,493	11,972
Marketable securities	800	-	7,823	7,823	7,823
Accounts receivable	1,052	1,579	5,482	4,854	6,233
Inventory	2,003	2,416	7,028	7,391	9,051
Prepaid expenses and other current assets	868	2,210	1,636	1,636	1,636
Total current assets	**16,108**	**30,952**	**101,964**	**55,197**	**36,714**
Non-current assets					
Net PPE	21,147	42,602	99,189	148,443	175,365
Finance lease right-of-use assets	1,686	1,260	1,182	1,182	1,182
Intangible assets, net	2,211	1,548	1,432	1,432	1,432
Digital assets	1,749	1,637	1,293	1,293	1,293
Goodwill	11,129	11,809	71,681	71,681	71,681
Deferred tax assets	696	141	35	435	345
Other assets	2,336	2,130	2,682	2,682	2,682
Total non-current assets	**40,954**	**61,127**	**177,495**	**227,147**	**253,980**
Total assets	**57,062**	**92,079**	**279,459**	**282,345**	**290,695**
Current liabilities					
Accounts payable	4,413	11,792	14,702	15,461	18,934
Deferred revenue, current	5,498	6,111	7,207	7,207	7,207
Debt and finance leases, current	372	928	1,538	1,538	1,538
Accrued expenses and other current liabilities	1,508	2,569	5,689	5,689	5,689
Total current liabilities	**11,791**	**21,400**	**29,136**	**29,895**	**33,368**
Long-term liabilities					
Deferred revenue, net of current	4,681	6,005	6,029	6,029	6,029
Debt and finance leases, net of current	13,421	21,968	53,727	33,702	33,702
Other liabilities	1,365	1,381	1,320	1,320	1,320
Total long-term liabilities	**19,467**	**29,354**	**61,076**	**41,051**	**41,051**
Total liabilities	**31,258**	**50,754**	**90,212**	**70,946**	**74,419**
Commitments and contingencies	-	-	-	-	-
Redeemable convertible preferred stock	20,941	38,752	7,049	7,049	7,049
Equity	0	0	0	0	0
Common stock	3	4	6	6	6
Additional paid-in capital	35,865	37,706	221,281	246,251	251,602
Accumulated deficit	(32,098)	(37,035)	(40,844)	(43,663)	(44,136)
Accumulated other comprehensive income	1,093	1,898	1,755	1,755	1,755
Total shareholders' equity	**4,863**	**2,573**	**182,198**	**204,349**	**209,227**
Total liabilities, stock & equity	**57,062**	**92,079**	**279,459**	**282,345**	**290,695**
Net debt (cash)	**1,608**	**(1,851)**	**(32,553)**	**(6,076)**	**15,445**

Source: HSBC estimates

Ongoing legal disputes

Category	Allegation	Fine (issues/potential)	Status
Unfair trade practices	Digital Services Act violation through X's deceptive blue checkmark, non-compliant ad repository, data restrictions	EUR120m (issued)	Under appeal
Copyright infringement	Patent infringement in video streaming technology by Vine/Periscope	USD105m + USD67m (issued)	Under appeal
Copyright infringement	Failure to remove infringing music from X		Discovery stage
Unfair trade practices	GDPR violation by X's MoPub ad exchange		Under appeal
Data privacy	Data breach by X under GDPR and DSA violation		Decision awaited
Copyright infringement	Patent infringement in mobile app/website by X	USD167m (potential)	Discovery stage
Data privacy	Generation of non-consensual deepfakes by Grok		Under investigation
Data privacy	Violation of DSA through ineffective content moderation on Grok	Up to 6% of global turnover (potential)	Under investigation
Data privacy	Inquiry into Grok (by X) processing personal data, including minors', under GDPR.	Up to 4% of global turnover (potential)	Under investigation
Data privacy	Generation of non-consensual deepfakes by Grok	EUR100k/day (potential)	Decision under review
Environmental law	Violation of Clean Air Act by xAI's unlawful operation of gas turbines	Up to USD124.4k/day (potential)	Injunction motion pending
Workplace safety	Worker fatality at Starbase		Under investigation
Workplace safety	Crane collapse and equipment failures at test sites	USD115.85k (issued)	Contested

Source: Company filings

Companies mentioned in this report

Company	RIC code	LCY	Mkt Cap (USDm)	CMP	TP	Rating
Adobe	ADBE.OQ	USD	86,798	218.36	308.00	Buy
Alphabet	GOOGL.N	USD	4,113,004	342.09	420.00	Buy
Amazon	AMZN.OQ	USD	2,633,878	244.85	310.00	Buy
Anthropic	private					
Apple	AAPL.O	USD	4,786,461	325.89	366.00	Buy
Arianespace	private					
AST SpaceMobile	ASTS.O	USD	24,044	61.95	NA	Not rated
AT&T	T.N	USD	157,879	23.04	27.00	Hold
Blue Origin	private					
Broadcom	AVGO.N	USD	1,887,856	396.81	600.00	Buy
CoreWeave	CRWV.N	USD	46,747	82.64	36.00	Reduce
Cursor	private					
Echostar	ECHO.O	USD	26,744	92.28	NA	Not rated
Eutelsat	ETL.PA	EUR	2,884	2.15	NA	Not rated
Globalstar	GSAT.O	USD	10,295	79.94	NA	Not rated
Intel	INTC.OQ	USD	515,768	102.62	200.00	Buy
Intuit	INTU.O	USD	77,813	284.47	707.00	Buy
Meta Platforms	META.O	USD	1,592,413	627.17	905.00	Buy
Micron	MU.N	USD	1,083,630	959.48	1,700.00	Buy
Microsoft	MSFT.O	USD	2,899,616	390.34	567.00	Buy
Nvidia	NVDA.OQ	USD	5,131,852	212.06	325.00	Buy
OpenAI	private					
Oracle	ORCL.OQ	USD	362,478	125.84	316.00	Buy
Palantir	PLTR.N	USD	286,022	124.57	151.00	Hold
Rocket Labs	RKLB.O	USD	41,723	69.75	NA	Not rated
Roscosmos	state-owned					
Salesforce	CRM.N	USD	133,497	163.00	356.00	Buy
Samsung Electronics	005930.KS	KRW	1,068,424	270,000.00	450,000.00	Buy
SAP	SAPG.DE	EUR	185,183	132.04	185.00	Buy
ServiceNow	NOW.N	USD	98,449	95.46	177.00	Buy
SK Hynix	000660.KS	KRW	925,728	1,919,000	4,000,000	Buy
Tesla	TSLA.N	USD	1,404,678	374.01	125.00	Reduce
T-Mobile US	TMUS.O	USD	206,636	190.94	285.00	Buy
TSMC	2330.TW	TWD	1,924,919	2,405.00	3,400.00	Buy
United Airlines	UAL.O	USD	38,061	117.26	NA	Not rated
United Launch Alliance	private					
Verizon	VZ.N	USD	184,935	44.29	46.30	Hold

Source: HSBC Research, LSEG DataStream. Priced as on 22 July 2026

Disclosure appendix

Analyst Certification

The following analyst(s), economist(s), or strategist(s) who is(are) primarily responsible for this report, including any analyst(s) whose name(s) appear(s) as author of an individual section or sections of the report and any analyst(s) named as the covering analyst(s) of a subsidiary company in a sum-of-the-parts valuation certifies(y) that the opinion(s) on the subject security(ies) or issuer(s), any views or forecasts expressed in the section(s) of which such individual(s) is(are) named as author(s), and any other views or forecasts expressed herein, including any views expressed on the back page of the research report, accurately reflect their personal view(s) and that no part of their compensation was, is or will be directly or indirectly related to the specific recommendation(s) or views contained in this research report: Nicolas Cote-Colisson, Charlie Rothbarth, Paul Rossington, Abhishek Shukla, CFA, Dylan Whitfield, FCA, Yaryna Kobel, Madhvendra Singh, CFA, Phani Kanumuri, Michael Tyndall, CFA, Pushkar Tendolkar, Mohammed Khallouf, CFA, Stephen Bersey, Corey Chan, Frank Lee and Ali Naqvi

Important disclosures

Equities: Stock ratings and basis for financial analysis

HSBC and its affiliates, including the issuer of this report ("HSBC") believes an investor's decision to buy or sell a stock should depend on individual circumstances such as the investor's existing holdings, risk tolerance and other considerations and that investors utilise various disciplines and investment horizons when making investment decisions. Ratings should not be used or relied on in isolation as investment advice. Different securities firms use a variety of ratings terms as well as different rating systems to describe their recommendations and therefore investors should carefully read the definitions of the ratings used in each research report. Further, investors should carefully read the entire research report and not infer its contents from the rating because research reports contain more complete information concerning the analysts' views and the basis for the rating.

From 23rd March 2015 HSBC has assigned ratings on the following basis:

The target price is based on the analyst's assessment of the stock's actual current value, although we expect it to take six to 12 months for the market price to reflect this. When the target price is more than 20% above the current share price, the stock will be classified as a Buy; when it is between 5% and 20% above the current share price, the stock may be classified as a Buy or a Hold; when it is between 5% below and 5% above the current share price, the stock will be classified as a Hold; when it is between 5% and 20% below the current share price, the stock may be classified as a Hold or a Reduce; and when it is more than 20% below the current share price, the stock will be classified as a Reduce.

Our ratings are re-calibrated against these bands at the time of any 'material change' (initiation or resumption of coverage, change in target price or estimates).

Upside/Downside is the percentage difference between the target price and the share price.

Prior to this date, HSBC's rating structure was applied on the following basis:

For each stock we set a required rate of return calculated from the cost of equity for that stock's domestic or, as appropriate, regional market established by our strategy team. The target price for a stock represented the value the analyst expected the stock to reach over our performance horizon. The performance horizon was 12 months. For a stock to be classified as Overweight, the potential return, which equals the percentage difference between the current share price and the target price, including the forecast dividend yield when indicated, had to exceed the required return by at least 5 percentage points over the succeeding 12 months (or 10 percentage points for a stock classified as Volatile*). For a stock to be classified as Underweight, the stock was expected to underperform its required return by at least 5 percentage points over the succeeding 12 months (or 10 percentage points for a stock classified as Volatile*). Stocks between these bands were classified as Neutral.

*A stock was classified as volatile if its historical volatility had exceeded 40%, if the stock had been listed for less than 12 months (unless it was in an industry or sector where volatility is low) or if the analyst expected significant volatility. However, stocks which we did not consider volatile may in fact also have behaved in such a way. Historical volatility was defined as the past month's average of the daily 365-day moving average volatilities. In order to avoid misleadingly frequent changes in rating, however, volatility had to move 2.5 percentage points past the 40% benchmark in either direction for a stock's status to change.

Rating distribution for long-term investment opportunities

As of 30 June 2026, the distribution of all independent ratings published by HSBC is as follows:

Buy	61%	(13% of these provided with Investment Banking Services in the past 12 months)
Hold	34%	(10% of these provided with Investment Banking Services in the past 12 months)
Sell	5%	(8% of these provided with Investment Banking Services in the past 12 months)

For the purposes of the distribution above the following mapping structure is used during the transition from the previous to current rating models: under our previous model, Overweight = Buy, Neutral = Hold and Underweight = Sell; under our current model Buy = Buy, Hold = Hold and Reduce = Sell. For rating definitions under both models, please see "Stock ratings and basis for financial analysis" above.

For the distribution of non-independent ratings published by HSBC, please see the disclosure page available at *http://www.hsbcnet.com/gbm/financial-regulation/investment-recommendations-disclosures*.

To view a list of all the independent fundamental ratings/recommendations disseminated by HSBC during the preceding 12-month period, and the location where we publish our quarterly distribution of non-fundamental recommendations (applicable to Fixed Income and Currencies research only), please use the following links to access the disclosure page:

Clients of HSBC Private Bank: *www.research.privatebank.hsbc.com/Disclosures*

All other clients: *www.research.hsbc.com/A/Disclosures*

HSBC & Analyst disclosures

Disclosure checklist

Company	Ticker	Recent price	Price date	Disclosure
SPACEX	SPCX.N	115.26	22 Jul 2026	7

Source: HSBC

1 HSBC has managed or co-managed a public offering of securities for this company within the past 12 months.
2 HSBC expects to receive or intends to seek compensation for investment banking services from this company in the next 3 months.
3 At the time of publication of this report, HSBC Securities (USA) Inc. is a Market Maker in securities issued by this company.
4 As of 30 June 2026, HSBC beneficially owned 1% or more of a class of common equity securities of this company.
5 This company was a client of HSBC or had during the preceding 12 month period been a client of and/or paid compensation to HSBC in respect of investment banking services.
6 This company was a client of HSBC or had during the preceding 12 month period been a client of and/or paid compensation to HSBC in respect of non-investment banking securities-related services.
7 This company was a client of HSBC or had during the preceding 12 month period been a client of and/or paid compensation to HSBC in respect of non-securities services.
8 A covering analyst/s has received compensation from this company in the past 12 months.
9 A covering analyst/s or a member of his/her household has a financial interest in the securities of this company, as detailed below.
10 A covering analyst/s or a member of his/her household is an officer, director or supervisory board member of this company, as detailed below.
11 At the time of publication of this report, HSBC is a non-US Market Maker in securities issued by this company and/or in securities in respect of this company.
12 As of 17 July 2026, HSBC beneficially held a net long position of more than 0.5% of this company's total issued share capital, calculated according to the SSR methodology.
13 As of 17 July 2026, HSBC beneficially held a net short position of more than 0.5% of this company's total issued share capital, calculated according to the SSR methodology.
14 HSBC Qianhai Securities Limited holds 1% or more of a class of common equity securities of this company.

HSBC and its affiliates will from time to time sell to and buy from customers the securities/instruments, both equity and debt (including derivatives) of companies covered in HSBC Research on a principal or agency basis or act as a market maker or liquidity provider in the securities/instruments mentioned in this report.

Analysts, economists, and strategists are paid in part by reference to the profitability of HSBC which includes investment banking, sales & trading, and principal trading revenues.

Whether, or in what time frame, an update of this analysis will be published is not determined in advance.

Non-U.S. analysts may not be associated persons of HSBC Securities (USA) Inc, and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with the subject company, public appearances and trading securities held by the analysts.

Economic sanctions laws imposed by certain jurisdictions such as the US, the EU, the UK, and others, may prohibit persons subject to those laws from making certain types of investments, including by transacting or dealing in securities of particular issuers, sectors, or regions. This report does not constitute advice in relation to any such laws and should not be construed as an inducement to transact in securities in breach of such laws.

For disclosures in respect of any company mentioned in this report, please see the most recently published report on that company available at *www.hsbcnet.com/research*. HSBC Private Bank clients should contact their Relationship Manager for queries regarding other research reports. In order to find out more about the proprietary models used to produce this report, please contact the authoring analyst.

Additional disclosures

1 This report is dated as at 23 July 2026.
2 All market data included in this report are dated as at close 22 July 2026, unless a different date and/or a specific time of day is indicated in the report.
3 HSBC has procedures in place to identify and manage any potential conflicts of interest that arise in connection with its Research business. HSBC's analysts and its other staff who are involved in the preparation and dissemination of Research operate and have a management reporting line independent of HSBC's Investment Banking business. Information Barrier procedures are in place between the Investment Banking, Principal Trading, and Research businesses to ensure that any confidential and/or price sensitive information is handled in an appropriate manner.
4 You are not permitted to use, for reference, any data in this document for the purpose of (i) determining the interest payable, or other sums due, under loan agreements or under other financial contracts or instruments, (ii) determining the price at which a financial instrument may be bought or sold or traded or redeemed, or the value of a financial instrument, and/or (iii) measuring the performance of a financial instrument or of an investment fund.

Production & distribution disclosures

1. This report was produced and signed off by the author on 23 Jul 2026 14:26 GMT.
2. In order to see when this report was first disseminated please see the disclosure page available at https://www.research.hsbc.com/R/34/JMv7VFB

Disclaimer

Legal entities as at 13 May 2026:
HSBC Bank plc; HSBC Continental Europe; HSBC Continental Europe SA, Germany; HSBC Bank Middle East Limited, DIFC; HSBC Bank Middle East Limited, Dubai branch; HSBC Yatirim Menkul Degerler AS, Istanbul; The Hongkong and Shanghai Banking Corporation Limited, Hong Kong; The Hongkong and Shanghai Banking Corporation Limited, Singapore Branch; The Hongkong and Shanghai Banking Corporation Limited, Seoul Securities Branch; The Hongkong and Shanghai Banking Corporation Limited, Seoul Branch; HSBC Qianhai Securities Limited; HSBC Securities (Taiwan) Corporation Limited; HSBC Securities and Capital Markets (India) Private Limited, Mumbai; HSBC Bank Australia Limited; HSBC Securities (USA) Inc., New York; HSBC México, SA, Institución de Banca Múltiple, Grupo Financiero HSBC; Banco HSBC SA

Issuer of report
HSBC Continental Europe
38 avenue Kléber
75116 Paris, France
Telephone: +33 1 40 70 70 40
Fax: +33 1 40 70 70 09
Website: www.research.hsbc.com

[1281160]

HSBC Global Investment Research

Issuer of report:
HSBC Continental Europe
38 avenue Kléber
75116 Paris, France
Telephone: +33 1 4070 7040
Fax: +33 1 4070 7009

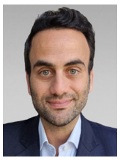
Main contributors

Nicolas Cote-Colisson*
MD, Head of Global Tech Platforms
HSBC Continental Europe
nicolas.cote-colisson@hsbcib.com
+44 20 7991 6826

Charlie Rothbarth*
Global Tech Platforms Analyst
HSBC Bank plc
charlie.rothbarth@hsbc.com
+44 20 3268 5284

Abhishek Shukla*, CFA
Senior Analyst, Technology
HSBC Bank Middle East Limited, DIFC
abhishek2.shukla@hsbc.com
+971 4 509 3343

Paul Rossington*
Senior Global Tech Platforms Analyst
HSBC Bank plc
paul.rossington@hsbcib.com
+44 20 7991 6734



Mohammed Khallouf*, CFA
Global Tech Platforms Analyste
HSBC Bank Middle East Limited, DIFC
mohammed.khallouf@hsbc.com
+971 4509 3337

Frank Lee*
Global Head of Tech Hardware & Semi Research
The Hongkong and Shanghai Banking Corporation Limited
frank.lee@hsbc.com.hk
+852 2996 6916

Madhvendra Singh*, CFA
Senior Analyst, TMT
HSBC Bank Middle East Limited, DIFC
madhvendra.singh@hsbc.com
+971 4509 3348

Phani Kanumuri
Analyst, TMT
HSBC Securities (USA) Inc.
phani.kanumuri@us.hsbc.com
+1 240 709 8135

Michael Tyndall*, CFA
Senior Global Autos Analyst
HSBC Bank plc
michael.tyndall@hsbc.com
+44 20 3359 6301

Pushkar Tendolkar*
Global Autos Analyst
HSBC Securities and Capital Markets (India) Private Limited
pushkarnarendratendolkar@hsbc.co.in
+91 80 4555 2752

Corey Chan* (Reg. No. S1700518100001)
Head, A-share Industrials & Renewables Research
HSBC Qianhai Securities Limited
corey.chan@hsbcqh.com.cn
+86 21 5066 2001

Dylan Whitfield*, FCA
Head, Accounting Valuations & Forensic Accounting
HSBC Bank plc
dylan.b.whitfield@hsbc.com
+44 20 3359 5903

Stephen Bersey
Head of US Technology Research
HSBC Securities (USA) Inc
stephen.bersey@us.hsbc.com
+1 212 525 4153

Yaryna Kobel
Corporate Governance Analyst
HSBC Bank plc
yaryna.kobel@hsbc.com
+44 20 3359 6152

* Employed by a non-US affiliate of HSBC Securities (USA) Inc, and is not registered / qualified pursuant to FINRA regulations

How to access our research

HSBC Global Investment Research app: